Mathematical Statistics II

Lecture notes based on Kim, *Mathematical Statistics*, Ch. 6–11

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Chapter 6. Estimation

Setting

$X_1, \dots, X_n$ is a random sample from a pdf $f(x;\theta)$ with unknown parameter $\theta \in \Omega$. Guessing the value of θ, or of a function $\eta = \eta(\theta)$, from the sample is **estimation**. A statistic $\hat{\eta} = \hat{\eta}(X_1, \dots, X_n)$ used for this is an **estimator**, and its value at the data is an **estimate**. We write P_θ, E_θ, Var_θ to stress that probabilities are computed under the model with parameter θ. The most basic requirement is that $\hat{\eta}_n$ approach its target as n grows.

Definition (Consistency)

$\hat{\eta}_n$ is **consistent** for $\eta(\theta)$ if $\hat{\eta}_n \xrightarrow{p} \eta(\theta)$ under P_θ for every $\theta \in \Omega$, i.e. $\lim_n P_\theta(|\hat{\eta}_n - \eta(\theta)| \geq \epsilon) = 0$ for all $\epsilon > 0$ and all θ.

6.1 The method of moments

To estimate the population mean we use the sample mean; likewise the population moment $m_r = EX_1^r$ is naturally estimated by the sample moment $\hat{m}_r = \frac{1}{n}\sum X_i^r$. If a parameter can be written as $\eta = g(m_1, \dots, m_k)$, the **method-of-moments estimator** (MME) is

$$\hat{\eta}^{\mathrm{MME}} = g(\hat{m}_1, \dots, \hat{m}_k).$$

For a multivariate population one uses joint moments $m_{r,s} = E(X_1^r Y_1^s)$ and their sample versions $\hat{m}_{r,s} = \frac{1}{n}\sum X_i^r Y_i^s$ in the same way.

Example 6.1

(a) $\sigma^2 = m_2 - m_1^2$ and $\sigma = \sqrt{m_2 - m_1^2}$, so

$$\hat{\sigma}^{2\,\mathrm{MME}} = \hat{m}_2 - \hat{m}_1^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X})^2, \qquad \hat{\sigma}^{\mathrm{MME}} = \sqrt{\frac{1}{n}\sum (X_i - \bar{X})^2}.$$

(b) The MME is not unique. For Poisson(λ), $\lambda = m_1 = m_2 - m_1^2$, which gives the two MMEs $\bar{X}$ and $\frac{1}{n}\sum (X_i - \bar{X})^2$. One normally uses the lowest moments possible.

(c) For a bivariate population, $\rho = \dfrac{m_{1,1} - m_{1,0}m_{0,1}}{\sqrt{m_{2,0} - m_{1,0}^2}\sqrt{m_{0,2} - m_{0,1}^2}}$. Substituting sample moments gives

$$\hat{\rho}^{\mathrm{MME}} = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2}\sqrt{\sum (Y_i - \bar{Y})^2}},$$

the sample correlation coefficient.

(d) For Gamma(α, β), $m_1 = \alpha\beta$ and $m_2 - m_1^2 = \alpha\beta^2$. Solving, $\hat{\beta} = \hat{\sigma}^2/\bar{X}$ and $\hat{\alpha} = \bar{X}^2/\hat{\sigma}^2$ with $\hat{\sigma}^2 = \frac{1}{n}\sum (X_i - \bar{X})^2$.

Theorem 6.1 (Consistency of MMEs)

If $E|X_1|^k < \infty$ and g is continuous at $(m_1, \dots, m_k)$, then $\hat{\eta}_n^{\mathrm{MME}} = g(\hat{m}_1, \dots, \hat{m}_k) \xrightarrow{p} g(m_1, \dots, m_k) = \eta$.

Proof. By the multivariate law of large numbers, $(\hat{m}_1, \dots, \hat{m}_k)' \xrightarrow{p} (m_1, \dots, m_k)'$ (Example 5.5 in Volume I). Continuous functions preserve convergence in probability (Theorem 5.7 in Volume I). ■

For example, $\hat{\sigma}^{2\,\mathrm{MME}}$ is consistent, and hence so is $S^2 = \frac{n}{n-1}\hat{\sigma}^{2\,\mathrm{MME}}$; the sample correlation is consistent for ρ. Consistency is the first property to check for any proposed estimator, and Theorem 6.1 shows that MMEs usually have it. They are also asymptotically normal:

Theorem 6.2 (Limiting distribution of MMEs)

If $EX_1^{2k} < \infty$ and g has continuous first partial derivatives at $\mathbf{m} = (m_1, \dots, m_k)'$, then

$$\sqrt{n}(\hat{\eta}_n^{\mathrm{MME}} - \eta) \xrightarrow{d} N\big(0,\ \dot{g}(\mathbf{m})'\Sigma\,\dot{g}(\mathbf{m})\big), \qquad \Sigma = (m_{r+s} - m_r m_s)_{r,s=1}^{k}.$$

Proof. The vectors $(X_i, X_i^2, \dots, X_i^k)'$ are i.i.d. with mean $\mathbf{m}$ and covariance matrix Σ, since $\mathrm{Cov}(X_1^r, X_1^s) = m_{r+s} - m_r m_s$. The multivariate CLT gives $\sqrt{n}(\hat{\mathbf{m}} - \mathbf{m}) \xrightarrow{d} N_k(\mathbf{0}, \Sigma)$, and the delta method (Theorem 5.12 in Volume I) finishes the proof. ■

Example 6.2

For $\hat{\sigma}^{2\,\mathrm{MME}} = g(\hat{m}_1, \hat{m}_2)$ with $g(m_1, m_2) = m_2 - m_1^2$, $\dot{g} = (-2m_1, 1)'$, and a direct computation of $\dot{g}'\Sigma\dot{g}$ gives $\mu_4 - \sigma^4$, with $\mu_4 = E(X_1 - \mu)^4$. This agrees with Example 5.10 in Volume I.

6.2 Maximum likelihood

Suppose the model is $\{f(x;\theta) : \theta \in \Omega\}$. The probability that the sample falls in a small box around the observed $\mathbf{x} = (x_1, \dots, x_n)'$ is about $\prod_i f(x_i;\theta)\,|\Delta\mathbf{x}|$. Comparing $\prod f(x_i;\theta)$ with $\prod f(x_i;\theta')$ tells us which of two parameter values makes the observed data more plausible. So we regard the joint pdf as a function of θ with the data fixed.

Definition (Likelihood and MLE)

(a) The **likelihood function** is $L(\theta) = L(\theta;\mathbf{x}) = \prod_{i=1}^n f(x_i;\theta)$, $\theta \in \Omega$, and $l(\theta) = \log L(\theta) = \sum \log f(x_i;\theta)$ is the **log-likelihood**.

(b) A value $\hat{\theta}(\mathbf{x}) \in \Omega$ with $L(\hat{\theta}(\mathbf{x});\mathbf{x}) = \max_{\theta\in\Omega} L(\theta;\mathbf{x})$ is a **maximum likelihood estimate**; the statistic $\hat{\theta}^{\mathrm{MLE}} = \hat{\theta}(\mathbf{X})$ is the **maximum likelihood estimator** (MLE).

The derivative $\dot{l}(\theta)$ is easier to handle than that of L. The equation $\dot{l}(\theta) = 0$ is the **likelihood equation**, and the MLE is usually found among its roots.

Example 6.3 (Poisson)

For a Poisson(λ) sample, $\lambda \geq 0$ (with $\lambda = 0$ the point mass at 0),

$$l(\lambda) = -n\lambda + n\bar{x}\log\lambda - \log(x_1!\cdots x_n!), \qquad \dot{l}(\lambda) = -n + \frac{n\bar{x}}{\lambda}, \qquad \ddot{l}(\lambda) = -\frac{n\bar{x}}{\lambda^2}.$$

If $\bar{x} > 0$, l increases up to $\bar{x}$ and decreases afterwards, so $\max_{\lambda>0} l = l(\bar{x})$. If $\bar{x} = 0$, $l(\lambda) = -n\lambda$ is decreasing and the supremum is the limit at $\lambda = 0$. Since L is continuous on $[0, \infty)$, in both cases $\max_{\lambda\geq 0} L(\lambda) = L(\bar{x})$, and $\hat{\lambda}^{\mathrm{MLE}} = \bar{X}$.

Theorem 6.3

Let l be twice differentiable with continuous second derivative on an open interval Ω_o. If $\ddot{l}(\theta) < 0$ for all $\theta \in \Omega_o$ and $\dot{l}(\hat{\theta}) = 0$ for some $\hat{\theta} \in \Omega_o$, then $\hat{\theta}$ is the unique maximizer of l on Ω_o.

Proof. By Taylor's theorem, for each $\theta \in \Omega_o$ there is θ^* between θ and $\hat{\theta}$ with

$$l(\theta) = l(\hat{\theta}) + \dot{l}(\hat{\theta})(\theta - \hat{\theta}) + \tfrac{1}{2}\ddot{l}(\theta^*)(\theta - \hat{\theta})^2 = l(\hat{\theta}) + \tfrac{1}{2}\ddot{l}(\theta^*)(\theta - \hat{\theta})^2 < l(\hat{\theta})$$

for $\theta \neq \hat{\theta}$. ■

Example 6.4 (Bernoulli)

For a Bernoulli(p) sample, $0 \le p \le 1$, with $\hat{p} = \bar{x}$,

$$l(p) = n\hat{p}\log p + n(1-\hat{p})\log(1-p), \qquad \dot{l}(p) = \frac{n\hat{p}}{p} - \frac{n(1-\hat{p})}{1-p}, \qquad \ddot{l}(p) = -\frac{n\hat{p}}{p^2} - \frac{n(1-\hat{p})}{(1-p)^2}.$$

If $0 < \hat{p} < 1$, then $\ddot{l} < 0$ on $(0,1)$ and $\dot{l}(\hat{p}) = 0$, so $\hat{p}$ maximizes l by Theorem 6.3. If $\hat{p} = 0$ (all $x_i = 0$), $L(p) = (1-p)^n$ is maximized at $p = 0$; if $\hat{p} = 1$, $L(p) = p^n$ is maximized at $p = 1$. In all cases $\hat{p}^{\mathrm{MLE}} = \bar{X}$.

The condition $\ddot{l} < 0$ says that l is **strictly concave**: $l(\alpha\theta_1 + (1-\alpha)\theta_2) > \alpha l(\theta_1) + (1-\alpha)l(\theta_2)$ for $0 < \alpha < 1$ and $\theta_1 \neq \theta_2$. When the likelihood equation cannot be solved explicitly, the following guarantees a unique root that is the MLE.

Theorem 6.4

Let l be twice continuously differentiable on an open interval Ω_o, with $\ddot{l}(\theta) < 0$ on Ω_o and $l(\theta) \to -\infty$ as θ tends to either end of Ω_o. Then the likelihood equation $\dot{l}(\theta) = 0$ has exactly one root $\hat{\theta}$ in Ω_o, and $\hat{\theta}$ is the unique maximizer of l.

Proof. Consider $\Omega_o = (-\infty, \infty)$; other intervals are similar. Since $l \to -\infty$ at both ends and l is continuous, l attains its maximum at some $\hat{\theta}$. If it were attained at two points $\hat{\theta}_1 < \hat{\theta}_2$, strict concavity would give

$$l(\tfrac{1}{2}\hat{\theta}_1 + \tfrac{1}{2}\hat{\theta}_2) > \tfrac{1}{2}l(\hat{\theta}_1) + \tfrac{1}{2}l(\hat{\theta}_2) = \max l,$$

a contradiction. At the interior maximum $\dot{l}(\hat{\theta}) = 0$. Since $\dot{l}$ is strictly decreasing ($\ddot{l} < 0$), it has no other zero. ■

When the root has no closed form, it can be computed by **Newton–Raphson**: linearize $0 = \dot{l}(\hat{\theta}) \approx \dot{l}(\theta^{(r)}) + \ddot{l}(\theta^{(r)})(\hat{\theta} - \theta^{(r)})$ and iterate

$$\hat{\theta}^{(r+1)} = \hat{\theta}^{(r)} + \big[-\ddot{l}(\hat{\theta}^{(r)})\big]^{-1}\dot{l}(\hat{\theta}^{(r)}), \qquad r = 0, 1, \ldots,$$

starting, for instance, from the MME $\hat{\theta}^{(0)}$.

Example 6.5 (Logistic location)

For the logistic model $f(x;\theta) = e^{-(x-\theta)}/(1+e^{-(x-\theta)})^2$, $\theta \in \mathbb{R}$,

$$l(\theta) = -n\bar{x} + n\theta - 2\sum_{i=1}^{n}\log\left(1 + e^{-(x_i-\theta)}\right), \qquad \dot{l}(\theta) = n - 2\sum_i \frac{e^{-(x_i-\theta)}}{1+e^{-(x_i-\theta)}},$$

$$\ddot{l}(\theta) = -2\sum_i \frac{e^{-(x_i-\theta)}}{(1+e^{-(x_i-\theta)})^2} < 0.$$

As $\theta \to -\infty$, $l(\theta) \approx n\theta - n\bar{x} \to -\infty$; as $\theta \to +\infty$, $l(\theta) \approx n\theta - 2\sum(\theta - x_i) - n\bar{x} = -n\theta + n\bar{x} \to -\infty$. By Theorem 6.4 the MLE is the unique root of $\dot{l}(\theta) = 0$. The logistic mean is θ, so Newton–Raphson can start at $\hat{\theta}^{(0)} = \bar{x}$.

Theorem 6.5 (Invariance)

If $\hat{\theta}^{\text{MLE}}$ exists and $\eta = g(\theta)$ is a one-to-one function on Ω, then $\hat{\eta}^{\text{MLE}} = g(\hat{\theta}^{\text{MLE}})$.

Proof. Reparametrizing does not change the family of densities: the density with parameter $\eta \in g(\Omega)$ is $f(\cdot; g^{-1}(\eta))$. So the likelihood in η is $L^*(\eta) = L(g^{-1}(\eta))$, and

$$\max_{\eta \in g(\Omega)} L^*(\eta) = \max_{\theta \in \Omega} L(\theta) = L(\hat{\theta}^{\text{MLE}}) = L^*(g(\hat{\theta}^{\text{MLE}})).$$

■

Example 6.6 (Exponential)

For $\text{Exp}(\theta)$, $f(x;\theta) = \theta^{-1}e^{-x/\theta}$, we have $l(\theta) = -n\log\theta - n\bar{x}/\theta$ and $\ddot{l}(\theta) = n/\theta^2 - 2n\bar{x}/\theta^3$, which is not negative everywhere; Theorem 6.4 does not apply directly. With $\lambda = 1/\theta$, however, $l(\lambda) = n\log\lambda - n\bar{x}\lambda$ has $\ddot{l}(\lambda) = -n/\lambda^2 < 0$, and $l \to -\infty$ as $\lambda \to 0$ or ∞ (when $\bar{x} > 0$). So $\hat{\lambda} = 1/\bar{X}$, and by invariance $\hat{\theta}^{\text{MLE}} = 1/\hat{\lambda} = \bar{X}$.

A good parametrization can make the log-likelihood concave. For exponential families there is a canonical choice.

Theorem 6.6 (MLE in a one-parameter exponential family)

Let

$$f(x;\eta) = \exp\{\eta T(x) - A(\eta) + S(x)\}, \qquad x \in \mathcal{X},\ \eta \in N,$$

where (i) the support $\mathcal{X}$ does not depend on η;
(ii) N is an open interval;
(iii) $\text{Var}_\eta\, T(X_1) > 0$. If the likelihood equation

$$\dot{A}(\eta) = \frac{1}{n}\sum_{i=1}^n T(x_i), \qquad \text{i.e.} \quad E_\eta T(X_1) = \frac{1}{n}\sum_{i=1}^n T(x_i),$$

has a root $\hat{\eta}$ in N, then $\hat{\eta}$ is the MLE of η.

Proof. The key fact is the cgf of $T(X_1)$. In the continuous case, for $|s|$ small enough that $s+\eta \in N$ (possible by (ii)),

$$E_\eta e^{sT(X_1)} = \int e^{(s+\eta)T(x) - A(\eta) + S(x)}dx = e^{A(s+\eta)-A(\eta)}\int f(x; s+\eta)\,dx = e^{A(s+\eta)-A(\eta)},$$

using (i) for the last integral. So the cgf is $A(s+\eta) - A(\eta)$, and differentiating at $s = 0$ gives $E_\eta T(X_1) = \dot{A}(\eta)$ and $\text{Var}_\eta\, T(X_1) = \ddot{A}(\eta)$, which is positive by (iii). The log-likelihood $l(\eta) = \eta\sum T(x_i) - nA(\eta) + \sum S(x_i)$ therefore has $\ddot{l}(\eta) = -n\ddot{A}(\eta) < 0$, and a root of $\dot{l}(\eta) = \sum T(x_i) - n\dot{A}(\eta) = 0$ is the MLE by Theorem 6.3. ■

If the family is written in another parameter, $f(x;\theta) = \exp\{g(\theta)T(x) - A(g(\theta)) + S(x)\}$ with $\eta = g(\theta)$ one-to-one, then by invariance the MLE of θ solves $E_\theta T(X_1) = \frac{1}{n}\sum T(x_i)$: *match the mean of T to its sample average.*

Example 6.7

(a) Bernoulli(p): $f = \exp\{x \log \frac{p}{1-p} + \log(1-p)\}$, $T(x) = x$. The equation $E_p X = p = \bar{x}$ gives $\hat{p} = \bar{x}$ when $0 < \bar{x} < 1$.

(b) Poisson(λ): $T(x) = x$, $\eta = \log \lambda$; $\hat{\lambda} = \bar{x}$ when $\bar{x} > 0$.

(c) Geometric, $f(x;p) = (1-p)^{x-1}p$, $x = 1, 2, \ldots$: $T(x) = x$ with $\eta = \log(1-p)$. Then $E_p X = 1/p = \bar{x}$ gives $\hat{p} = 1/\bar{x}$ when $\bar{x} > 1$.

(d) Exponential(θ): $T(x) = x$, $\eta = -1/\theta$; $\hat{\theta} = \bar{x}$.

(e) Pareto, $f(x;\theta) = \theta x^{-\theta-1}$, $x > 1$: $f = \exp\{-\theta \log x + \log \theta - \log x\}$, so $T(x) = -\log x$. Since $X^{-\theta} \sim U(0,1)$, $\theta \log X \sim \text{Exp}(1)$ and $E_\theta(-\log X) = -1/\theta$. The equation gives $\hat{\theta} = 1/(\frac{1}{n}\sum \log x_i)$.

In the examples so far the likelihood was differentiable. Otherwise one studies the likelihood directly, and the MLE need not be unique.

Example 6.8 (Double exponential)

For $f(x;\theta) = \frac{1}{2}e^{-|x-\theta|}$ and $n = 2m+1$, $l(\theta) = -\sum|x_i - \theta| - n \log 2$. With order statistics $x_{(1)} < \cdots < x_{(n)}$ and $x_{(0)} = -\infty$, $x_{(n+1)} = \infty$, for $x_{(r)} \le \theta < x_{(r+1)}$,

$$l(\theta) = (n-2r)\theta + \sum_{i \le r} x_{(i)} - \sum_{i > r} x_{(i)} - n \log 2.$$

The slope $n - 2r$ is positive for $r \le m$ and negative for $r \ge m+1$, so l increases up to $x_{(m+1)}$ and decreases afterwards. The MLE is the sample median $X_{(m+1)}$.

Example 6.9 (Uniform)

(a) For $U(0,\theta)$, $L(\theta) = \theta^{-n}\mathbf{1}\{\theta \ge x_{(n)}\}$, which is decreasing on $[x_{(n)}, \infty)$ and zero before it. So $\hat{\theta}^{\text{MLE}} = X_{(n)}$; the likelihood equation plays no role.
(b) For $U(\theta-1, \theta+1)$, $L(\theta) = 2^{-n}\mathbf{1}\{x_{(n)} - 1 \le \theta \le x_{(1)} + 1\}$. Every point of $[x_{(n)} - 1, x_{(1)} + 1]$ maximizes L, so every statistic between $X_{(n)} - 1$ and $X_{(1)} + 1$, for instance $\alpha(X_{(n)} - 1) + (1 - \alpha)(X_{(1)} + 1)$ with $0 \le \alpha \le 1$, is an MLE. Other criteria (efficiency, simplicity) must choose among them.

6.3 MLE for vector parameters

For $\theta = (\theta_1, \ldots, \theta_k)' \in \Omega \subseteq \mathbb{R}^k$ the definitions are the same: $L(\theta) = \prod f(x_i;\theta)$, and $\hat{\theta}^{\text{MLE}}$ maximizes L over Ω. Its components are the MLEs of the components of θ. One can maximize one variable at a time, or solve the likelihood equations $\partial l / \partial \theta_j = 0$.

Example 6.10 (Double exponential with scale)

For $f(x;\mu,\sigma) = \frac{1}{2\sigma}e^{-|x-\mu|/\sigma}$ and $n = 2m+1$,

$$l(\mu,\sigma) = -\frac{1}{\sigma}\sum|x_i - \mu| - n \log \sigma - n \log 2.$$

For each σ, l is maximized in μ by the median $\hat{\mu} = x_{(m+1)}$ (Example 6.8). Then

$$\frac{\partial}{\partial \sigma} l(\hat{\mu}, \sigma) = \frac{\sum|x_i - \hat{\mu}|}{\sigma^2} - \frac{n}{\sigma} > 0 \iff \sigma < \frac{1}{n}\sum|x_i - \hat{\mu}|,$$

so $\hat{\sigma} = \frac{1}{n}\sum|x_i - \hat{\mu}|$. Since $\max_\sigma \max_\mu l = \max_\sigma l(\hat{\mu}, \sigma)$, the MLE is $(X_{(m+1)}, \frac{1}{n}\sum|X_i - X_{(m+1)}|)$.

Example 6.11 (Normal)

For $N(\mu, \sigma^2)$,

$$l(\mu, \sigma^2) = -\frac{1}{2\sigma^2}\sum(x_i - \mu)^2 - \frac{n}{2}\log\sigma^2 - \frac{n}{2}\log 2\pi.$$

Since $\sum(x_i - \mu)^2 = \sum(x_i - \bar{x})^2 + n(\bar{x} - \mu)^2$, l is maximized in μ at $\bar{x}$ for every σ^2. Then $\theta_2 = \sigma^2 \mapsto l(\bar{x}, \theta_2)$ has derivative $\frac{1}{2\theta_2^2}\sum(x_i - \bar{x})^2 - \frac{n}{2\theta_2}$, positive iff $\theta_2 < \frac{1}{n}\sum(x_i - \bar{x})^2$. Hence $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$.

Theorem 6.7 (Invariance for functions)

If $\hat{\theta}^{\text{MLE}}$ exists and $\eta_1 = g_1(\theta)$ can be completed to a one-to-one map $\eta = (g_1(\theta), g_2(\theta))$, then $\hat{\eta}_1^{\text{MLE}} = g_1(\hat{\theta}^{\text{MLE}})$.

Proof. By Theorem 6.5 (whose proof works verbatim for vectors), $\hat{\eta}^{\text{MLE}} = (g_1(\hat{\theta}^{\text{MLE}}), g_2(\hat{\theta}^{\text{MLE}}))$, and the MLE of a component is the corresponding component of the MLE. ■

Example 6.12

(a) In $N(\mu, \sigma^2)$, $(|\mu|, \operatorname{sgn}\mu, \sigma^2)$ is a one-to-one function of (μ, σ^2), so $\widehat{|\mu|}^{\text{MLE}} = |\bar{X}|$.
(b) In Bernoulli(p), $(p(1-p), \operatorname{sgn}(p - \frac{1}{2}))$ is one-to-one in p, so the MLE of the variance $p(1-p)$ is $\hat{p}(1-\hat{p})$ with $\hat{p} = \bar{X}$.

The one-dimensional results on concavity extend; their proofs are the same and are omitted.

Theorem 6.8

Let $\Omega_o = (a_1, b_1) \times \cdots \times (a_k, b_k)$ and let l have continuous second partial derivatives, with Hessian $\ddot{l}(\theta) = (\partial^2 l / \partial\theta_i \partial\theta_j)$.

(a) If $\mathbf{c}'\ddot{l}(\theta)\mathbf{c} < 0$ for all $\mathbf{c} \neq \mathbf{0}$ and $\theta \in \Omega_o$, and $\dot{l}(\hat{\theta}) = \mathbf{0}$ for some $\hat{\theta} \in \Omega_o$, then $\hat{\theta}$ is the unique maximizer of l.

(b) If, moreover, $l(\theta) \to -\infty$ as θ tends to the boundary of Ω_o, then $\dot{l}(\theta) = \mathbf{0}$ has exactly one root in Ω_o, and it is the MLE.

Theorem 6.9 (Multiparameter exponential family)

Let $f(x; \eta) = \exp\{\sum_{j=1}^k \eta_j T_j(x) - A(\eta) + S(x)\}$, $x \in \mathcal{X}$, $\eta \in N$, where (i) $\mathcal{X}$ does not depend on η;
(ii) N contains an open box in $\mathbb{R}^k$;
(iii) $\operatorname{Var}_\eta(\mathbf{c}'\mathbf{T}(X_1)) > 0$ for every $\mathbf{c} \neq \mathbf{0}$. If the likelihood equations

$$E_\eta T_j(X_1) = \frac{1}{n}\sum_{i=1}^n T_j(x_i), \qquad j = 1, \ldots, k,$$

have a root $\hat{\eta}$ in N, then $\hat{\eta}$ is the MLE.

Proof. As in Theorem 6.6, $\mathbf{T}(X_1)$ has cgf $A(\mathbf{s}+\eta)-A(\eta)$, so $E_\eta \mathbf{T} = \dot{A}(\eta)$ and $\operatorname{Var}_\eta \mathbf{T} = \ddot{A}(\eta)$. Then $\mathbf{c}'\ddot{l}(\eta)\mathbf{c} = -n\mathbf{c}'\ddot{A}(\eta)\mathbf{c} = -n\operatorname{Var}_\eta(\mathbf{c}'\mathbf{T}) < 0$, and Theorem 6.8(a) applies. ∎

Example 6.13

(a) *Normal.* $N(\mu,\sigma^2)$ has $\eta_1 = \mu/\sigma^2$, $\eta_2 = -1/(2\sigma^2)$, $T = (x, x^2)$. The equations $EX = \bar{x}$, $EX^2 = \frac{1}{n}\sum x_i^2$ give $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 + \hat{\mu}^2 = \frac{1}{n}\sum x_i^2$, the same answer as Example 6.11.
(b) *Bivariate normal.* The exponent is linear in $T = (x, y, xy, x^2, y^2)$. Matching the five means gives $\hat{\mu}_1 = \bar{x}$, $\hat{\mu}_2 = \bar{y}$, $\hat{\sigma}_j^2 = \frac{1}{n}\sum(\cdot - \text{mean})^2$, and $\hat{\rho}^{\text{MLE}}$ equal to the sample correlation coefficient.
(c) *Multinomial.* For Multi$(n, \mathbf{p})$ observed once, drop the redundant last cell: with $\eta_j = \log(p_j/p_k)$, $j < k$, the pmf is $\exp\{\sum_{j<k}\eta_j x_j - nA(\eta) + S(x)\}$ with $A(\eta) = \log(1 + e^{\eta_1} + \cdots + e^{\eta_{k-1}})$. Matching $EX_j = np_j$ to x_j gives $\hat{p}_j = x_j/n$ (when all $0 < x_j < n$).

6.4 Asymptotic properties of the MLE

Motivation. Maximum likelihood is the default estimator because, under regularity conditions, it is consistent and asymptotically normal with variance equal to the Cramér–Rao bound: in large samples no regular estimator does better. These results also give Wald, score and likelihood ratio tests.

Example 6.14

In the examples above the MLE coincided with an MME, so the results of Chapter 5 apply:

- Bernoulli: $\hat{p} \xrightarrow{p} p$ and $\sqrt{n}(\hat{p}-p) \xrightarrow{d} N(0, p(1-p))$.
- Poisson: $\sqrt{n}(\hat{\lambda}-\lambda) \xrightarrow{d} N(0,\lambda)$. Exponential: $\sqrt{n}(\hat{\theta}-\theta) \xrightarrow{d} N(0,\theta^2)$.
- Normal: $\sqrt{n}(\hat{\mu}-\mu) \sim N(0,\sigma^2)$ exactly, $n\hat{\sigma}^2/\sigma^2 \sim \chi^2(n-1)$, and $\sqrt{n}(\hat{\sigma}^2-\sigma^2) \xrightarrow{d} N(0, 2\sigma^4)$.
- Bivariate normal: $\sqrt{n}(\hat{\rho}-\rho) \xrightarrow{d} N(0, (1-\rho^2)^2)$.

In general the MLE is not an MME, and we need a direct argument. Write $\bar{l}_n(\theta) = \frac{1}{n}\sum_i \log f(X_i;\theta)$ for the average log-likelihood and let θ^0 be the true value. By the law of large numbers $\bar{l}_n(\theta) \xrightarrow{p} E_{\theta^0}\log f(X_1;\theta)$ for each θ. So we expect $\hat{\theta}_n = \arg\max \bar{l}_n$ to approach $\arg\max_\theta E_{\theta^0}\log f(X_1;\theta)$. The next theorem shows that this maximizer is θ^0.

Theorem 6.10 (Kullback–Leibler divergence)

Assume (R0) *identifiability*: $f(\cdot;\theta) = f(\cdot;\theta^0)$ implies $\theta = \theta^0$; and (R1) the support $\{x : f(x;\theta) > 0\} = \mathcal{X}$ does not depend on θ. Let

$$KL(\theta,\theta^0) = -E_{\theta^0}\log\frac{f(X;\theta)}{f(X;\theta^0)}.$$

Then (i) $KL(\theta,\theta^0) \geq 0$ for all θ, and (ii) $KL(\theta,\theta^0) = 0$ iff $\theta = \theta^0$. Equivalently, θ^0 is the unique maximizer of $E_{\theta^0}\log f(X;\theta)$.

Proof. For $t > 0$, $-\log t \geq 1 - t$, with equality iff $t = 1$. Put $t = f(X;\theta)/f(X;\theta^0)$ and take E_{θ^0}:

$$KL(\theta,\theta^0) \geq E_{\theta^0}\Big[1 - \frac{f(X;\theta)}{f(X;\theta^0)}\Big] = 1 - \int_{\mathcal{X}} f(x;\theta)\,dx = 0,$$

using (R1). Equality holds iff $f(X;\theta) = f(X;\theta^0)$ with probability one, i.e. (up to null sets) $f(\cdot;\theta) = f(\cdot;\theta^0)$, which by (R0) means $\theta = \theta^0$. ■

To turn the heuristic into a proof one needs more than pointwise convergence: e.g. uniform convergence of $\bar{l}_n$ on compact sets and continuity of $E_{\theta^0} \log f(X_1;\theta)$. When the log-likelihood is strictly concave and tends to $-\infty$ at the boundary, these hold, and one obtains:

Theorem 6.11 (Consistency of the MLE)

Let Ω be an open box in $\mathbb{R}^k$ and assume (R0), (R1). Suppose l_n has continuous second partial derivatives with $\mathbf{c}'\ddot{l}_n(\theta)\mathbf{c} < 0$ for $\mathbf{c} \neq \mathbf{0}$ and $l_n(\theta) \to -\infty$ at the boundary of Ω, and that $E_{\theta^0} \log f(X_1;\theta)$ exists and is continuous in θ. Then the MLE $\hat{\theta}_n$ is the unique root of $\dot{l}_n(\theta) = \mathbf{0}$, and $\hat{\theta}_n \xrightarrow{p} \theta$ under P_θ for every $\theta \in \Omega$.

(The proof is beyond these notes.) For example, the logistic MLE of Example 6.5 and the gamma MLE (the root of $-\Psi(\alpha) - \log\beta + \overline{\log x} = 0$, $-\alpha/\beta + \bar{x}/\beta^2 = 0$ with $\Psi = (\log\Gamma)'$) are consistent.

Heuristics for the limit law. Expanding the likelihood equation around θ,

$$0 = \dot{l}_n(\hat{\theta}_n) \approx \dot{l}_n(\theta) + \ddot{l}_n(\theta)(\hat{\theta}_n - \theta), \qquad \text{so} \qquad \sqrt{n}(\hat{\theta}_n - \theta) \approx \{-\ddot{l}_n(\theta)\}^{-1}\sqrt{n}\,\dot{l}_n(\theta).$$

By the LLN, $-\ddot{l}_n(\theta) \xrightarrow{p} I(\theta) := E_\theta[-\partial_\theta^2 \log f(X_1;\theta)]$, and by the CLT,

$$\sqrt{n}\,\dot{l}_n(\theta) \xrightarrow{d} N\big(0, \operatorname{Var}_\theta \partial_\theta \log f(X_1;\theta)\big),$$

provided $E_\theta \partial_\theta \log f(X_1;\theta) = 0$. We now make this rigorous.

Regularity conditions. In addition to (R0) and (R1):

(R2) Ω is an open subset of $\mathbb{R}^k$.

(R3) For every $x \in \mathcal{X}$, $\log f(x;\theta)$ has continuous first and second partial derivatives in θ.

(R4) For statistics $u(\mathbf{X})$ with $E_\theta u(\mathbf{X})$ finite, derivatives in θ up to second order may be taken under the integral (or sum) sign: $\partial_\theta^r E_\theta u(\mathbf{X}) = \int u(\mathbf{x}) \partial_\theta^r \prod f(x_i;\theta)\, d\mathbf{x}$, $r = 1, 2$.

(R5) The **Fisher information** $I(\theta) = \operatorname{Var}_\theta\big(\partial_\theta \log f(X_1;\theta)\big)$ is finite and nonsingular.

The derivative $\dot{l}_n(\theta)$ is called the **score function**.

Theorem 6.12 (Properties of the score)

Under (R0)–(R5),

(a) $E_\theta\{\partial_\theta \log f(X_1;\theta)\} = 0$;

(b) $I(\theta) = \operatorname{Var}_\theta\{\partial_\theta \log f(X_1;\theta)\} = E_\theta\{-\partial_\theta^2 \log f(X_1;\theta)\}$.

Proof. We treat a continuous one-dimensional θ. For all θ, $\int_{\mathcal{X}} \exp\{\log f(x;\theta)\}\, dx = 1$ (R1). Differentiating under the integral (R3, R4),

$$\int \{\partial_\theta \log f(x;\theta)\} f(x;\theta)\, dx = 0, \qquad \text{i.e.} \quad E_\theta \partial_\theta \log f(X_1;\theta) = 0.$$

Differentiating again,

$$\int \Big\{\partial_\theta^2 \log f(x;\theta) + \big[\partial_\theta \log f(x;\theta)\big]^2\Big\} f(x;\theta)\, dx = 0,$$

so $E_\theta[(\partial_\theta \log f)^2] = -E_\theta[\partial_\theta^2 \log f]$. By (a), the left side is the variance. ■

Theorem 6.13 (Asymptotic normality of the MLE)

Assume (R0)–(R5) and also:

(R6) the MLE $\hat\theta_n$ is the unique root of $\dot l_n(\theta)=0$ and is consistent;

(R7) the third derivatives $\partial_\theta^3 \log f(x;\theta)$ exist and are continuous, and there is $M(x)$ with $\max_{\theta\in\Omega}\|\partial_\theta^3 \log f(x;\theta)\| \le M(x)$ and $E_\theta M(X_1)<\infty$.

Then $\sqrt{n}(\hat\theta_n-\theta)\xrightarrow{d} N(0, I(\theta)^{-1})$.

Proof. We give the one-dimensional proof. Write $\dot l_n = \dot l_n/n$, $\ddot l_n = \ddot l_n/n$ and $\bar l_n^{(3)} = l_n^{(3)}/n$.

Step 1 (expansion). By Taylor's theorem there is θ_n^* between θ and $\hat\theta_n$ with

$$0=\dot{\bar l}_n(\hat\theta_n)=\dot{\bar l}_n(\theta)+\ddot{\bar l}_n(\theta)(\hat\theta_n-\theta)+\tfrac12 \bar l_n^{(3)}(\theta_n^*)(\hat\theta_n-\theta)^2.$$

Step 2 (the remainder is negligible). Let $R_n=\frac12\bar l_n^{(3)}(\theta_n^*)(\hat\theta_n-\theta)$. For $\epsilon, K>0$,

$$P_\theta(|R_n|\ge\epsilon)\le P_\theta(|\bar l_n^{(3)}(\theta_n^*)|>K)+P_\theta(|\hat\theta_n-\theta|\ge 2\epsilon/K).$$

By (R7) and Markov's inequality, $P_\theta(|\bar l_n^{(3)}(\theta_n^*)|>K)\le P_\theta(\frac1n\sum M(X_i)>K)\le E_\theta M(X_1)/K$. By (R6), the second term tends to 0. So $\limsup_n P_\theta(|R_n|\ge\epsilon)\le E_\theta M(X_1)/K$ for every K, and $R_n\xrightarrow{p}0$.

Step 3 (linearization). Step 1 reads $0=\dot{\bar l}_n(\theta)+\{\ddot{\bar l}_n(\theta)+R_n\}(\hat\theta_n-\theta)$. By the LLN and Theorem 6.12(b), $-\ddot{\bar l}_n(\theta)\xrightarrow{p} I(\theta)$. Hence

$$\sqrt{n}(\hat\theta_n-\theta)=\big(I(\theta)+r_n\big)^{-1}\sqrt{n}\,\dot{\bar l}_n(\theta),\qquad r_n\xrightarrow{p}0.$$

Step 4 (limit). $\sqrt{n}\,\dot{\bar l}_n(\theta)=\sqrt{n}(\frac1n\sum\partial_\theta\log f(X_i;\theta)-0)\xrightarrow{d}N(0,I(\theta))$ by the CLT and Theorem 6.12. By Slutsky's theorem, $\sqrt{n}(\hat\theta_n-\theta)\xrightarrow{d} I(\theta)^{-1}Z$ with $Z\sim N(0,I(\theta))$, i.e. $N(0,I(\theta)^{-1})$. ∎

Condition (R7) is stronger than necessary but keeps the proof short. Step 3 also shows

$$\sqrt{n}(\hat\theta_n-\theta)=I(\theta)^{-1}\sqrt{n}\,\dot{\bar l}_n(\theta)+o_p(1):$$

the error of the MLE is driven by the score, and its precision grows with the information $I(\theta)$.

When the MLE has no closed form, one Newton–Raphson step from a good starting value is already as good asymptotically.

Theorem 6.14 (One-step estimator)

Assume (R0)–(R7), and let $\hat\theta_n^{(0)}$ be an estimator with $\sqrt{n}(\hat\theta_n^{(0)}-\theta)\xrightarrow{d}Z$ for some random variable Z (R8). Then

$$\hat\theta_n^{(1)}=\hat\theta_n^{(0)}+\big[-\ddot l_n(\hat\theta_n^{(0)})\big]^{-1}\dot l_n(\hat\theta_n^{(0)})$$

satisfies $\sqrt{n}(\hat\theta_n^{(1)}-\theta)\xrightarrow{d}N(0,I(\theta)^{-1})$.

The proof follows the same steps as Theorem 6.13, expanding $\dot l_n(\hat\theta_n^{(0)})$ around θ.

Example 6.15 (Logistic location)

In Example 6.5, start from $\hat\theta^{(0)} = \bar X$, which satisfies $\sqrt{n}(\bar X - \theta) \xrightarrow{d} N(0, \pi^2/3)$ (the logistic variance). The information is

$$I(\theta) = E_\theta\big[-\partial_\theta^2 \log f(X;\theta)\big] = 2E\Big[\frac{e^{-(X-\theta)}}{(1+e^{-(X-\theta)})^2}\Big] = 2E[U(1-U)] = \frac{1}{3},$$

since $U = 1/(1+e^{-(X-\theta)}) \sim U(0,1)$ by the probability integral transform. So the one-step estimator has $\sqrt{n}(\hat\theta^{(1)} - \theta) \xrightarrow{d} N(0,3)$, an improvement over $\pi^2/3 \approx 3.29$ for $\bar X$.

Example 6.16 (Gamma)

For Gamma(α,β), with $\Psi = (\log\Gamma)'$,

$$\dot l_n = \begin{pmatrix} -n\Psi(\alpha) - n\log\beta + n\overline{\log x} \\ -n\alpha/\beta + n\bar x/\beta^2 \end{pmatrix}, \qquad I(\theta) = \begin{pmatrix} \dot\Psi(\alpha) & 1/\beta \\ 1/\beta & \alpha/\beta^2 \end{pmatrix}.$$

Start from the MME $(\hat\alpha^{(0)}, \hat\beta^{(0)})$ of Example 6.1(d), which is $\sqrt{n}$-consistent by Theorem 6.2. One Newton step gives an estimator with limit $N_2(\mathbf{0}, I(\theta)^{-1})$, where

$$I(\theta)^{-1} = \frac{1}{\alpha\dot\Psi(\alpha) - 1}\begin{pmatrix} \alpha & -\beta \\ -\beta & \beta^2\dot\Psi(\alpha) \end{pmatrix}.$$

6.5 Least squares

In the **linear regression model**

$$Y_i = x_{i0}\beta_0 + x_{i1}\beta_1 + \cdots + x_{ip}\beta_p + e_i, \qquad Ee_i = 0,\ \operatorname{Var} e_i = \sigma^2,\ \operatorname{Cov}(e_i, e_j) = 0\ (i \neq j),$$

the mean response is a linear function of the explanatory variables. The **least squares estimator** (LSE) $\hat\beta$ minimizes $\sum_i\{Y_i - (x_{i0}\beta_0 + \cdots + x_{ip}\beta_p)\}^2$, and the mean response at $(x_0,\dots,x_p)$ is estimated by $x_0\hat\beta_0 + \cdots + x_p\hat\beta_p$. In matrix form, $\mathbf{Y} = X\beta + \mathbf{e}$ with X an $n \times (p+1)$ matrix of rank $p+1$, $E\mathbf{e} = \mathbf{0}$, $\operatorname{Var}\mathbf{e} = \sigma^2 I_n$, and

$$\hat\beta^{\mathrm{LSE}} = \operatorname*{arg\,min}_{\beta\in\mathbb{R}^{p+1}} \|\mathbf{Y} - X\beta\|^2.$$

Theorem 6.15 (Least squares)

(a) $\Pi = X(X'X)^{-1}X'$ satisfies $\Pi' = \Pi$, $\Pi'(I-\Pi) = 0$ and $\Pi X = X$.

(b) $\|\mathbf{Y} - X\beta\|^2 = \|\Pi(\mathbf{Y} - X\beta)\|^2 + \|(I-\Pi)\mathbf{Y}\|^2$; hence $X\hat\beta^{\mathrm{LSE}} = \Pi\mathbf{Y}$ and $\hat\beta^{\mathrm{LSE}} = (X'X)^{-1}X'\mathbf{Y}$.

Proof. (a) Since $X'X$ is symmetric and invertible, $\Pi' = X\{(X'X)^{-1}\}'X' = \Pi$ and

$$\Pi'\Pi = X(X'X)^{-1}X'X(X'X)^{-1}X' = \Pi,$$

so $\Pi'(I-\Pi) = 0$, and $\Pi X = X$.

(b) By (a), $\mathbf{Y} - X\beta = \Pi(\mathbf{Y} - X\beta) + (I-\Pi)\mathbf{Y}$, and the cross term $2\{\Pi(\mathbf{Y} - X\beta)\}'(I-\Pi)\mathbf{Y}$ vanishes. The second term does not involve β, and the first is zero iff $X\beta = \Pi\mathbf{Y}$, i.e. $\beta = (X'X)^{-1}X'\mathbf{Y}$ (X has full column rank). ∎

Π is the orthogonal projection onto the column space of X. The error variance is estimated by the mean squared residual

$$\hat\sigma^2 = \frac{\|\mathbf{Y} - X\hat\beta\|^2}{n-p-1}.$$

Theorem 6.16 (Properties of the LSE)

(a) $E\hat\beta = \beta$ and $\operatorname{Var}\hat\beta = \sigma^2(X'X)^{-1}$.

(b) $E\hat\sigma^2 = \sigma^2$.

(c) If $\mathbf{e} \sim N_n(\mathbf{0}, \sigma^2 I)$, then $\hat\beta \sim N(\beta, \sigma^2(X'X)^{-1})$, $(n-p-1)\hat\sigma^2/\sigma^2 \sim \chi^2(n-p-1)$, and they are independent.

Proof. (a) $E\hat\beta = (X'X)^{-1}X'X\beta = \beta$ and $\operatorname{Var}\hat\beta = (X'X)^{-1}X'(\sigma^2 I)X(X'X)^{-1} = \sigma^2(X'X)^{-1}$.

(b) $(I-\Pi)\mathbf{Y} = (I-\Pi)\mathbf{e}$ since $(I-\Pi)X = 0$, so $(n-p-1)\hat\sigma^2 = \mathbf{e}'(I-\Pi)\mathbf{e}$. Using $\operatorname{tr}(AB) = \operatorname{tr}(BA)$,

$$E\,\mathbf{e}'(I-\Pi)\mathbf{e} = E\operatorname{tr}\{(I-\Pi)\mathbf{e}\mathbf{e}'\} = \operatorname{tr}\{(I-\Pi)\sigma^2 I\} = \sigma^2\{n - \operatorname{tr}((X'X)^{-1}X'X)\} = \sigma^2(n-p-1).$$

(c) is Theorem 4.19 in Volume I. ∎

Orthogonalization. Computation and interpretation are easiest when the columns of X are orthogonal. Partition $X = (X_0, X_1)$ with X_0 of size $n \times p_0$ and X_1 of size $n \times p_1$, and let $\Pi_0 = X_0(X_0'X_0)^{-1}X_0'$. Then $X_{1|0} = (I-\Pi_0)X_1$ is orthogonal to the columns of X_0, and

$$X\beta = X_0\beta_0 + X_1\beta_1 = X_0\gamma_0 + X_{1|0}\beta_1, \qquad \gamma_0 = \beta_0 + (X_0'X_0)^{-1}X_0'X_1\beta_1.$$

With $\Pi_{1|0} = X_{1|0}(X_{1|0}'X_{1|0})^{-1}X_{1|0}'$, the projection onto the column space of X splits as $\Pi = \Pi_0 + \Pi_{1|0}$ with $\Pi_0\Pi_{1|0} = 0$.

Theorem 6.17

In this notation,

(a) $X_0\hat\gamma_0 = \Pi_0\mathbf{Y}$, $X_{1|0}\hat\beta_1 = \Pi_{1|0}\mathbf{Y}$, i.e. $\hat\gamma_0 = (X_0'X_0)^{-1}X_0'\mathbf{Y}$ and $\hat\beta_1 = (X_{1|0}'X_{1|0})^{-1}X_{1|0}'\mathbf{Y}$, and $\hat\beta_0 = \hat\gamma_0 - (X_0'X_0)^{-1}X_0'X_1\hat\beta_1$;

(b) $\operatorname{Cov}(\hat\gamma_0, \hat\beta_1) = 0$, $\operatorname{Var}\hat\gamma_0 = \sigma^2(X_0'X_0)^{-1}$ and $\operatorname{Var}\hat\beta_1 = \sigma^2(X_{1|0}'X_{1|0})^{-1}$.

Proof. (a) In the parametrization (γ_0, β_1) the design matrix is $(X_0, X_{1|0})$ with $X_0'X_{1|0} = 0$, so its Gram matrix is block diagonal and $(X'X)^{-1}X'\mathbf{Y}$ splits into the two stated pieces; the relation for $\hat\beta_0$ inverts the definition of γ_0.

(b) $\operatorname{Cov}(\hat\gamma_0, \hat\beta_1) = \sigma^2(X_0'X_0)^{-1}X_0'X_{1|0}(X_{1|0}'X_{1|0})^{-1} = 0$, and the variances follow from Theorem 6.16(a) for each block. ∎

Example 6.17 (Centring the explanatory variables)

With an intercept, $EY_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} = \gamma_0 + \sum_j \beta_j(x_{ij} - \bar x_{\cdot j})$, where $\gamma_0 = \beta_0 + \sum_j \beta_j \bar x_{\cdot j}$. Here $X_0 = \mathbf{1}$ and $X_{1|0}$ has entries $x_{ij} - \bar x_{\cdot j}$. With $S_{jk} = \sum_i (x_{ij} - \bar x_{\cdot j})(x_{ik} - \bar x_{\cdot k})$ and $S_{jY} = \sum_i (x_{ij} - \bar x_{\cdot j})(Y_i - \bar Y)$,

$$\hat\gamma_0 = \bar Y, \qquad \begin{pmatrix} S_{11} & \cdots & S_{1p} \\ \vdots & & \vdots \\ S_{p1} & \cdots & S_{pp} \end{pmatrix} \begin{pmatrix} \hat\beta_1 \\ \vdots \\ \hat\beta_p \end{pmatrix} = \begin{pmatrix} S_{1Y} \\ \vdots \\ S_{pY} \end{pmatrix}.$$

For $p = 1$ this is $\hat{\beta}_1 = S_{1Y}/S_{11}$ and $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$, with $\operatorname{Var} \hat{\beta}_1 = \sigma^2/S_{11}$ and $\operatorname{Cov}(\bar{Y}, \hat{\beta}_1) = 0$.

Chapter 7. Hypothesis Testing

7.1 The logic of testing

A model is proposed on the basis of experience; we then ask whether the data agree with it. If, under the model, the observed result would be very unlikely, we reject the model in favour of an alternative. Deciding between a hypothesis and its alternative from data in this way is **statistical hypothesis testing**.

Example 7.1

The time until a standard painkiller takes effect is $N(20, 4^2)$ minutes. A new drug is claimed to act faster. In a trial with $n = 64$ patients the mean time is $\bar{X}$, and $\bar{X} \sim N(\mu, 4^2/64) = N(\mu, 0.5^2)$. The question is whether to accept $H_1 : \mu < 20$ (faster) over $H_0 : \mu = 20$ (no change).
Results with $\bar{X}$ well below 20 are hard to explain under H_0. For instance,

$$P(\bar{X} \le 19 \mid H_0) = P\Big(\frac{\bar{X} - 20}{0.5} \le -2\Big) = 0.023 :$$

such a result happens about twice in a hundred trials if H_0 holds. Observing it is strong evidence against H_0.

H_0 is the **null hypothesis** and H_1 the **alternative**. A set C of sample values for which we reject H_0 is the **rejection** (critical) region, and its complement is the **acceptance region**. A statistic used to describe C, such as $\bar{X}$ above, is a **test statistic**. Two kinds of error are possible:

	H_0 true	H_1 true
accept H_0	correct	type II error
reject H_0	type I error	correct

Since H_0 is the hypothesis we reject only on clear evidence, we bound the probability of a type I error by a small preassigned number α, the **significance level** (typically 0.10, 0.05, 0.01). A test with $P(\text{reject} \mid H_0) \le \alpha$ is a **level-α test**. Among level-α tests we want the smallest probability of a type II error, so we make the rejection region as large as the level permits.

Example 7.2

In Example 7.1, both "$\bar{X} \le 19$" (probability 0.023 under H_0) and "$\bar{X} \le 20 - z_{0.05} \cdot 0.5 = 19.1775$" (probability exactly 0.05) are level-0.05 tests. If the true mean is μ,

$$P_\mu(\bar{X} \le 19) = \Phi(38 - 2\mu), \qquad P_\mu(\bar{X} \le 19.1775) = \Phi(38.355 - 2\mu).$$

The second is larger for every $\mu < 20$, i.e. its type II error probability is smaller. In general the level-α rejection region here is $\bar{X} \le 20 - z_\alpha \cdot 0.5$.

Example 7.3 (Randomization)

Let $X_1, \dots, X_{50}$ be a Poisson(θ) sample, and test $H_0 : \theta = 0.1$ against $H_1 : \theta < 0.1$ at $\alpha = 0.05$. Small values of $S = \sum X_i \sim$ Poisson(50θ) speak against H_0, and under H_0, $S \sim$ Poisson(5):

$$P_{0.1}(S \le 0) = 0.0067, \qquad P_{0.1}(S \le 1) = 0.0404, \qquad P_{0.1}(S \le 2) = 0.1247.$$

No region of the form $S \le k$ has probability exactly 0.05. We can reach it by rejecting for

$S \le 1$, rejecting with probability γ when $S = 2$, and accepting for $S \ge 3$, where

$$0.0404 + \gamma \cdot P_{0.1}(S = 2) = 0.05, \qquad \gamma = \frac{0.05 - 0.0404}{0.0842} = 0.114.$$

This is a **randomized test**.

General formulation. Let the model be $\{f(x;\theta) : \theta \in \Omega\}$ and test

$$H_0 : \theta \in \Omega_0 \quad \text{vs} \quad H_1 : \theta \in \Omega_1, \qquad \Omega_0 \cap \Omega_1 = \varnothing,\ \Omega_0 \cup \Omega_1 = \Omega.$$

A (possibly randomized) test is described by a **test function** $\phi(\mathbf{x}) \in [0,1]$, the probability of rejecting H_0 when $\mathbf{X} = \mathbf{x}$; a nonrandomized test has $\phi = \mathbf{1}_C$. Its **power function** is $\gamma_\phi(\theta) = E_\theta\phi(\mathbf{X})$, the probability of rejection under θ. Its **size** is $\max_{\theta\in\Omega_0} E_\theta\phi(\mathbf{X})$, and ϕ has level α if its size is at most α. We seek tests with size exactly α whose power on Ω_1 is as large as possible.

Example 7.4 (Composite null)

In Example 7.1 take $H_0 : \mu \ge 20$ against $H_1 : \mu < 20$. For a region $\bar X \le c$, $P_\mu(\bar X \le c) = \Phi((c-\mu)/0.5)$ decreases in μ, so its maximum over $\mu \ge 20$ is attained at $\mu = 20$. Setting it equal to α gives the same region $\bar X \le 20 - 0.5 z_\alpha$ as before. Its power function $\gamma(\mu) = \Phi(38.355 - 2\mu)$ (for $\alpha = 0.05$) decreases from 1 to 0, passing through 0.05 at $\mu = 20$. In Example 7.3 the power function is

$$\gamma(\theta) = \sum_{x=0}^{1} e^{-50\theta}\frac{(50\theta)^x}{x!} + 0.114\, e^{-50\theta}\frac{(50\theta)^2}{2},$$

which is also decreasing in θ.

7.2 Likelihood ratio tests

To test $H_0 : \theta \in \Omega_0$ against $H_1 : \theta \in \Omega_1$, compare how well the best parameter value in each hypothesis explains the data:

$$\frac{\max_{\theta\in\Omega_1} L(\theta;\mathbf{x})}{\max_{\theta\in\Omega_0} L(\theta;\mathbf{x})}.$$

Large values are evidence against H_0. Since $\max_\Omega L = \max\{\max_{\Omega_0} L, \max_{\Omega_1} L\}$, we have

$$\frac{\max_\Omega L}{\max_{\Omega_0} L} = \max\left\{1, \frac{\max_{\Omega_1} L}{\max_{\Omega_0} L}\right\},$$

and large values of one ratio correspond to large values of the other. So we use the full maximum. With $\hat\theta^{\Omega}$ and $\hat\theta^{\Omega_0}$ the MLEs over Ω and Ω_0:

Definition (Likelihood ratio test)

The **likelihood ratio test** (LRT) of level α rejects H_0 when

$$2\{l(\hat\theta^{\Omega}) - l(\hat\theta^{\Omega_0})\} \ge c, \qquad \text{with } c \text{ chosen so that} \quad \max_{\theta\in\Omega_0} P_\theta(\text{reject}) = \alpha$$

(randomizing if necessary).

Example 7.5 (Two-sided t test)

Let $X_1, \dots, X_n$ ($n \geq 2$) be $N(\mu, \sigma^2)$ and test $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$. With

$$l(\mu, \sigma^2) = -\frac{1}{2\sigma^2}\sum (x_i - \mu)^2 - \frac{n}{2}\log \sigma^2 - \frac{n}{2}\log 2\pi,$$

the MLEs are $\hat\theta^{\Omega} = (\bar x, \hat\sigma^2)$, $\hat\sigma^2 = \frac{1}{n}\sum (x_i - \bar x)^2$, and $\hat\theta^{\Omega_0} = (\mu_0, \hat\sigma_0^2)$, $\hat\sigma_0^2 = \frac{1}{n}\sum (x_i - \mu_0)^2 = \hat\sigma^2 + (\bar x - \mu_0)^2$. Hence

$$2\{l(\hat\theta^{\Omega}) - l(\hat\theta^{\Omega_0})\} = n \log \frac{\hat\sigma_0^2}{\hat\sigma^2} = n \log \Big(1 + \frac{(\bar x - \mu_0)^2}{\hat\sigma^2}\Big),$$

an increasing function of $|\bar x - \mu_0|/(s/\sqrt{n})$ with $s^2 = \frac{1}{n-1}\sum (x_i - \bar x)^2$. Under H_0, $\sqrt{n}(\bar X - \mu_0)/S \sim t(n-1)$, so the level-$\alpha$ LRT is the **two-sided t test**:

$$\text{reject } H_0 \iff \Big|\frac{\bar X - \mu_0}{S/\sqrt{n}}\Big| \geq t_{\alpha/2}(n-1).$$

Example 7.6 (One-sided t test)

Same model, $H_0 : \mu \leq \mu_0$ vs $H_1 : \mu > \mu_0$. Over Ω_0, $\sum (x_i - \mu)^2 = \sum (x_i - \bar x)^2 + n(\bar x - \mu)^2$ is minimized by $\hat\mu^{\Omega_0} = \min(\bar x, \mu_0)$, and $\hat\sigma^{2,\Omega_0} = \frac{1}{n}\sum (x_i - \hat\mu^{\Omega_0})^2$. Thus

$$2\{l(\hat\theta^{\Omega}) - l(\hat\theta^{\Omega_0})\} = n \log \Big(1 + \frac{(\bar x - \min(\bar x, \mu_0))^2}{\hat\sigma^2}\Big),$$

which is 0 when $\bar x \leq \mu_0$ and increases in $(\bar x - \mu_0)/(s/\sqrt{n})$ when $\bar x > \mu_0$. For $\alpha < 1$ the region is $\{\sqrt{n}(\bar x - \mu_0)/s \geq c\}$ with $c > 0$. To find the size, write $Z = \sqrt{n}(\bar X - \mu)/\sigma$ and $V = (n-1)S^2/\sigma^2$, which are independent $N(0,1)$ and $\chi^2(n-1)$. Then

$$P_{\mu,\sigma^2}\Big(\frac{\bar X - \mu_0}{S/\sqrt{n}} \geq c\Big) = P\Big(c\sqrt{V/(n-1)} - Z \leq \frac{\mu - \mu_0}{\sigma/\sqrt{n}}\Big),$$

which increases in μ. Over Ω_0 the maximum is at $\mu = \mu_0$, where the statistic is $t(n-1)$. The level-α LRT is the **one-sided t test**: reject when $\sqrt{n}(\bar X - \mu_0)/S \geq t_\alpha(n-1)$.

Example 7.7 (Exponential mean)

For an $\mathrm{Exp}(\theta)$ sample and $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$: $l(\theta) = -n\log\theta - n\bar x/\theta$, $\hat\theta^{\Omega} = \bar x$, so

$$2\{l(\hat\theta^{\Omega}) - l(\theta_0)\} = 2n\Big(\frac{\bar x}{\theta_0} - 1 - \log\frac{\bar x}{\theta_0}\Big).$$

The function $t \mapsto t - 1 - \log t$ decreases on $(0,1)$ and increases on $(1,\infty)$, so the region has the form

$$\frac{\bar x}{\theta_0} \leq c_1 \quad \text{or} \quad \frac{\bar x}{\theta_0} \geq c_2, \qquad c_1 - \log c_1 = c_2 - \log c_2.$$

Under H_0, $2n\bar X/\theta_0 \sim \chi^2(2n)$, so the size condition is $\int_{2nc_1}^{2nc_2} f_{\chi^2(2n)}(y)\,dy = 1 - \alpha$. Solving the two equations gives c_1, c_2. For large n, $c_{1,2} \approx 1 \mp z_{\alpha/2}/\sqrt{n}$, and the test is approximately "$n(\bar X/\theta_0 - 1)^2 \geq \chi^2_\alpha(1)$".

Example 7.8 (Two-sample t test)

Independent samples $X_{i1},\dots,X_{in_i}$ from $N(\mu_i,\sigma^2)$, $i=1,2$; $H_0:\mu_1=\mu_2$ vs $H_1:\mu_1\neq\mu_2$. Under Ω, $\hat\mu_i=\bar x_i$ and $\hat\sigma^2=\sum_{i,j}(x_{ij}-\bar x_i)^2/(n_1+n_2)$; under Ω_0, the common mean is $\hat\mu=(n_1\bar x_1+n_2\bar x_2)/(n_1+n_2)$ and $\hat\sigma_0^2=\sum_{i,j}(x_{ij}-\hat\mu)^2/(n_1+n_2)$. Since $\hat\sigma_0^2=\hat\sigma^2+\frac{n_1n_2}{(n_1+n_2)^2}(\bar x_1-\bar x_2)^2$,

$$2\{l(\hat\theta^{\Omega})-l(\hat\theta^{\Omega_0})\}=(n_1+n_2)\log\Big(1+\frac{n_1n_2}{(n_1+n_2)^2}\frac{(\bar x_1-\bar x_2)^2}{\hat\sigma^2}\Big),$$

increasing in $|\bar x_1-\bar x_2|/(s_p\sqrt{1/n_1+1/n_2})$, where $s_p^2=\sum_{i,j}(x_{ij}-\bar x_i)^2/(n_1+n_2-2)$. Under H_0 this statistic is $t(n_1+n_2-2)$ (Corollary 4.10 in Volume I). So the LRT rejects when

$$\frac{|\bar X_1-\bar X_2|}{S_p\sqrt{1/n_1+1/n_2}}\geq t_{\alpha/2}(n_1+n_2-2).$$

Example 7.9 (Comparing two variances)

Independent samples from $N(\mu_i,\sigma_i^2)$; $H_0:\sigma_1^2=\sigma_2^2$ vs $H_1:\sigma_1^2\neq\sigma_2^2$. With $\hat\sigma_i^2=\frac{1}{n_i}\sum_j(x_{ij}-\bar x_i)^2$ and the pooled $\hat\sigma^2=(n_1\hat\sigma_1^2+n_2\hat\sigma_2^2)/(n_1+n_2)$,

$$2\{l(\hat\theta^{\Omega})-l(\hat\theta^{\Omega_0})\}=(n_1+n_2)\log\hat\sigma^2-n_1\log\hat\sigma_1^2-n_2\log\hat\sigma_2^2.$$

In terms of $t=n_1\hat\sigma_1^2/(n_1\hat\sigma_1^2+n_2\hat\sigma_2^2)$ this equals $-n_1\log t-n_2\log(1-t)+\text{const}$, which decreases for $t<n_1/(n_1+n_2)$ and increases afterwards. Since t is an increasing function of $F=S_1^2/S_2^2$, the region is $F\leq d_1$ or $F\geq d_2$. Under H_0, $F\sim F(n_1-1,n_2-1)$, and d_1,d_2 are determined by $\int_{d_1}^{d_2}f_{F(r_1,r_2)}=1-\alpha$ together with equality of the LRT statistic at d_1 and d_2. In practice one uses $d_1=F_{1-\alpha/2}$, $d_2=F_{\alpha/2}$.

Example 7.10 (Poisson, one-sided)

In the setting of Example 7.3, $H_0:\theta\geq 0.1$ vs $H_1:\theta<0.1$. Here $\hat\theta^{\Omega}=\bar x$ and $\hat\theta^{\Omega_0}=\max(\bar x,0.1)$, and

$$2\{l(\hat\theta^{\Omega})-l(\hat\theta^{\Omega_0})\}=2n\{\bar x\log\bar x-\bar x(1+\log 0.1)+0.1\}\mathbf{1}\{\bar x<0.1\},$$

which decreases in $\bar x$ on $\bar x\leq 0.1$. So the LRT rejects for small $S=\sum X_i$. Since $P_\theta(S\leq k)=\int_{n\theta}^{\infty}z^ke^{-z}/k!\,dz$ decreases in θ, the size is attained at $\theta=0.1$, and the randomized test of Example 7.3 is the level-0.05 LRT.

7.3 Large-sample approximations

When exact null distributions are unavailable, we use the asymptotic theory of the MLE. Let $\Omega\subseteq\mathbb{R}^k$ be open, $l(\theta)=\sum\log f(X_i;\theta)$, $\bar l=l/n$, and $I(\theta)$ the Fisher information (matrix). From Chapter 6, $\sqrt{n}(\hat\theta_n-\theta)\approx I(\theta)^{-1}\sqrt{n}\,\dot{\bar l}(\theta)$.

Simple null hypothesis. Let $H_0:\theta=\theta_0$. Expanding l about $\hat\theta_n$, where $\dot l(\hat\theta_n)=0$,

$$l(\theta_0)=l(\hat\theta_n)+\tfrac12(\theta_0-\hat\theta_n)'\ddot l(\theta_n^*)(\theta_0-\hat\theta_n),$$

and $-\frac1n\ddot l(\theta_n^*)\xrightarrow{p}I(\theta_0)$ under H_0. Hence

$$2\{l(\hat\theta_n)-l(\theta_0)\}\approx\sqrt{n}(\hat\theta_n-\theta_0)'I(\theta_0)\sqrt{n}(\hat\theta_n-\theta_0).$$

Since $\sqrt{n}(\hat\theta_n-\theta_0)\xrightarrow{d}\mathbf{Z}\sim N_k(\mathbf{0},I(\theta_0)^{-1})$, the right side converges to $\mathbf{Z}'I(\theta_0)\mathbf{Z}\sim\chi^2(k)$ (continuous mapping, and Theorem 4.18 in Volume I).

Theorem 7.1 (Simple null hypothesis)

Under the conditions (R0)–(R7) of Theorem 6.13, for $H_0:\theta=\theta_0$ vs $H_1:\theta\neq\theta_0$, the following hold when H_0 is true.

(a) $2\{l(\hat\theta_n)-l(\theta_0)\}\xrightarrow{d}\chi^2(k)$. So the test rejecting when this is $\geq\chi^2_\alpha(k)$ has asymptotic level α.

(b) $2\{l(\hat\theta_n)-l(\theta_0)\}=W_n(\theta_0)+o_p(1)$, where $W_n(\theta_0)=n(\hat\theta_n-\theta_0)'I(\theta_0)(\hat\theta_n-\theta_0)$.

(c) $2\{l(\hat\theta_n)-l(\theta_0)\}=R_n(\theta_0)+o_p(1)$, where $R_n(\theta_0)=n\,\dot l(\theta_0)'I(\theta_0)^{-1}\dot l(\theta_0)$.

Sketch. (a), (b) were derived above; the remainder is controlled as in the proof of Theorem 6.13. For (c), substitute $\sqrt{n}(\hat\theta_n-\theta_0)=I(\theta_0)^{-1}\sqrt{n}\,\dot l(\theta_0)+o_p(1)$ into W_n. ■

W_n is **Wald's statistic** and R_n is **Rao's (score) statistic**. All three are asymptotically $\chi^2(k)$ under H_0, and rejecting when any of them exceeds $\chi^2_\alpha(k)$ is an asymptotic level-α test. The score test needs only θ_0, not the MLE.

Example 7.11 (Exponential)

For $\mathrm{Exp}(\theta)$, $H_0:\theta=\theta_0$: $\dot l(\theta)=-\frac{1}{\theta}+\frac{\bar x}{\theta^2}$, $\hat\theta=\bar x$, $I(\theta)=1/\theta^2$. The three statistics are

$$2\{l(\hat\theta)-l(\theta_0)\}=2n\Big(\frac{\bar x}{\theta_0}-1-\log\frac{\bar x}{\theta_0}\Big),\qquad W_n=\frac{n(\bar x-\theta_0)^2}{\bar x^2},\qquad R_n=\frac{n(\bar x-\theta_0)^2}{\theta_0^2}.$$

Each is compared with $\chi^2_\alpha(1)$.

Example 7.12 (Multinomial goodness of fit)

Let $\mathbf{Z}_1,\dots,\mathbf{Z}_n$ be i.i.d. $\mathrm{Multi}(1,\mathbf{p})$ with counts $X_j=\sum_i Z_{ij}$, and test $H_0:\mathbf{p}=\mathbf{p}_0$. Use $\theta=(p_1,\dots,p_r)'$, $r=k-1$, and write $\theta_\cdot=\sum_{j\leq r}\theta_j=1-p_k$. Then

$$l(\theta)=\sum_{j\leq r}x_j\log\theta_j+x_k\log(1-\theta_\cdot),\qquad \hat\theta_j=x_j/n,$$

$$I(\theta)=\mathrm{diag}(1/\theta_j)+\frac{1}{p_k}\mathbf{1}\mathbf{1}'.$$

A direct computation gives $I(\theta_0)(\hat\theta-\theta_0)=\dot l(\theta_0)$, with j-th entry $\frac{x_j-np_{0j}}{np_{0j}}-\frac{x_k-np_{0k}}{np_{0k}}$, and therefore

$$W_n(\theta_0)=R_n(\theta_0)=\sum_{j=1}^{k}\frac{(X_j-np_{0j})^2}{np_{0j}},$$

Pearson's chi-square statistic, while the LRT statistic is $2\sum_j X_j\log\frac{X_j}{np_{0j}}$. All are asymptotically $\chi^2(k-1)$ under H_0 (compare Example 5.12 in Volume I).

Composite null hypotheses. Often H_0 fixes only part of θ, e.g. the mean of a normal model with unknown variance. Write $\theta=(\xi',\eta')'$ with $\xi\in\mathbb{R}^{k_1}$, $\eta\in\mathbb{R}^{k_0}$, $k_0+k_1=k$, and $H_0:\xi=\xi_0$.

Partition

$$\dot{l} = \begin{pmatrix} \dot{l}_1 \\ \dot{l}_2 \end{pmatrix}, \qquad I = \begin{pmatrix} I_{11} & I_{12} \\ I_{21} & I_{22} \end{pmatrix}, \qquad I_{11\cdot 2} = I_{11} - I_{12} I_{22}^{-1} I_{21}.$$

Let $\hat{\theta}_n$ be the unrestricted MLE and $\hat{\theta}_n^0 = (\xi_0', \hat{\eta}_n^{0\prime})'$ the MLE under H_0.

Theorem 7.2 (Composite null hypothesis)

Under the conditions of Theorem 6.13, if $H_0 : \xi = \xi_0$ is true with true value $\theta_0 = (\xi_0', \eta_0')'$, then

(a) $2\{l(\hat{\theta}_n) - l(\hat{\theta}_n^0)\} \xrightarrow{d} \chi^2(k - k_0)$;

(b) $2\{l(\hat{\theta}_n) - l(\hat{\theta}_n^0)\} = n(\hat{\theta}_n - \hat{\theta}_n^0)' I(\theta_0)(\hat{\theta}_n - \hat{\theta}_n^0) + o_p(1)$;

(c) $2\{l(\hat{\theta}_n) - l(\hat{\theta}_n^0)\} = n\, \dot{l}(\hat{\theta}_n^0)' I(\theta_0)^{-1} \dot{l}(\hat{\theta}_n^0) + o_p(1)$.

Sketch. As in Theorem 7.1,

$$2\{l(\hat{\theta}_n) - l(\theta_0)\} \approx \sqrt{n}\, \dot{l}(\theta_0)' I^{-1} \sqrt{n}\, \dot{l}(\theta_0).$$

Applying the same argument to the submodel with $\xi = \xi_0$ fixed, where only η is estimated, gives

$$2\{l(\hat{\theta}_n^0) - l(\theta_0)\} \approx \sqrt{n}\, \dot{l}_2(\theta_0)' I_{22}^{-1} \sqrt{n}\, \dot{l}_2(\theta_0).$$

Subtracting, and using the formula for the inverse of a partitioned matrix,

$$\begin{pmatrix} I_{11} & I_{12} \\ I_{21} & I_{22} \end{pmatrix}^{-1} = \begin{pmatrix} I_{11\cdot 2}^{-1} & -I_{11\cdot 2}^{-1} I_{12} I_{22}^{-1} \\ -I_{22}^{-1} I_{21} I_{11\cdot 2}^{-1} & I_{22}^{-1} + I_{22}^{-1} I_{21} I_{11\cdot 2}^{-1} I_{12} I_{22}^{-1} \end{pmatrix},$$

one finds

$$2\{l(\hat{\theta}_n) - l(\hat{\theta}_n^0)\} \approx \sqrt{n}\, \mathbf{Z}_n' I_{11\cdot 2}^{-1} \sqrt{n}\, \mathbf{Z}_n, \qquad \mathbf{Z}_n = \dot{l}_1(\theta_0) - I_{12} I_{22}^{-1} \dot{l}_2(\theta_0).$$

By the CLT, $\sqrt{n}\, \mathbf{Z}_n \xrightarrow{d} N_{k_1}(\mathbf{0}, I_{11\cdot 2})$, so the limit is $\chi^2(k_1) = \chi^2(k - k_0)$. This proves (a). For (b) one checks $\hat{\eta}_n^0 - \eta_0 \approx I_{22}^{-1} \dot{l}_2(\theta_0)$ and $I_{11\cdot 2}(\hat{\xi}_n - \xi_0) \approx \mathbf{Z}_n$, so that $n(\hat{\theta}_n - \hat{\theta}_n^0)' I(\hat{\theta}_n - \hat{\theta}_n^0) \approx n \mathbf{Z}_n' I_{11\cdot 2}^{-1} \mathbf{Z}_n$; (c) is similar. ■

The statistics in (b) and (c), with $I(\theta_0)$ replaced by $I(\hat{\theta}_n^0)$ (since η_0 is unknown), are the Wald and Rao statistics for composite hypotheses. A null hypothesis of the form $H_0 : g_1(\theta) = \cdots = g_{k_1}(\theta) = 0$ with smooth g_j is handled by reparametrizing so that $\xi_j = g_j(\theta)$; the limit is $\chi^2(k_1)$, where k_1 is the number of constraints, i.e. $\dim \Omega - \dim \Omega_0$.

Example 7.13 (Independence in a contingency table)

Counts X_{ij} ($i \le r$, $j \le c$) of an $r \times c$ table are Multi$(n, (p_{ij}))$. Independence is $H_0 : p_{ij} = p_{i\cdot} p_{\cdot j}$ for all i, j. The MLEs are $\hat{p}_{ij} = x_{ij}/n$ in general and $\hat{p}_{ij}^0 = \frac{x_{i\cdot}}{n} \frac{x_{\cdot j}}{n}$ under H_0, so

$$2\{l(\hat{p}) - l(\hat{p}^0)\} = 2 \sum_{i,j} X_{ij} \log \frac{X_{ij}}{n \hat{p}_{i\cdot} \hat{p}_{\cdot j}}.$$

The dimensions are $\dim \Omega = rc - 1$ and $\dim \Omega_0 = (r-1) + (c-1)$, so the limit is $\chi^2((r-1)(c-1))$. If the marginals were known, Example 7.12 would give Pearson's statistic with $np_{i\cdot}p_{\cdot j}$; substituting their estimates gives the usual **chi-square test of independence**:

$$\sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}} \ge \chi_\alpha^2((r-1)(c-1)), \qquad O_{ij} = X_{ij},\ \hat{E}_{ij} = \frac{X_{i\cdot} X_{\cdot j}}{n}.$$

Example 7.14 (A worked 2×3 table)

A survey of 200 students classified by year (lower, upper) and preferred study place (library, home, cafe) gives

	library	home	cafe	total
lower	40	30	30	100
upper	20	50	30	100
total	60	80	60	200

The expected counts are $\hat{E}_{1j} = \hat{E}_{2j} = (30, 40, 30)$. Pearson's statistic is

$$2\Big(\frac{10^2}{30} + \frac{10^2}{40} + 0\Big) = 11.67 > \chi^2_{0.05}(2) = 5.99,$$

so independence is rejected at the 5% level.

Non-identically distributed observations. The same results hold, under suitable conditions, when observations are independent but not identically distributed. Then $\bar{I}_n(\theta) = \frac{1}{n} E_\theta[-\ddot{l}(\theta)]$ replaces $I(\theta)$, and $W_n = n(\hat{\theta} - \theta_0)' \bar{I}_n(\theta_0)(\hat{\theta} - \theta_0)$, $R_n = n\,\dot{l}(\theta_0)' \bar{I}_n(\theta_0)^{-1} \dot{l}(\theta_0)$.

Example 7.15 (Homogeneity of multinomial populations)

Let $\mathbf{X}_i \sim \text{Multi}(n_i, \mathbf{p}_i)$, $i = 1, \dots, r$, be independent, each with c cells, and test $H_0 : \mathbf{p}_1 = \cdots = \mathbf{p}_r$. Under H_0 the common $\mathbf{p}$ is estimated by pooling, $\hat{p}^0_j = X_{\cdot j}/n$, $n = \sum n_i$. With $O_{ij} = X_{ij}$ and $\hat{E}_{ij} = n_i \hat{p}^0_j$, the LRT statistic is $2 \sum O_{ij} \log(O_{ij}/\hat{E}_{ij})$. Expanding $2u \log u \approx 2(u-1) + (u-1)^2$ around $u = O_{ij}/\hat{E}_{ij} = 1$, it is approximately Pearson's $\sum \sum (O_{ij} - \hat{E}_{ij})^2 / \hat{E}_{ij}$. The degrees of freedom are $r(c-1) - (c-1) = (r-1)(c-1)$, the same as for independence, although the sampling scheme differs: here the row totals are fixed. (One shows the limit by writing the statistic as a quadratic form in the asymptotically normal vectors $\sqrt{n_i}(\hat{\mathbf{p}}_i - \hat{\mathbf{p}}^0)$ with an idempotent matrix of rank $(r-1)(c-1)$.)

7.4 Selected exercises

Motivation. These are textbook exercises for this chapter (numbered as in the book). They derive likelihood ratio tests in models not treated above, show that the power of the t test grows with the standardized effect, and explain why the Poisson homogeneity statistic is approximately Pearson's χ^2.

Exercise 7.6 (Power of the two-sided t test)

Let $X_1, \dots, X_n$ $(n \geq 2)$ be $N(\mu, \sigma^2)$ and consider the two-sided t test of Example 7.5 with statistic $t = (\bar{X} - \mu_0)/(S/\sqrt{n})$.

(a) Show that $t \overset{d}{=} (Z + \delta)/\sqrt{V/(n-1)}$, where $\delta = (\mu - \mu_0)/(\sigma/\sqrt{n})$, $Z \sim N(0,1)$, $V \sim \chi^2(n-1)$, and Z, V are independent.

(b) Show that the power $P_{\mu,\sigma^2}\{|t| \geq t_{\alpha/2}(n-1)\}$ is a function of $|\delta|$ alone and is increasing in $|\delta|$.

Solution.

(a) Put $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n})$ and $V = (n-1)S^2/\sigma^2$. For a normal sample $\bar{X}$ and S^2 are independent, $Z \sim N(0,1)$ and $V \sim \chi^2(n-1)$ (the normal-sample theorem of Chapter 4). Dividing numerator

and denominator of t by $\sigma/\sqrt{n}$,

$$t = \frac{(\bar{X}-\mu)/(\sigma/\sqrt{n}) + (\mu-\mu_0)/(\sigma/\sqrt{n})}{S/\sigma} = \frac{Z+\delta}{\sqrt{V/(n-1)}}.$$

(b)

Step 1. Let $c = t_{\alpha/2}(n-1)$ and, for $k>0$, $g_k(\delta) = P(|Z+\delta| \ge k) = 1 - \Phi(k-\delta) + \Phi(-k-\delta)$. By (a) and independence of Z and V,

$$\gamma(\delta) = P(|t| \ge c) = E\big[g_K(\delta)\big], \qquad K = c\sqrt{V/(n-1)}.$$

Step 2. Since $-Z \overset{d}{=} Z$, $g_k(-\delta) = P(|-Z+\delta| \ge k) = g_k(\delta)$, so $\gamma(-\delta) = \gamma(\delta)$: the power depends on $|\delta|$ only.

Step 3. $g_k'(\delta) = \phi(k-\delta) - \phi(k+\delta)$. For $\delta > 0$, $|k-\delta| < k+\delta$, hence $\phi(k-\delta) > \phi(k+\delta)$ and g_k is strictly increasing on $[0,\infty)$ for every $k>0$.

Step 4. $K>0$ with probability 1, so for $0 \le \delta_1 < \delta_2$ we have $g_K(\delta_1) < g_K(\delta_2)$ pointwise, and taking expectations gives $\gamma(\delta_1) < \gamma(\delta_2)$. The law of t is the *noncentral t distribution* with noncentrality δ. ■

Exercise 7.9 (A one-sided test for Beta$(\theta,1)$)

Let $X_1,\dots,X_n$ have density $f(x;\theta) = \theta x^{\theta-1}$ on $(0,1)$, $\theta>0$. Find the level-α LRT of $H_0:\theta \le 1$ against $H_1:\theta>1$ $(0<\alpha<\frac{1}{2})$.

Solution.

Step 1. With $T = -\sum_i \log X_i > 0$, the log-likelihood is $l(\theta) = n\log\theta - (\theta-1)T$, strictly concave with maximum at $\hat\theta = n/T$. The MLE over $\Omega_0 = (0,1]$ is $\min(\hat\theta, 1)$ by concavity.

Step 2. If $\hat\theta \le 1$ the two maxima agree and the statistic is 0. If $\hat\theta > 1$, i.e. $T<n$,

$$2\{l(\hat\theta) - l(1)\} = 2\{n\log(n/T) - n + T\} =: h(T), \qquad h'(T) = 2\Big(1 - \frac{n}{T}\Big) < 0.$$

So h decreases from $+\infty$ to 0 on $(0,n)$, and for every $c>0$ the event $\{\text{statistic} \ge c\}$ equals $\{T \le d\}$ for some $d \in (0,n)$. The LRT rejects H_0 for small T.

Step 3. Under θ, $P(-\log X_i > s) = P(X_i < e^{-s}) = e^{-\theta s}$, so the $-\log X_i$ are i.i.d. exponential with rate θ and $2\theta T \sim \chi^2(2n)$. Hence $P_\theta(T \le d) = P\{\chi^2(2n) \le 2\theta d\}$ increases with θ, and the size is attained at $\theta = 1$.

Step 4. The level-α LRT therefore rejects H_0 iff

$$-2\sum_{i=1}^{n} \log X_i \le \chi^2_{1-\alpha}(2n),$$

the lower α point of $\chi^2(2n)$. Since $\alpha < \frac{1}{2}$, $\chi^2_{1-\alpha}(2n)$ is below the median of $\chi^2(2n)$, which is known to lie below its mean $2n$ (the median of a gamma law is smaller than its mean); so $d = \chi^2_{1-\alpha}(2n)/2 < n$ as required in Step 2. ■

Exercise 7.15(a) (Testing the endpoint of a uniform law)

Let $X_1,\dots,X_n$ be $U[0,\theta]$, $\theta>0$. Find the level-α LRT of $H_0:\theta=\theta_0$ against $H_1:\theta\ne\theta_0$.

Solution.

Step 1. With $M = \max_i X_i$, $L(\theta) = \theta^{-n} I(\theta \geq M)$, so the MLE is $\hat{\theta} = M$ and

$$\frac{L(\theta_0)}{L(\hat{\theta})} = \begin{cases} (M/\theta_0)^n, & M \leq \theta_0, \\ 0, & M > \theta_0. \end{cases}$$

The LRT rejects when this ratio is small, i.e. when $M > \theta_0$ or $M \leq k\theta_0$ for a constant $k \in (0,1)$.

Step 2. Under H_0, $P(M > \theta_0) = 0$ and $P(M \leq k\theta_0) = k^n$. Setting $k^n = \alpha$ gives the level-α LRT

$$\text{reject } H_0 \iff M > \theta_0 \text{ or } M \leq \alpha^{1/n}\theta_0,$$

whose size is exactly α. Unlike the regular cases of Section 7.3, the statistic $2\{l(\hat{\theta}) - l(\theta_0)\} = -2n\log(M/\theta_0)$ is exactly $\chi^2(2)$ under H_0, not $\chi^2(1)$: $-n\log(M/\theta_0)$ is standard exponential because $P(M/\theta_0 \leq e^{-s/n}) = e^{-s}$. ■

Exercise 7.17(a)(b) (Homogeneity of Poisson means)

Let $X_{i1}, \ldots, X_{in_i}$ be a sample from Poisson(θ_i), $i = 1, \ldots, k$ $(k \geq 2)$, the samples being independent. Put $X_{i\cdot} = \sum_j X_{ij}$, $n = n_1 + \cdots + n_k$, $\hat{\theta}_i = X_{i\cdot}/n_i$ and $\hat{\theta}^0 = \sum_i X_{i\cdot}/n$. For $H_0 : \theta_1 = \cdots = \theta_k$:
(a) Show that the LRT statistic is $2\sum_i n_i\{\hat{\theta}_i \log(\hat{\theta}_i/\hat{\theta}^0) - \hat{\theta}_i + \hat{\theta}^0\}$ (with $0\log 0 = 0$).
(b) Show that if $n_i \to \infty$ with $n_i/n \to \gamma_i \in (0,1)$, then under H_0 (with common value $\theta > 0$) the statistic is approximately $\sum_i (X_{i\cdot} - n_i\hat{\theta}^0)^2/(n_i\hat{\theta}^0)$, in the sense that the ratio of the two tends to 1 in probability.

Solution.

(a) Up to a term free of θ, $l(\theta) = \sum_i \{X_{i\cdot}\log\theta_i - n_i\theta_i\}$. Each summand is maximized at $\hat{\theta}_i$, and under H_0 the function $(\sum_i X_{i\cdot})\log\vartheta - n\vartheta$ is maximized at $\hat{\theta}^0$. Using $X_{i\cdot} = n_i\hat{\theta}_i$,

$$2\{l(\hat{\theta}) - l(\hat{\theta}^0)\} = 2\sum_i \{n_i\hat{\theta}_i \log(\hat{\theta}_i/\hat{\theta}^0) - n_i\hat{\theta}_i + n_i\hat{\theta}^0\}.$$

(The linear terms sum to 0, since $\sum_i n_i\hat{\theta}_i = n\hat{\theta}^0$.)

(b)

Step 1. For $a > 0$ let $g_a(x) = x\log(x/a) - x + a$, $x \geq 0$. Then $g_a(a) = 0$, $g_a'(a) = \log(x/a)|_{x=a} = 0$ and $g_a''(x) = 1/x$. By Taylor's theorem with the Lagrange remainder, for $x > 0$

$$g_a(x) = \frac{(x-a)^2}{2\xi}, \qquad \xi \text{ between } x \text{ and } a.$$

Step 2. Under H_0, the law of large numbers gives $\hat{\theta}_i \xrightarrow{p} \theta$ and $\hat{\theta}^0 \xrightarrow{p} \theta$, so $P(\hat{\theta}_i > 0) \to 1$ and the intermediate points satisfy $\xi_i \xrightarrow{p} \theta$. Hence

$$2\{l(\hat{\theta}) - l(\hat{\theta}^0)\} = \sum_i n_i \frac{(\hat{\theta}_i - \hat{\theta}^0)^2}{\hat{\theta}^0} \cdot \frac{\hat{\theta}^0}{\xi_i},$$

and each factor $\hat{\theta}^0/\xi_i \xrightarrow{p} 1$.

Step 3. A sum $\sum_i a_i b_i$ with $a_i \geq 0$, $\sum_i a_i > 0$ lies between $\min_i b_i \sum_i a_i$ and $\max_i b_i \sum_i a_i$; with k fixed and every $b_i \xrightarrow{p} 1$, the ratio $\sum_i a_i b_i / \sum_i a_i \xrightarrow{p} 1$. Since $n_i(\hat{\theta}_i - \hat{\theta}^0)^2/\hat{\theta}^0 = (X_{i\cdot} - n_i\hat{\theta}^0)^2/(n_i\hat{\theta}^0)$, the statistic is the Pearson-type sum times factors tending to 1, which is the claim. (Its limit law is $\chi^2(k-1)$ by Theorem 7.2.) ■

Chapter 8. Comparing Estimators

8.1 Criteria for comparing estimators

A good estimator of $\eta = \eta(\theta)$ gives values close to the truth. The most common measure of performance is the **mean squared error**

$$MSE(\hat{\eta}, \theta) = E_\theta[\{\hat{\eta}(X_1, \ldots, X_n) - \eta(\theta)\}^2] = \operatorname{Var}_\theta \hat{\eta} + \{E_\theta \hat{\eta} - \eta(\theta)\}^2,$$

the variance plus the squared **bias**.

Example 8.1 (A clear winner)

For a $U(0, \theta)$ sample ($n \geq 2$), compare $\hat{\theta}^{\mathrm{MLE}} = X_{(n)}$ with $\hat{\theta}^{\mathrm{MME}} = 2\bar{X}$. Using $X_{(n)}/\theta \sim \mathrm{Beta}(n, 1)$,

$$MSE(X_{(n)}, \theta) = \frac{2\theta^2}{(n+1)(n+2)}, \qquad MSE(2\bar{X}, \theta) = 4 \operatorname{Var} \bar{X} = \frac{\theta^2}{3n}.$$

The MLE is better for every θ and every n, and its MSE decays like n^{-2} rather than n^{-1}.

Example 8.2 (No uniform winner)

For $N(\theta, 1)$, compare $\bar{X}$ (MSE $1/n$) with the silly estimator $\hat{\theta}^0 \equiv 0$ (MSE θ^2). If $|\theta| < 1/\sqrt{n}$, the constant has smaller MSE. Neither dominates the other. A more interesting competitor is the "pretest" estimator

$$\hat{\theta}^H = \begin{cases} 0, & |\bar{X}| < 1.96/\sqrt{n}, \\ \bar{X}, & |\bar{X}| \geq 1.96/\sqrt{n}. \end{cases}$$

Its MSE, $n^{-1}E[(Z + \sqrt{n}\theta)^2 \mathbf{1}\{|Z + \sqrt{n}\theta| \geq 1.96\}] + \theta^2 P(|Z + \sqrt{n}\theta| < 1.96)$ with $Z \sim N(0, 1)$, depends on θ only through $\sqrt{n}\theta$. Times n, it is below 1 near $\theta = 0$ but rises to about 2.5 when $\sqrt{n}|\theta| \approx 2$. For fixed $\theta \neq 0$, $n \cdot MSE \to 1$.

Since MSEs typically cross, we need criteria that do not depend on the unknown θ:

- **Maximum MSE**: compare $\max_\theta MSE(\hat{\eta}, \theta)$; an estimator minimizing it is **minimax**.
- **Bayes MSE**: for a weight (prior) function $\pi \geq 0$ on Ω, compare $r(\pi, \hat{\eta}) = \int_\Omega MSE(\hat{\eta}, \theta)\pi(\theta)\, d\theta$; a minimizer is a **Bayes estimator** for π.

For a positive target one sometimes uses the relative MSE $E_\theta(\hat{\eta}/\eta - 1)^2$ instead.

Example 8.3

For $N(\mu, \sigma^2)$, consider $\hat{\sigma}_c^2 = c\sum(X_i - \bar{X})^2$, $c > 0$. Since $V = \sum(X_i - \bar{X})^2/\sigma^2 \sim \chi^2(n-1)$,

$$E_\theta(\hat{\sigma}_c^2/\sigma^2 - 1)^2 = E(cV - 1)^2 = c^2 \cdot 2(n-1) + \{c(n-1) - 1\}^2,$$

which does not depend on θ. Minimizing in c gives $c = 1/(n+1)$: the minimax relative-MSE estimator is $\frac{1}{n+1}\sum(X_i - \bar{X})^2$, neither the MLE ($c = 1/n$) nor S^2 ($c = 1/(n-1)$).

Another approach restricts the class of estimators. An estimator is **unbiased** if $E_\theta \hat{\eta} = \eta(\theta)$ for all $\theta \in \Omega$; then $MSE = \operatorname{Var}_\theta \hat{\eta}$.

Definition (UMVUE)

$\hat{\eta}^*$ is a **uniformly minimum variance unbiased estimator** (UMVUE) of $\eta(\theta)$ if (i) it is unbiased, and (ii) $\operatorname{Var}_\theta \hat{\eta}^* \le \operatorname{Var}_\theta \hat{\eta}$ for all θ and every unbiased $\hat{\eta}$.

Theorem 8.1 (Uniqueness)

If the UMVUE of $\eta(\theta)$ has finite variance, it is unique: if $\hat{\eta}_1, \hat{\eta}_2$ are both UMVUEs, then $P_\theta(\hat{\eta}_1 = \hat{\eta}_2) = 1$ for all θ.

Proof. $\hat{\eta}_3 = (\hat{\eta}_1 + \hat{\eta}_2)/2$ is unbiased, so $\operatorname{Var} \hat{\eta}_3 \ge \operatorname{Var} \hat{\eta}_1 = \operatorname{Var} \hat{\eta}_2$, i.e.

$$\tfrac{1}{4}\{\operatorname{Var} \hat{\eta}_1 + 2 \operatorname{Cov}(\hat{\eta}_1, \hat{\eta}_2) + \operatorname{Var} \hat{\eta}_2\} \ge \operatorname{Var} \hat{\eta}_2,$$

so $\operatorname{Cov}(\hat{\eta}_1, \hat{\eta}_2) \ge \operatorname{Var} \hat{\eta}_2$. Then $\operatorname{Var}(\hat{\eta}_1 - \hat{\eta}_2) = \operatorname{Var} \hat{\eta}_1 + \operatorname{Var} \hat{\eta}_2 - 2 \operatorname{Cov}(\hat{\eta}_1, \hat{\eta}_2) \le 0$. A variable with zero variance and zero mean is 0 with probability one (Theorem 1.11 in Volume I). ■

8.2 Sufficient statistics

Motivation. A sample of size n often carries all its information about θ in a few numbers, such as $\sum X_i$. Sufficiency makes this precise, and the factorization theorem lets us find such statistics by inspecting the likelihood.

Can a part of the data, a statistic, carry all the information about θ?

Example 8.4

Two Bernoulli(θ) trials X_1, X_2, summarized by $Y = X_1 + X_2$. If $Y = 0$ or 2, the data are determined. If $Y = 1$,

$$P_\theta(X_1 = 0, X_2 = 1 \mid Y = 1) = \frac{(1-\theta)\theta}{2\theta(1-\theta)} = \frac{1}{2} = P_\theta(X_1 = 1, X_2 = 0 \mid Y = 1).$$

The original data can be "recreated" from Y by a coin flip, without knowing θ. So Y is as informative as (X_1, X_2).

Definition (Sufficiency)

A statistic $Y = u(X_1, \ldots, X_n)$ is **sufficient** for $\theta \in \Omega$ if the conditional distribution of $(X_1, \ldots, X_n)$ given $Y = y$ does not depend on θ, i.e. $P_{\theta_1}(\mathbf{X} \in A \mid Y = y) = P_{\theta_2}(\mathbf{X} \in A \mid Y = y)$ for all A, y, θ_1, θ_2.

Example 8.5

For n Bernoulli(θ) trials and $Y = \sum X_i \sim B(n, \theta)$,

$$P_\theta(X_1 = x_1, \ldots, X_n = x_n \mid Y = y) = \frac{\theta^y (1-\theta)^{n-y}}{\binom{n}{y} \theta^y (1-\theta)^{n-y}} = \binom{n}{y}^{-1} \quad (\textstyle\sum x_i = y),$$

free of θ. So $\sum X_i$ is sufficient: all arrangements with y successes are equally likely.

Theorem 8.2 (Factorization)

$Y = u(\mathbf{X})$ is sufficient for $\theta \in \Omega$ if and only if there are functions k_1, k_2 with

$$\prod_{i=1}^{n} f(x_i;\theta) = k_1(u(\mathbf{x}),\theta)\, k_2(\mathbf{x}) \qquad \text{for all } \mathbf{x} \text{ and } \theta \in \Omega.$$

Proof. We prove the discrete case.

($\Leftarrow$) For $\mathbf{x}$ with $u(\mathbf{x}) = y$,

$$P_\theta(\mathbf{X}=\mathbf{x} \mid Y=y) = \frac{P_\theta(\mathbf{X}=\mathbf{x})}{\sum_{\mathbf{z}:u(\mathbf{z})=y} P_\theta(\mathbf{X}=\mathbf{z})} = \frac{k_1(y,\theta)k_2(\mathbf{x})}{k_1(y,\theta)\sum_{\mathbf{z}:u(\mathbf{z})=y} k_2(\mathbf{z})} = \frac{k_2(\mathbf{x})}{\sum_{u(\mathbf{z})=y} k_2(\mathbf{z})},$$

which does not depend on θ; for $u(\mathbf{x}) \neq y$ it is 0.

($\Rightarrow$) If Y is sufficient, $P_\theta(\mathbf{X}=\mathbf{x} \mid Y = u(\mathbf{x})) =: k_2(\mathbf{x})$ does not depend on θ, and with $k_1(y,\theta) = P_\theta(Y=y)$,

$$P_\theta(\mathbf{X}=\mathbf{x}) = P_\theta(\mathbf{X}=\mathbf{x}, Y=u(\mathbf{x})) = k_2(\mathbf{x})\, k_1(u(\mathbf{x}),\theta).$$

■

Example 8.6

(a) Poisson(θ): $\prod f = e^{-n\theta}\theta^{\sum x_i} \cdot \prod(1/x_i!)$, so $\sum X_i$ is sufficient.

(b) Exp(θ): $\prod f = \theta^{-n} e^{-\sum x_i/\theta} \cdot \prod \mathbf{1}\{x_i > 0\}$, so $\sum X_i$ is sufficient.

(c) Gamma(α,β): $\prod f = \{\Gamma(\alpha)\beta^\alpha\}^{-n}(\prod x_i)^{\alpha-1} e^{-\sum x_i/\beta} \cdot \prod \mathbf{1}\{x_i>0\}$, so $(\sum X_i, \prod X_i)$ is sufficient for (α,β).

Theorem 8.3 (Exponential families)

If $f(x;\theta) = \exp\{\sum_{j=1}^{k} g_j(\theta)T_j(x) - A(g(\theta)) + S(x)\}$ for $x \in \mathcal{X}$, with support $\mathcal{X}$ not depending on θ, then $(\sum_i T_1(X_i), \ldots, \sum_i T_k(X_i))$ is sufficient for θ.

Proof. $\prod f(x_i;\theta) = \exp\{\sum_j g_j(\theta)\sum_i T_j(x_i) - nA(g(\theta))\} \cdot \exp\{\sum S(x_i)\} \prod \mathbf{1}_{\mathcal{X}}(x_i)$; apply the factorization theorem. ■

Theorem 8.4

If Y is sufficient for θ and $W = g(Y)$ with g one-to-one, then W is sufficient.

Proof. $\{W=w\} = \{Y = g^{-1}(w)\}$, so $P_\theta(\mathbf{X}\in A \mid W=w) = P_\theta(\mathbf{X}\in A \mid Y = g^{-1}(w))$, which does not depend on θ. ■

Example 8.7

(a) $N(\mu,\sigma^2)$ is a two-parameter exponential family with $T = (x, x^2)$, so $(\sum X_i, \sum X_i^2)$ is sufficient, and so is the one-to-one image $(\bar{X}, S^2)$.
(b) Beta(α, 1): $f = \exp\{(\alpha-1)\log x + \log\alpha\}$ on $(0,1)$, so $\sum \log X_i$ is sufficient.
(c) When the support depends on θ, use indicators. For the two-parameter exponential $\frac{1}{\sigma}e^{-(x-\mu)/\sigma}\mathbf{1}\{x \geq \mu\}$,

$$\prod f(x_i;\mu,\sigma) = \sigma^{-n} e^{-(\sum x_i - n\mu)/\sigma} \mathbf{1}\{\min x_i \geq \mu\},$$

so $(\sum X_i, X_{(1)})$ is sufficient. For $U[\theta_1,\theta_2]$, $\prod f = (\theta_2-\theta_1)^{-n}\mathbf{1}\{\theta_1 \le x_{(1)}, x_{(n)} \le \theta_2\}$, so $(X_{(1)}, X_{(n)})$ is sufficient.

Theorem 8.5

If the MLE $\hat\theta$ exists uniquely, it is a function of any sufficient statistic $S = u(\mathbf{X})$.

Proof. By factorization, $\hat\theta(\mathbf{x}) = \arg\max_\theta k_1(u(\mathbf{x}),\theta)k_2(\mathbf{x}) = \arg\max_\theta k_1(u(\mathbf{x}),\theta)$, which depends on $\mathbf{x}$ only through $u(\mathbf{x})$. ■

If the unique MLE is itself sufficient, then by this theorem it is a function of every sufficient statistic; such a statistic is called **minimal sufficient**. In Examples 8.5–8.7 the MLEs are of this kind.

Theorem 8.6 (Rao–Blackwell)

Let Y be sufficient for θ, $\hat\eta$ an estimator of $\eta(\theta)$ with finite variance, and $\hat\eta^{RB} = E[\hat\eta \mid Y]$. Then $\hat\eta^{RB}$ is a statistic, $E_\theta\hat\eta^{RB} = E_\theta\hat\eta$, and

$$MSE(\hat\eta^{RB},\theta) \le MSE(\hat\eta,\theta) \qquad \text{for all } \theta.$$

Proof. Since the conditional law of $\mathbf{X}$ given Y is free of θ, so is $E[\hat\eta(\mathbf{X}) \mid Y]$; thus $\hat\eta^{RB}$ is a statistic. By the properties of conditional expectation, $E\hat\eta^{RB} = E\hat\eta$ and

$$\operatorname{Var}\hat\eta = E[\operatorname{Var}(\hat\eta \mid Y)] + \operatorname{Var}(E[\hat\eta \mid Y]) \ge \operatorname{Var}\hat\eta^{RB}.$$

Since the bias is the same, $MSE = \text{Var} + \text{bias}^2$ gives the inequality. ■

Example 8.8

For $U[0,\theta]$, $Y = X_{(n)}$ is sufficient (Example 8.7(c) with $\theta_1 = 0$). Improve $\hat\theta = 2\bar X$, whose MSE is $\theta^2/(3n)$. Given $X_{(n)} = y$, the other $n-1$ order statistics are distributed as order statistics of $n-1$ uniforms on $(0,y)$, so $E(X_{(r)} \mid X_{(n)} = y) = \frac{r}{n}y$, $r \le n-1$. Hence

$$\hat\theta^{RB} = \frac{2}{n}E\Big(\sum_{r=1}^{n} X_{(r)} \,\Big|\, X_{(n)}\Big) = \frac{2}{n}\Big(\sum_{r=1}^{n-1}\frac{r}{n} + 1\Big)X_{(n)} = \frac{n+1}{n}X_{(n)},$$

with $MSE = \int_0^\theta \left(\frac{n+1}{n}x - \theta\right)^2 \frac{nx^{n-1}}{\theta^n}dx = \frac{\theta^2}{n(n+2)} \le \frac{\theta^2}{3n}$.

8.3 Minimum variance unbiased estimation

Motivation. Among unbiased estimators we want the one with the smallest variance. Rao–Blackwell says conditioning on a sufficient statistic never hurts, and with completeness (Lehmann–Scheffé) the result is the unique best unbiased estimator.

Rao–Blackwell improves an estimator by conditioning on a sufficient statistic. If the statistic is also complete, the result is the UMVUE.

Definition (Completeness)

A statistic Y is **complete** for $\theta \in \Omega$ if $E_\theta g(Y) = 0$ for all $\theta \in \Omega$ implies $P_\theta(g(Y) = 0) = 1$ for all θ. A statistic that is complete and sufficient is a **complete sufficient statistic**.

Theorem 8.7 (Lehmann–Scheffé)

Let Y be complete sufficient for $\theta \in \Omega$.

(a) If $\hat{\eta}_0$ is any unbiased estimator of $\eta(\theta)$, then $E[\hat{\eta}_0 \mid Y]$ is the UMVUE of $\eta(\theta)$.

(b) If $\delta(Y)$ is unbiased for $\eta(\theta)$, then $\delta(Y)$ is the UMVUE.

Proof. (a) $\hat{\eta}_0^{RB} = E[\hat{\eta}_0 \mid Y]$ is unbiased. Let $\hat{\eta}_1$ be any other unbiased estimator and $\hat{\eta}_1^{RB} = E[\hat{\eta}_1 \mid Y]$. By Rao–Blackwell, $\operatorname{Var}\hat{\eta}_1^{RB} \le \operatorname{Var}\hat{\eta}_1$. Both $\hat{\eta}_0^{RB}$ and $\hat{\eta}_1^{RB}$ are functions of Y with mean $\eta(\theta)$, so $E_\theta[\hat{\eta}_0^{RB} - \hat{\eta}_1^{RB}] = 0$ for all θ, and by completeness $\hat{\eta}_0^{RB} = \hat{\eta}_1^{RB}$ with probability one. Hence $\operatorname{Var}\hat{\eta}_0^{RB} = \operatorname{Var}\hat{\eta}_1^{RB} \le \operatorname{Var}\hat{\eta}_1$.

(b) Apply (a) with $\hat{\eta}_0 = \delta(Y)$, for which $E[\delta(Y) \mid Y] = \delta(Y)$. ■

So, to find a UMVUE: find a complete sufficient statistic Y, then either find a function of Y that is unbiased, or condition a simple unbiased estimator on Y.

Example 8.9 (Bernoulli)

$Y = \sum X_i \sim B(n, \theta)$. If $E_\theta g(Y) = \sum_y g(y)\binom{n}{y}\theta^y(1-\theta)^{n-y} = 0$ for all $0 < \theta < 1$, divide by $(1-\theta)^n$ and set $\lambda = \theta/(1-\theta)$: $\sum_y g(y)\binom{n}{y}\lambda^y = 0$ for all $\lambda > 0$. A polynomial vanishing on an interval has zero coefficients, so $g \equiv 0$. Thus Y is complete sufficient, and $\bar{X} = Y/n$ is the UMVUE of θ.

Example 8.10 (Poisson and exponential)

(a) Poisson: $Y = \sum X_i \sim \text{Poisson}(n\theta)$, and $\sum_y g(y)\lambda^y/y! = 0$ for all $\lambda > 0$ forces $g \equiv 0$ (uniqueness of power series). So $\bar{X}$ is the UMVUE of θ.
(b) Exponential: $Y = \sum X_i \sim \text{Gamma}(n, \theta)$, and $\int_0^\infty g(y)y^{n-1}e^{-\lambda y}dy = 0$ for all $\lambda > 0$ forces $g(y)y^{n-1} = 0$ (uniqueness of the Laplace transform). So $\bar{X}$ is the UMVUE of θ.

The general result for exponential families also rests on the uniqueness of Laplace transforms; we state it without proof.

Theorem 8.8

Let $f(x;\eta) = \exp\{\sum_{j=1}^k \eta_j T_j(x) - A(\eta) + S(x)\}$ satisfy:
(i) the support does not depend on η;
(ii) the parameter space N contains an open box $(a_1, b_1) \times \cdots \times (a_k, b_k)$;
(iii) $\mathbf{c}'\mathbf{T}(X)$ is not constant for $\mathbf{c} \neq \mathbf{0}$. Then $(\sum_i T_1(X_i), \ldots, \sum_i T_k(X_i))$ is complete sufficient.

Example 8.11

(a) For $N(\mu, \sigma^2)$ the natural parameter $(\mu/\sigma^2, -1/(2\sigma^2))$ ranges over $\mathbb{R} \times (-\infty, 0)$, so $(\sum X_i, \sum X_i^2)$, and hence $(\bar{X}, S^2)$, is complete sufficient. Since $E\bar{X} = \mu$ and $ES^2 = \sigma^2$, they are the UMVUEs.
(b) For $N(\mu, \mu^2)$, $\mu > 0$, the natural parameter $(1/\mu, -1/(2\mu^2))$ lies on a curve, which contains

no open box, so Theorem 8.8 does not apply. Indeed $(Y_1, Y_2) = (\sum X_i, \sum X_i^2)$ is not complete: $E_\mu Y_2 = 2n\mu^2$ and $E_\mu Y_1^2 = n(n+1)\mu^2$, so $g = \frac{Y_2}{2n} - \frac{Y_1^2}{n(n+1)}$ has mean 0 for all μ without being 0.
(c) For $U[0,\theta]$, $Y = X_{(n)}$ has pdf ny^{n-1}/θ^n on $(0,\theta)$. If $\int_0^\theta g(y)y^{n-1}dy = 0$ for all $\theta > 0$, differentiating gives $g(\theta)\theta^{n-1} = 0$. So Y is complete, and $\frac{n+1}{n}X_{(n)}$ is the UMVUE of θ (Example 8.8).

Example 8.12 (Bernoulli variance)

To estimate $\eta = \theta(1-\theta)$ from $n \geq 2$ trials: $X_1(1-X_2)$ is unbiased. Conditioning on $Y = \sum X_i = y$,

$$E[X_1(1-X_2) \mid Y = y] = \frac{P(X_1 = 1, X_2 = 0, \sum_{i\geq 3} X_i = y-1)}{P(Y=y)} = \frac{\binom{n-2}{y-1}}{\binom{n}{y}} = \frac{y(n-y)}{n(n-1)}.$$

With $\hat\theta = Y/n$, the UMVUE is $\frac{n}{n-1}\hat\theta(1-\hat\theta)$.

Example 8.13 (A function of a Poisson mean)

Find the UMVUE of $\eta = e^{-2\theta}$ from a Poisson(θ) sample. We need δ with $E_\theta\delta(Y) = e^{-2\theta}$, $Y \sim$ Poisson$(n\theta)$:

$$\sum_{y\geq 0}\delta(y)\frac{e^{-n\theta}(n\theta)^y}{y!} = e^{-2\theta} \iff \sum_y \frac{\delta(y)}{y!}\lambda^y = e^{(1-2/n)\lambda} = \sum_y \frac{(1-2/n)^y}{y!}\lambda^y,$$

with $\lambda = n\theta$. Comparing coefficients, $\delta(y) = (1-2/n)^y$. For large n, $(1-2/n)^{n\bar X} \approx e^{-2\bar X}$, the MLE. But for $n = 1$ the UMVUE is $(-1)^{X_1}$, which takes the value -1 for an estimand in $(0,1)$: insisting on unbiasedness can give absurd answers.

Example 8.14 (Reliability for a normal mean)

For $N(\theta, 1)$, $n \geq 2$, estimate $\eta = P_\theta(X_1 > a)$. The indicator $\mathbf{1}\{X_1 > a\}$ is unbiased, and $\bar X$ is complete sufficient, so the UMVUE is $P(X_1 > a \mid \bar X)$. Since $X_1 - \bar X \sim N(0, \frac{n-1}{n})$ is independent of $\bar X$ (Theorem 4.5 in Volume I),

$$P(X_1 > a \mid \bar X = y) = P(X_1 - \bar X > a - y) = 1 - \Phi\Big(\sqrt{\tfrac{n}{n-1}}\,(a-y)\Big),$$

so the UMVUE is $1 - \Phi\big(\sqrt{n/(n-1)}\,(a - \bar X)\big)$.

In this example the independence of $X_1 - \bar X$ and $\bar X$ made the conditional expectation easy. A general source of such independence:

Theorem 8.9 (Basu)

If Y is complete sufficient for θ, and the distribution of a statistic Z does not depend on θ (Z is **ancillary**), then Y and Z are independent.

Proof. Fix B. By sufficiency, $P(Z \in B \mid Y)$ does not depend on θ; and $P_\theta(Z \in B) =: p_B$ does not depend on θ by ancillarity. Then $E_\theta[P(Z \in B \mid Y) - p_B] = 0$ for all θ, so by completeness $P(Z \in B \mid Y) = p_B$ with probability one. That is, $P(Z \in B \mid Y) = P(Z \in B)$ for every B. ∎

Example 8.15 (Reliability for an exponential mean)

For $\mathrm{Exp}(\theta)$, estimate $\eta = P_\theta(X_1 > a) = e^{-a/\theta}$. With $Y = \sum X_i$ complete sufficient, the UMVUE is $P(X_1 > a \mid Y)$. Writing $X_i = \theta Z_i$ with $Z_i \overset{\text{i.i.d.}}{\sim} \mathrm{Exp}(1)$, the ratio $X_1/Y = Z_1/\sum Z_i \sim \mathrm{Beta}(1, n-1)$ is ancillary, hence independent of Y by Basu's theorem. Therefore

$$P(X_1 > a \mid Y = y) = P\Big(\frac{X_1}{Y} > \frac{a}{y}\Big) = \int_{a/y}^{1} (n-1)(1-z)^{n-2} dz = \Big(1 - \frac{a}{y}\Big)^{n-1} \mathbf{1}\{y > a\},$$

and the UMVUE is $\big(1 - \frac{a}{n\bar{X}}\big)^{n-1} \mathbf{1}\{n\bar{X} > a\}$, close to the MLE $e^{-a/\bar{X}}$ for large n.

8.4 Asymptotic comparison

For consistent estimators the MSE is Var $+\text{bias}^2$, and typically the squared bias is $O(n^{-2})$ while the variance is of order n^{-1}. So for large n we compare the variances of the limiting distributions.

Definition (Asymptotic relative efficiency)

If $\sqrt{n}(\hat{\theta}_n^i - \theta) \xrightarrow{d} N(0, \sigma_i^2(\theta))$, $i = 1, 2$, the **asymptotic relative efficiency** of $\hat{\theta}^1$ with respect to $\hat{\theta}^2$ is

$$ARE(\{\hat{\theta}_n^1\}, \{\hat{\theta}_n^2\}) = \frac{\sigma_2^2(\theta)}{\sigma_1^2(\theta)}.$$

A value above 1 means $\hat{\theta}^1$ is more precise. Equivalently, to reach the same accuracy, $\hat{\theta}^2$ needs about ARE times as many observations as $\hat{\theta}^1$: if a 95% error bound $1.96\sigma_i/\sqrt{n_i}$ is to equal d, then $n_i \approx (1.96/d)^2 \sigma_i^2$, and $n_2/n_1 = \sigma_2^2/\sigma_1^2$.

Example 8.16 (Location families)

Let $f(x - \theta)$ be a location model with f symmetric about 0. From Chapter 5 and Theorem 6.13,

$$\sqrt{n}(\hat{\theta}^{\text{med}} - \theta) \xrightarrow{d} N\Big(0, \frac{1}{4f(0)^2}\Big), \qquad \sqrt{n}(\bar{X} - \theta) \xrightarrow{d} N(0, \sigma_f^2), \qquad \sqrt{n}(\hat{\theta}^{\text{MLE}} - \theta) \xrightarrow{d} N(0, I_f^{-1}),$$

where $\sigma_f^2 = \int x^2 f$ and $I_f = \int (f'/f)^2 f$. For three standard cases:

	$N(\theta, 1)$	$L(\theta, 1)$	$DE(\theta, 1)$
$1/(4f(0)^2)$ (median)	$\pi/2$	4	1
σ_f^2 (mean)	1	$\pi^2/3$	2
I_f^{-1} (MLE)	1	3	1
ARE(mean, median)	$\pi/2$	$12/\pi^2$	1/2
ARE(median, MLE)	$2/\pi$	3/4	1

For logistic data, the MLE beats the mean, which beats the median. The mean is about 1.2 times as efficient as the median, so the median needs about 20% more observations for the same accuracy.

Example 8.17 (Beta$(\alpha,1)$)

Here $m_1 = \alpha/(\alpha+1)$, so $\hat\alpha^{\mathrm{MME}} = \bar X/(1-\bar X)$. With $\sigma^2 = \operatorname{Var} X = \frac{\alpha}{\alpha+2} - \frac{\alpha^2}{(\alpha+1)^2}$ and $g(m) = m/(1-m)$, $\dot g(m_1) = (\alpha+1)^2$, the delta method gives

$$\sqrt{n}(\hat\alpha^{\mathrm{MME}} - \alpha) \xrightarrow{d} N\Big(0, \frac{\alpha(\alpha+1)^2}{\alpha+2}\Big).$$

The family is exponential with $T = \log x$; the MLE solves $\sum \log X_i + n/\alpha = 0$, so $\hat\alpha^{\mathrm{MLE}} = -1/\overline{\log X}$, with $I(\alpha) = 1/\alpha^2$ and $\sqrt{n}(\hat\alpha^{\mathrm{MLE}} - \alpha) \xrightarrow{d} N(0, \alpha^2)$. Hence

$$ARE(\{\hat\alpha^{\mathrm{MME}}\}, \{\hat\alpha^{\mathrm{MLE}}\}) = \frac{\alpha^2(\alpha+2)}{\alpha(\alpha+1)^2} = \frac{\alpha(\alpha+2)}{(\alpha+1)^2} < 1.$$

In these examples the MLE had the smallest limiting variance. The next inequality explains why.

Theorem 8.10 (Information inequality)

Assume the regularity conditions (R0)–(R5) of Chapter 6. For any estimator $\hat\eta_n = \hat\eta_n(X_1, \dots, X_n)$ of a real quantity with finite variance,

$$\operatorname{Var}_\theta(\hat\eta_n) \geq \Big(\frac{\partial}{\partial\theta} E_\theta \hat\eta_n\Big)' [nI(\theta)]^{-1} \Big(\frac{\partial}{\partial\theta} E_\theta \hat\eta_n\Big).$$

Proof. One-dimensional θ first. By the Cauchy–Schwarz inequality for covariances,

$$\operatorname{Var}_\theta(\hat\eta_n) \operatorname{Var}_\theta(\dot l_n(\theta)) \geq \{\operatorname{Cov}_\theta(\hat\eta_n, \dot l_n(\theta))\}^2.$$

By Theorem 6.12, $E_\theta \dot l_n(\theta) = nE_\theta \partial_\theta \log f(X_1;\theta) = 0$ and $\operatorname{Var}_\theta \dot l_n(\theta) = nI(\theta)$. By (R4),

$$\operatorname{Cov}_\theta(\hat\eta_n, \dot l_n(\theta)) = E_\theta[\hat\eta_n \dot l_n(\theta)] = \int \hat\eta_n(\mathbf{x}) \frac{\partial}{\partial\theta} \prod f(x_i;\theta)\, d\mathbf{x} = \frac{\partial}{\partial\theta} E_\theta \hat\eta_n.$$

Substituting gives the inequality. For vector θ, let $\Sigma_{11} = \operatorname{Var} \hat\eta_n$, $\Sigma_{12} = \operatorname{Cov}(\hat\eta_n, \dot l_n(\theta))' = (\partial_\theta E_\theta \hat\eta_n)'$, $\Sigma_{22} = \operatorname{Var} \dot l_n(\theta) = nI(\theta)$. Then

$$\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} = \operatorname{Var}\left(\hat\eta_n - \Sigma_{12}\Sigma_{22}^{-1}\dot l_n(\theta)\right) \geq 0.$$

■

Corollary 8.11 (Cramér–Rao bound)

If $\hat\theta^{UE}$ is unbiased for $\theta \in \mathbb{R}^k$, then $\mathbf{c}' \operatorname{Var}_\theta(\hat\theta^{UE})\mathbf{c} \geq \mathbf{c}' I(\theta)^{-1}\mathbf{c}/n$ for all $\mathbf{c}$; i.e. $\operatorname{Var}_\theta(\hat\theta^{UE}) - I(\theta)^{-1}/n$ is nonnegative definite. For $k = 1$, $\operatorname{Var}_\theta(\hat\theta^{UE}) \geq 1/\{nI(\theta)\}$.

Proof. Apply Theorem 8.10 to $\hat\eta = \mathbf{c}'\hat\theta^{UE}$, for which $\partial_\theta E_\theta \hat\eta = \mathbf{c}$. ■

Example 8.18

In Beta$(\alpha,1)$ (Example 8.17), $Y = \sum \log X_i$ is complete sufficient, and $-\alpha \log X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Exp}(1)$, so $-\alpha Y \sim \mathrm{Gamma}(n,1)$. Hence

$$E\Big[\frac{1}{-Y}\Big] = \frac{\alpha}{n-1}, \qquad \operatorname{Var}\Big[\frac{1}{-Y}\Big] = \frac{\alpha^2}{(n-1)^2(n-2)}.$$

So $\hat{\alpha}^{\mathrm{UMVUE}} = (n-1)/(-Y) = \frac{n-1}{n}\hat{\alpha}^{\mathrm{MLE}}$, with Var $= \alpha^2/(n-2)$. This exceeds the Cramér–Rao bound α^2/n, but the ratio tends to 1, and $\sqrt{n}(\hat{\alpha}^{\mathrm{UMVUE}} - \alpha) \xrightarrow{d} N(0, \alpha^2)$, the same limit as the MLE.

By the corollary, $\mathrm{Var}\{\sqrt{n}(\hat{\theta}^{UE} - \theta)\} \geq 1/I(\theta)$. This suggests that any asymptotically normal estimator has limiting variance at least $1/I(\theta)$, the limiting variance of the MLE (Theorem 6.13). This is **Fisher's conjecture**. It is not true in complete generality, but it holds under additional conditions such as locally uniform convergence. In this sense the MLE is **asymptotically efficient**.

Example 8.19 (Gamma)

For Gamma(α, β), the MME $(\hat{\alpha}, \hat{\beta})$ (Example 6.1(d)) satisfies

$$\sqrt{n}(\hat{\theta}^{\mathrm{MME}} - \theta) \xrightarrow{d} N(\mathbf{0}, \Sigma(\theta)), \qquad \Sigma(\theta) = \begin{pmatrix} 2\alpha(\alpha+1) & -2(\alpha+1)\beta \\ -2(\alpha+1)\beta & (2+3/\alpha)\beta^2 \end{pmatrix},$$

while the MLE (or its one-step version, Example 6.16) has limit covariance $I(\theta)^{-1}$. Thus

$$ARE(\hat{\alpha}^{\mathrm{MME}}, \hat{\alpha}^{\mathrm{MLE}}) = \frac{1}{2(\alpha+1)\{\alpha\dot{\Psi}(\alpha) - 1\}},$$

$$ARE(\hat{\beta}^{\mathrm{MME}}, \hat{\beta}^{\mathrm{MLE}}) = \frac{\alpha\dot{\Psi}(\alpha)}{(2\alpha+3)\{\alpha\dot{\Psi}(\alpha) - 1\}}.$$

For $\alpha = 1$ these are about 0.39 and 0.51; they increase towards 1 as $\alpha \to \infty$ but are always below 1. Generally, the MME loses efficiency relative to the MLE.

Chapter 9. Comparing Tests

9.1 Criteria for comparing tests

The simplest setting is a **simple** null and alternative, each specifying one distribution: $\Omega = \{\theta_0, \theta_1\}$ and

$$H_0 : \theta = \theta_0 \quad \text{vs} \quad H_1 : \theta = \theta_1.$$

A test ϕ has two error probabilities, $E_{\theta_0}\phi(\mathbf{X})$ (type I) and $E_{\theta_1}[1 - \phi(\mathbf{X})]$ (type II). Making one small makes the other large, so, as in estimation, we combine them:

- **maximum error**: $\max\{E_{\theta_0}\phi, E_{\theta_1}(1-\phi)\}$;
- **Bayes error**: $\pi_0 E_{\theta_0}\phi + \pi_1 E_{\theta_1}(1-\phi)$, with prior weights $\pi_0, \pi_1 > 0$, $\pi_0 + \pi_1 = 1$.

Write $p_\theta(\mathbf{x}) = \prod f(x_i;\theta)$ for the joint pdf.

Theorem 9.1 (Bayes test for simple hypotheses)

The test

$$\phi^\pi(\mathbf{x}) = \begin{cases} 1, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) > \pi_0/\pi_1, \\ 0, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) < \pi_0/\pi_1 \end{cases}$$

(arbitrary on equality; a positive number divided by 0 counts as $+\infty$) minimizes the Bayes error.

Proof. The Bayes error equals

$$\pi_0 \int \phi p_{\theta_0} + \pi_1\Big(1 - \int \phi p_{\theta_1}\Big) = \pi_1 - \int \phi(\mathbf{x})\{\pi_1 p_{\theta_1}(\mathbf{x}) - \pi_0 p_{\theta_0}(\mathbf{x})\}\, d\mathbf{x}.$$

Since $0 \le \phi \le 1$, this is minimized by taking $\phi = 1$ where the brace is positive and $\phi = 0$ where it is negative. ∎

Example 9.1

For $N(\mu, 1)$ and $H_0 : \mu = \mu_0$ vs $H_1 : \mu = \mu_1$,

$$\frac{p_{\mu_1}(\mathbf{x})}{p_{\mu_0}(\mathbf{x})} = \exp\Big\{n(\mu_1 - \mu_0)\Big(\bar{x} - \frac{\mu_0 + \mu_1}{2}\Big)\Big\}.$$

If $\mu_1 > \mu_0$, the Bayes test rejects when $\bar{x} > \frac{\mu_0+\mu_1}{2} + \frac{\log(\pi_0/\pi_1)}{n(\mu_1-\mu_0)}$; if $\mu_1 < \mu_0$, when $\bar{x} < \frac{\mu_0+\mu_1}{2} - \frac{\log(\pi_0/\pi_1)}{n(\mu_0-\mu_1)}$. With equal priors, the cut-off is the midpoint.

These criteria treat H_0 and H_1 symmetrically. The classical theory of Chapter 7 does not: it fixes the type I error at most α and then makes the power $\gamma_\phi(\theta_1) = E_{\theta_1}\phi$ as large as possible.

Definition (Most powerful test)

For simple H_0 and H_1, a test ϕ^{MP} is a **most powerful (MP) test** of level α if (i) $E_{\theta_0}\phi^{MP} \le \alpha$, and (ii) $E_{\theta_1}\phi^{MP} \ge E_{\theta_1}\phi$ for every ϕ with $E_{\theta_0}\phi \le \alpha$.

Theorem 9.2 (Neyman–Pearson lemma)

A test of the form

$$\phi^*(\mathbf{x}) = \begin{cases} 1, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) > k, \\ \gamma, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) = k, \\ 0, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) < k, \end{cases} \qquad k \ge 0,\ 0 \le \gamma \le 1,$$

with size $E_{\theta_0}\phi^* = \alpha$ is a most powerful level-α test.

Proof. For any test ϕ,

$$E_{\theta_1}\phi - kE_{\theta_0}\phi = \int \phi\,(p_{\theta_1} - kp_{\theta_0})\,d\mathbf{x}.$$

Hence

$$\{E_{\theta_1}\phi^* - kE_{\theta_0}\phi^*\} - \{E_{\theta_1}\phi - kE_{\theta_0}\phi\} = \int (\phi^* - \phi)(p_{\theta_1} - kp_{\theta_0})\,d\mathbf{x} \ge 0,$$

because where $p_{\theta_1} > kp_{\theta_0}$ we have $\phi^* = 1 \ge \phi$, and where $p_{\theta_1} < kp_{\theta_0}$ we have $\phi^* = 0 \le \phi$. So if $E_{\theta_0}\phi \le \alpha$,

$$E_{\theta_1}\phi^* - E_{\theta_1}\phi \ge k\{E_{\theta_0}\phi^* - E_{\theta_0}\phi\} = k\{\alpha - E_{\theta_0}\phi\} \ge 0.$$

■

Such k and γ always exist, and conversely every MP test has this form (except on a set of probability zero). γ may depend on $\mathbf{x}$.

Example 9.2

Poisson(θ), $n = 50$, $H_0 : \theta = 0.1$ vs $H_1 : \theta = 0.05$. The likelihood ratio is $e^{-n(\theta_1-\theta_0)}(\theta_1/\theta_0)^{\sum x_i} = e^{2.5}(1/2)^{\sum x_i}$, decreasing in $S = \sum x_i$. So the MP test rejects for small S: reject when $S \le c-1$, randomize with probability γ when $S = c$, with $E_{0.1}\phi^* = 0.05$. As in Example 7.3, $c = 2$ and $\gamma = 0.114$.

9.2 Uniformly most powerful tests

Motivation. Neyman–Pearson gives the best test for one simple alternative. For one-sided composite alternatives the same test is often best for every alternative at once, which happens for families with monotone likelihood ratio.

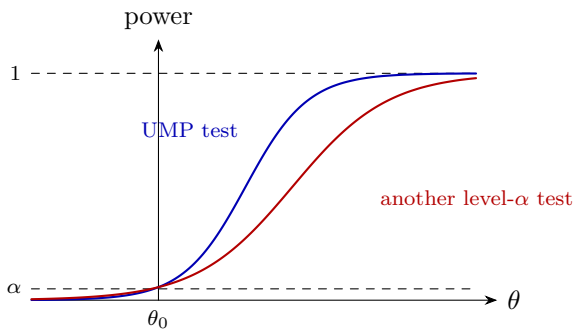


Figure. *Power functions for $H_0 : \theta \le \theta_0$ versus $H_1 : \theta > \theta_0$. Both tests have size α at θ_0; the uniformly most powerful test has the larger power at every alternative $\theta > \theta_0$.*

For composite hypotheses $H_0 : \theta \in \Omega_0$ vs $H_1 : \theta \in \Omega_1$, we want large power at every $\theta_1 \in \Omega_1$.

Definition (UMP test)

ϕ^{UMP} is a **uniformly most powerful (UMP) test** of level α if (i) $\max_{\theta\in\Omega_0} E_\theta\phi^{UMP} \le \alpha$, and (ii) $E_{\theta_1}\phi^{UMP} \ge E_{\theta_1}\phi$ for all $\theta_1 \in \Omega_1$ and every ϕ with $\max_{\Omega_0} E_\theta\phi \le \alpha$.

Example 9.3

For $N(\mu,1)$, $H_0:\mu=\mu_0$ vs $H_1:\mu>\mu_0$. Fix $\mu_1>\mu_0$. By Example 9.1 the likelihood ratio increases in $\bar x$, so the MP test of μ_0 vs μ_1 rejects when $\bar x \ge c$, with $c=\mu_0+z_\alpha/\sqrt{n}$. This test does not depend on μ_1, so it is MP against every $\mu_1>\mu_0$, i.e. UMP:

$$\phi^*(\mathbf{x}) = \mathbf{1}\{\bar x - \mu_0 \ge z_\alpha/\sqrt{n}\}.$$

Example 9.4

The same test is UMP for $H_0^-:\mu\le\mu_0$ vs $H_1:\mu>\mu_0$. First, its power $E_\mu\phi^* = P(Z \ge \sqrt{n}(\mu_0-\mu)+z_\alpha)$ increases in μ, so $\max_{\mu\le\mu_0} E_\mu\phi^* = \alpha$: it has level α for H_0^-. Second, any level-α test for H_0^- is also level α for $\mu=\mu_0$, and among those ϕ^* has the largest power at each $\mu_1>\mu_0$.

Theorem 9.3 (UMP one-sided tests in exponential families)

Let $f(x;\theta) = \exp\{g(\theta)T(x) - B(\theta) + S(x)\}$, $x\in\mathcal{X}$, $\theta\in\Omega\subseteq\mathbb{R}$, with g increasing. For $H_0:\theta\le\theta_0$ vs $H_1:\theta>\theta_0$, the test

$$\phi^*(\mathbf{x}) = \begin{cases} 1, & \sum T(x_i) > c, \\ \gamma, & \sum T(x_i) = c, \\ 0, & \sum T(x_i) < c, \end{cases} \qquad E_{\theta_0}\phi^* = \alpha,$$

is UMP of level α. (For $H_0:\theta\ge\theta_0$ vs $H_1:\theta<\theta_0$, reverse the inequalities.)

Proof.

Step 1. For $\theta_1>\theta_0$,

$$\frac{p_{\theta_1}(\mathbf{x})}{p_{\theta_0}(\mathbf{x})} = \exp\Big\{(g(\theta_1)-g(\theta_0))\sum T(x_i) - n(B(\theta_1)-B(\theta_0))\Big\}$$

is increasing in $\sum T(x_i)$, so ϕ^* has the Neyman–Pearson form for θ_0 vs θ_1 and is MP. It does not depend on θ_1, so it is UMP for $H_0:\theta=\theta_0$ vs $H_1:\theta>\theta_0$.

Step 2. The power $E_\theta\phi^*$ is nondecreasing in θ. Take $\theta'<\theta''$ and $\alpha'=E_{\theta'}\phi^*$. By the argument of Step 1, ϕ^* is MP of level α' for θ' vs θ''. The trivial test $\phi\equiv\alpha'$ also has level α', so $E_{\theta''}\phi^*\ge\alpha' = E_{\theta'}\phi^*$.

Step 3. By Step 2, $\max_{\theta\le\theta_0} E_\theta\phi^* = E_{\theta_0}\phi^* = \alpha$. Any level-$\alpha$ test of $H_0:\theta\le\theta_0$ has level α for $\theta=\theta_0$, so by Step 1 its power at each $\theta_1>\theta_0$ is at most that of ϕ^*. ∎

Example 9.5

(a) Poisson: $f=\exp\{x\log\theta-\theta-\log x!\}$ with $g=\log$ increasing. For $H_0:\theta\ge0.1$ vs $H_1:\theta<0.1$ with $n=50$, the UMP level-0.05 test is the randomized test of Example 9.2.
(b) Exponential: $f=\exp\{-x/\theta-\log\theta\}$ with $g(\theta)=-1/\theta$ increasing. For $H_0:\theta\le\theta_0$ vs $H_1:\theta>\theta_0$, the UMP test rejects when $\sum x_i \ge c$. Since $2\sum X_i/\theta_0 \sim \chi^2(2n)$ at θ_0, the

rejection region is $\bar{x}/\theta_0 \geq \chi^2_\alpha(2n)/(2n)$.

Example 9.6 (No UMP two-sided test)

For $N(\mu, 1)$ and $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$, suppose a UMP test ϕ^* existed. It would be MP for μ_0 vs $\mu_1 < \mu_0$ and for μ_0 vs $\mu_2 > \mu_0$. By the converse of the Neyman–Pearson lemma, it would have to be $\mathbf{1}\{\bar{x} \leq c_1\}$ and also $\mathbf{1}\{\bar{x} \geq c_2\}$, both with size α, which is impossible. So no UMP test exists.

Example 9.7 (Optimality of the two-sided test)

In the setting of Example 9.6, the LRT is $\phi^* = \mathbf{1}\{\sqrt{n}|\bar{x} - \mu_0| \geq z_{\alpha/2}\}$, with power $1 - \Phi(z_{\alpha/2} - \delta) + \Phi(-z_{\alpha/2} - \delta)$, $\delta = \sqrt{n}(\mu - \mu_0)$. Hence (a) $E_{\mu_0}\phi^* = \alpha$ and $\frac{d}{d\mu}E_\mu\phi^*\big|_{\mu_0} = 0$. We claim that (b) among all tests with $E_{\mu_0}\phi = \alpha$ and $\frac{d}{d\mu}E_\mu\phi|_{\mu_0} = 0$, ϕ^* has the largest power at every $\mu_1 \neq \mu_0$.

Write p_μ for the joint pdf and $\dot{p}_{\mu_0} = \partial_\mu p_\mu|_{\mu_0}$, so $\frac{d}{d\mu}E_\mu\phi|_{\mu_0} = E_{\mu_0}[\phi\,\dot{p}_{\mu_0}/p_{\mu_0}]$. For constants k_1, k_2,

$$E_{\mu_1}\phi - k_1 E_{\mu_0}\phi - k_2 E_{\mu_0}\Big[\phi\frac{\dot{p}_{\mu_0}}{p_{\mu_0}}\Big] = \int \phi\,\{p_{\mu_1} - k_1 p_{\mu_0} - k_2\dot{p}_{\mu_0}\}\,d\mathbf{x}$$

is maximized by $\phi^{**} = \mathbf{1}\{p_{\mu_1} \geq k_1 p_{\mu_0} + k_2\dot{p}_{\mu_0}\}$. Since $\dot{p}_{\mu_0}/p_{\mu_0} = n(\bar{x} - \mu_0)$, this region is $\{e^{a(\bar{x}-\mu_0)} \geq b_1 + b_2(\bar{x} - \mu_0)\}$, the set where a convex exponential lies above a line, i.e. outside an interval. Choosing k_1, k_2 so that ϕ^{**} satisfies the two side conditions makes the interval symmetric, $\phi^{**} = \phi^*$. As in the Neyman–Pearson proof, $E_{\mu_1}\phi^* \geq E_{\mu_1}\phi$ for every ϕ satisfying the side conditions. Since ϕ^* does not depend on μ_1, it is uniformly most powerful among them.

A test with $\max_{\Omega_0} E_\theta\phi \leq \alpha \leq \min_{\Omega_1} E_\theta\phi$ is **unbiased**. For smooth power functions, unbiasedness implies the side conditions above, so the two-sided test is the **UMP unbiased** test.

9.3 Nonparametric tests and asymptotic comparison

We now drop the assumption of a specific form for f. Throughout, $X_1, \ldots, X_n$ is a sample from a continuous location model $f(x - \theta)$, where f is symmetric about 0 ($f(-x) = f(x)$) with strictly increasing cdf F. So θ is the median (and centre of symmetry).

Example 9.8 (Sign test)

To test $H_0 : \theta = \theta_0$ vs $H_1 : \theta > \theta_0$, use $S_n = \sum_i \mathbf{1}\{X_i > \theta_0\}$. Then $S_n \sim B(n, p(\theta))$ with $p(\theta) = P_\theta(X_1 > \theta_0) = 1 - F(\theta_0 - \theta)$, and H_0 corresponds to $p = \frac{1}{2}$. The test

$$\phi_S = \begin{cases} 1, & S_n \geq c+1, \\ \gamma, & S_n = c, \\ 0, & S_n \leq c-1, \end{cases} \qquad \sum_{k=c+1}^{n}\binom{n}{k}2^{-n} + \gamma\binom{n}{c}2^{-n} = \alpha,$$

has size α for every f. Writing $X_i = Z_i + \theta$ with $Z_i \overset{\text{i.i.d.}}{\sim} f$, $E_\theta\phi_S = E\phi_S(Z_1 + \theta, \ldots, Z_n + \theta)$ increases in θ since ϕ_S is nondecreasing in each x_i. So $\max_{\theta \leq \theta_0} E_\theta\phi_S = \alpha$, and ϕ_S is also a level-α test of $H_0 : \theta \leq \theta_0$. This is the **sign test**.

A test whose size does not depend on the form of the population is **nonparametric** (distribution-

free). Its robustness may cost efficiency, and we measure the cost by the sample size needed for a given power.

Example 9.9 (Power of the sign test)

By the normal approximation, the level-α sign test rejects when $S_n \ge np(\theta_0) + \sqrt{n}\sigma(\theta_0)z_\alpha$, with $p(\theta_0) = \frac{1}{2}$, $\sigma^2(\theta) = p(\theta)(1-p(\theta))$. Its power at $\theta_1 > \theta_0$ is

$$\gamma_n(\theta_1) \approx 1 - \Phi\Big(- \sqrt{n}\frac{p(\theta_1) - p(\theta_0)}{\sigma(\theta_1)} + \frac{\sigma(\theta_0)}{\sigma(\theta_1)} z_\alpha \Big) \to 1.$$

For alternatives approaching H_0, $\theta_{1n} = \theta_0 + K/\sqrt{n}$, we have $p(\theta_{1n}) - p(\theta_0) \approx (\theta_{1n} - \theta_0)\dot{p}(\theta_0)$ with $\dot{p}(\theta_0) = f(0)$, and

$$\gamma_n(\theta_{1n}) \approx 1 - \Phi\big(- 2\sqrt{n}(\theta_{1n} - \theta_0) f(0) + z_\alpha \big).$$

Setting this equal to γ gives the required sample size $n \approx (2f(0))^{-2}\big(\frac{z_\alpha + z_{1-\gamma}}{\theta_{1n} - \theta_0}\big)^2$.

Theorem 9.4 (Approximate power and sample size)

For $H_0 : \theta = \theta_0$ vs $H_1 : \theta > \theta_0$, let a test reject when $\sqrt{n}(T_n - \mu(\theta_0))/\sigma(\theta_0) \ge t_n$, with size α, where

$$\sqrt{n}\,(T_n - \mu(\theta))/\sigma(\theta) \xrightarrow{d} N(0,1)$$

(uniformly for θ near θ_0) and μ, σ are continuously differentiable. Then:

(a) for $\theta_{1n} = \theta_0 + K/\sqrt{n}$, $\gamma_n(\theta_{1n}) \approx 1 - \Phi\big(- \sqrt{n}(\theta_{1n} - \theta_0)\dot{\mu}(\theta_0)/\sigma(\theta_0) + z_\alpha\big)$;

(b) the sample size needed for power γ at θ_{1n} is

$$N(T; \gamma, \theta_{1n}) \approx \Big(\frac{\dot{\mu}(\theta_0)}{\sigma(\theta_0)}\Big)^{-2}\Big(\frac{z_\alpha + z_{1-\gamma}}{\theta_{1n} - \theta_0}\Big)^2.$$

Proof. Since the size is α and the statistic is asymptotically $N(0,1)$ at θ_0, $t_n \approx z_\alpha$. Under θ_1,

$$P_{\theta_1}\Big(\sqrt{n}\frac{T_n - \mu(\theta_1)}{\sigma(\theta_1)} \ge -\sqrt{n}\frac{\mu(\theta_1) - \mu(\theta_0)}{\sigma(\theta_1)} + \frac{\sigma(\theta_0)}{\sigma(\theta_1)} t_n\Big) \approx 1 - \Phi\Big(- \sqrt{n}\frac{\mu(\theta_1) - \mu(\theta_0)}{\sigma(\theta_1)} + \frac{\sigma(\theta_0)}{\sigma(\theta_1)} z_\alpha\Big).$$

For $\theta_1 = \theta_{1n}$, $\mu(\theta_{1n}) - \mu(\theta_0) \approx (\theta_{1n} - \theta_0)\dot{\mu}(\theta_0)$ and $\sigma(\theta_{1n}) \to \sigma(\theta_0)$, which gives (a). For (b), solve $\sqrt{N}(\theta_{1n} - \theta_0)\dot{\mu}(\theta_0)/\sigma(\theta_0) - z_\alpha = z_{1-\gamma}$. ∎

Definition (Asymptotic relative efficiency of tests)

If two tests based on T_{1n} and T_{2n} satisfy the conditions of Theorem 9.4, the ARE of the first relative to the second is

$$ARE(\{T_{1n}\}, \{T_{2n}\}) = \lim \frac{N(T_2; \gamma, \theta_{1n})}{N(T_1; \gamma, \theta_{1n})} = \frac{(\dot{\mu}_1(\theta_0)/\sigma_1(\theta_0))^2}{(\dot{\mu}_2(\theta_0)/\sigma_2(\theta_0))^2}.$$

Example 9.10 (Sign test vs t test)

For the location–scale model $\frac{1}{\sigma} f(\frac{x-\mu}{\sigma})$, the sign test needs $N(S; \gamma, \mu_{1n}) \approx (2f(0)/\sigma)^{-2}\big(\frac{z_\alpha + z_{1-\gamma}}{\mu_{1n} - \mu_0}\big)^2$. The t statistic $T_n = \sqrt{n}(\bar{X} - \mu_0)/S$ satisfies the conditions

with $\dot\mu/\sigma = 1/\sqrt{\operatorname{Var} X_1}$, $\operatorname{Var} X_1 = \sigma^2 \int z^2 f(z)dz$. Therefore

$$ARE(\{S_n\},\{T_n\}) = 4f(0)^2 \int z^2 f(z)\,dz,$$

the same as the ARE of the median relative to the mean (Example 8.16):

	$N(\mu,\sigma^2)$	$L(\mu,\sigma)$	$DE(\mu,\sigma)$
$ARE(S,T)$	$2/\pi = 0.637$	$\pi^2/12 = 0.822$	2

The sign test uses only the signs of $X_i - \theta_0$. Using their ranks as well gives a much more efficient nonparametric test.

Definition (Signed rank statistic)

Let $R(|X_i-\theta_0|) = 1 + \sum_j \mathbf{1}\{|X_j-\theta_0| < |X_i-\theta_0|\}$ be the rank of $|X_i-\theta_0|$ among $|X_1-\theta_0|,\dots,|X_n-\theta_0|$. The **signed rank statistic** is

$$W_n = \sum_{i=1}^n \operatorname{sgn}(X_i-\theta_0)\, R(|X_i-\theta_0|),$$

and $W_n^+ = \sum_i \mathbf{1}\{X_i > \theta_0\} R(|X_i-\theta_0|)$ is the sum of the ranks of the positive deviations.

Theorem 9.5 (Null distribution)

Under $H_0 : \theta = \theta_0$:

(a) $W_n \overset{d}{=} \sum_{j=1}^n j\, S(j)$, with $S(1),\dots,S(n)$ i.i.d., $P(S(j)=1) = P(S(j)=-1) = \frac12$;

(b) $EW_n = 0$, $\operatorname{Var} W_n = \sum j^2 = n(n+1)(2n+1)/6$, and $W_n/\sqrt{\operatorname{Var} W_n} \xrightarrow{d} N(0,1)$.

Proof. (a) Let $S = (S(1),\dots,S(n))$ with $S(i) = \operatorname{sgn}(X_i-\theta_0)$, and $R = (R(1),\dots,R(n))$ with $R(i) = R(|X_i-\theta_0|)$. By symmetry about θ_0,

$$P(|X_1-\theta_0| \le x,\ \operatorname{sgn}(X_1-\theta_0) = \pm 1) = \tfrac12 P(|X_1-\theta_0| \le x),$$

so $|X_i-\theta_0|$ and $\operatorname{sgn}(X_i-\theta_0)$ are independent; hence the sign vector S and the rank vector R are independent, and $S(1),\dots,S(n)$ are i.i.d. fair signs. With the inverse ranks $R^{-1}(j)$ (the index whose rank is j),

$$W_n = \sum_i S(i)R(i) = \sum_{j=1}^n S(R^{-1}(j))\, j.$$

For each permutation π, by independence of S and R and exchangeability of S,

$$P(S(R^{-1}(j)) = s_j\ \forall j,\ R=\pi) = P(S(\pi^{-1}(j)) = s_j\ \forall j)P(R=\pi) = P(S(j)=s_j\ \forall j)P(R=\pi).$$

Summing over π, $(S(R^{-1}(1)),\dots,S(R^{-1}(n))) \overset{d}{=} (S(1),\dots,S(n))$, which gives (a).

(b) The moments follow from (a). With $\sigma_n^2 = \sum j^2$, the cgf of $Z_n = W_n/\sigma_n$ is

$$\sum_{j=1}^n \log \frac{e^{jt/\sigma_n} + e^{-jt/\sigma_n}}{2} = \sum_j \log\Big(1 + \frac{j^2t^2}{2\sigma_n^2} + \dots\Big) = \frac{t^2}{2} + O\Big(\sum_j \frac{j^4}{\sigma_n^4}\Big),$$

and $\sum j^4/\sigma_n^4 = O(n^5/n^6) \to 0$. So the cgf tends to $t^2/2$. ∎

Theorem 9.6

(a) $W_n = 2W_n^+ - n(n+1)/2$.
(b) Under H_0, $W_n^+ \overset{d}{=} \sum_{j=1}^n jB_j$ with $B_j \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(\frac{1}{2})$, so $EW_n^+ = n(n+1)/4$, $\text{Var}\, W_n^+ = n(n+1)(2n+1)/24$, and $(W_n^+ - n(n+1)/4)/\sqrt{n(n+1)(2n+1)/24} \xrightarrow{d} N(0,1)$.

Proof. (a) $\text{sgn}(X_i - \theta_0) = 2\mathbf{1}\{X_i > \theta_0\} - 1$ (no ties), and $\sum R(i) = n(n+1)/2$.

(b) From (a) and Theorem 9.5(a), $W_n^+ = (W_n + n(n+1)/2)/2 \overset{d}{=} \sum j(S(j)+1)/2$, with $(S(j)+1)/2 \sim \text{Bernoulli}(\frac{1}{2})$. ■

The null distribution of W_n does not depend on f, and large values of W_n (or W_n^+) are evidence for $\theta > \theta_0$. The **signed rank test** of level α is

$$\phi_{SR} = \begin{cases} 1, & W_n^+ \geq c+1, \\ \gamma, & W_n^+ = c, \\ 0, & W_n^+ \leq c-1, \end{cases} \qquad E_{\theta_0}\phi_{SR} = \alpha,$$

and for large n it rejects when $(W_n^+ - n(n+1)/4)/\sqrt{n(n+1)(2n+1)/24} \geq z_\alpha$.

Example 9.11

For $n = 10$ and the differences $X_i - \theta_0$: $1.8, -0.4, 2.5, 0.9, -1.1, 3.2, 0.6, 1.5, -0.2, 2.1$. The ranks of the absolute values are $7, 2, 9, 4, 5, 10, 3, 6, 1, 8$, so $W^+ = 7+9+4+10+3+6+8 = 47$. The null mean and standard deviation are 27.5 and $\sqrt{96.25} = 9.81$, so $z = (47 - 27.5)/9.81 = 1.99 > 1.645$, and H_0 is rejected at the 5% level. The sign test, with $S = 7$ and $P(S \geq 7) = 0.172$ under H_0, does not reject.

Theorem 9.7 (Walsh averages)

(a) $W_n^+ = \sum_{1 \leq i \leq j \leq n} \mathbf{1}\{X_i + X_j > 2\theta_0\}$.
(b) The power function of ϕ_{SR} increases in θ; hence $\max_{\theta \leq \theta_0} E_\theta \phi_{SR} = \alpha$, and ϕ_{SR} is a level-α test of $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$.

Proof. (a) Let $Y_i = X_i - \theta_0$, and sort by absolute value, $|Y_{(1)}| < \cdots < |Y_{(n)}|$ (here $Y_{(i)}$ has the i-th smallest absolute value). Then

$$W_n^+ = \sum_i \mathbf{1}\{Y_{(i)} > 0\} \cdot i = \sum_i \mathbf{1}\{Y_{(i)} > 0\} + \sum_i \sum_{j<i} \mathbf{1}\{Y_{(i)} > 0\}.$$

For $j < i$, $|Y_{(j)}| < |Y_{(i)}|$, so $Y_{(i)} > 0$ iff $Y_{(i)} + Y_{(j)} > 0$. Hence $W_n^+ = \sum_i \mathbf{1}\{Y_i > 0\} + \sum_{j<i} \mathbf{1}\{Y_i + Y_j > 0\} = \sum_{i \leq j} \mathbf{1}\{Y_i + Y_j > 0\}$.

(b) By (a), W_n^+ is nondecreasing in each X_i, hence so is ϕ_{SR}. With $X_i = Z_i + \theta$, $E_\theta \phi_{SR} = E\phi_{SR}(Z_1 + \theta, \ldots, Z_n + \theta)$ is nondecreasing in θ. ■

For the efficiency of the signed rank test we need its behaviour under alternatives. The following is stated without proof.

Theorem 9.8

Let $\mu(\theta) = \frac{1}{2}P_\theta(X_1 + X_2 > 2\theta_0)$ and $\sigma^2(\theta) = \text{Cov}_\theta\left(\mathbf{1}\{X_1 + X_2 > 2\theta_0\}, \mathbf{1}\{X_1 + X_3 > 2\theta_0\}\right)$.
Then $E_\theta W_n^+ \approx n^2\mu(\theta)$, $\text{Var}_\theta\, W_n^+ \approx n^3\sigma^2(\theta)$, and $(W_n^+ - E_\theta W_n^+)/\sqrt{\text{Var}_\theta\, W_n^+} \xrightarrow{d} N(0,1)$.

Theorem 9.9 (Power of the signed rank test)

For $\theta_{1n} = \theta_0 + K/\sqrt{n}$,

$$\gamma_{\phi_{SR}}(\theta_{1n}) \approx 1 - \Phi\Big(- \sqrt{n}(\theta_{1n} - \theta_0)\frac{\dot\mu(\theta_0)}{\sigma(\theta_0)} + z_\alpha\Big), \qquad \dot\mu(\theta_0) = \int f^2(x)\,dx, \quad \sigma^2(\theta_0) = \frac{1}{12},$$

and the required sample size is $N(W^+;\gamma,\theta_{1n}) \approx \big(\sqrt{12}\int f^2\big)^{-2}\big(\frac{z_\alpha + z_{1-\gamma}}{\theta_{1n}-\theta_0}\big)^2$.

Proof. Apply Theorem 9.4 to $T_n = W_n^+/n^2$. With Z having cdf F and symmetric pdf f,

$$\mu(\theta) = \frac{1}{2}P(Z_1 + Z_2 > 2(\theta_0 - \theta)) = \frac{1}{2}\int\{1 - F(2\theta_0 - 2\theta - x)\}f(x)\,dx,$$

so $\dot\mu(\theta_0) = \int f(-x)f(x)\,dx = \int f^2$. At θ_0, $\mathbf{1}\{Z_1 + Z_2 > 0\}$ and $\mathbf{1}\{Z_1 + Z_3 > 0\}$ are conditionally independent given Z_1 with conditional mean $1 - F(-Z_1) = F(Z_1) =: U \sim U(0,1)$. So $\sigma^2(\theta_0) = EU^2 - (EU)^2 = \frac{1}{12}$. ■

Example 9.12 (Signed rank test vs t test)

For $\frac{1}{\sigma}f(\frac{x-\mu}{\sigma})$, $\int(\frac{1}{\sigma}f(\frac{x}{\sigma}))^2dx = \frac{1}{\sigma}\int f^2$. Combining Theorem 9.9 with Example 9.10,

$$ARE(\{W_n^+\},\{T_n\}) = 12\Big(\int f^2(z)\,dz\Big)^2\int z^2 f(z)\,dz.$$

	$N(\mu,\sigma^2)$	$L(\mu,\sigma)$	$DE(\mu,\sigma)$
$ARE(W^+,T)$	$3/\pi = 0.955$	$\pi^2/9 = 1.097$	1.5

Even for normal data the signed rank test loses less than 5% efficiency against the t test, and for heavier tails it wins. Together with its distribution-free size, this makes it a very useful test.

Chapter 10. Analysis of Variance and Regression

10.1 One-way analysis of variance

The **one-way normal model** (Chapter 4) is

$$X_{ij} = \mu_i + e_{ij}, \qquad e_{ij} \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2), \qquad i = 1, \dots, k,\ j = 1, \dots, n_i,$$

for the effect of a single **factor** with k **levels**; applying level i is called **treatment** i. With $n = \sum n_i$, $\bar\mu = \sum_i n_i \mu_i / n$ (overall effect) and $\alpha_i = \mu_i - \bar\mu$ (effect of treatment i), it can be written

$$X_{ij} = \bar\mu + \alpha_i + e_{ij}, \qquad \sum_i n_i \alpha_i = 0.$$

Theorem 10.1 (UMVUEs)

In the one-way normal model, the UMVUEs are

$$\hat\mu_i = \bar X_i = \frac{1}{n_i}\sum_j X_{ij}, \qquad \hat{\bar\mu} = \sum_i \frac{n_i}{n}\hat\mu_i, \qquad \hat\alpha_i = \hat\mu_i - \hat{\bar\mu}, \qquad \hat\sigma^2 = \frac{1}{n-k}\sum_{i,j}(X_{ij} - \bar X_i)^2.$$

Proof. The joint pdf is $\exp\{-\frac{1}{2\sigma^2}\sum_{i,j}(x_{ij}-\mu_i)^2 - \frac{n}{2}\log(2\pi\sigma^2)\}$, a $(k+1)$-parameter exponential family satisfying the conditions of Theorem 8.8. So $(\bar X_1, \dots, \bar X_k, \sum_{i,j}(X_{ij} - \bar X_i)^2)$ is complete sufficient. The listed estimators are unbiased functions of it (Theorem 4.9 in Volume I for $\hat\sigma^2$), hence UMVUEs by Lehmann–Scheffé. ■

The MLE of σ^2 is $\sum(X_{ij} - \bar X_i)^2/n$; the other MLEs coincide with the UMVUEs.

Theorem 10.2 (Sampling distributions)

(a) $\hat\mu_i = \bar X_i$ are independent, $\bar X_i \sim N(\mu_i, \sigma^2/n_i)$.
(b) $\hat\sigma^2$ is independent of $(\bar X_1, \dots, \bar X_k)$.
(c) $(n-k)\hat\sigma^2/\sigma^2 \sim \chi^2(n-k)$.

Proof. Apply Theorem 4.5 in Volume I to each group: $\bar X_i \sim N(\mu_i, \sigma^2/n_i)$ is independent of $SS_i = \sum_j (X_{ij} - \bar X_i)^2$, and $SS_i/\sigma^2 \sim \chi^2(n_i - 1)$. The groups are independent and $(n-k)\hat\sigma^2 = \sum_i SS_i$; add the chi-squares. ■

Simultaneous confidence intervals

Write $\hat\mu = (\hat\mu_1, \dots, \hat\mu_k)'$, $D = \operatorname{Var}(\hat\mu)/\sigma^2 = \operatorname{diag}(1/n_i)$, and for a $k \times r$ matrix C of rank r let $\operatorname{col}(C) = \{C\mathbf{a} : \mathbf{a} \in \mathbb{R}^r\}$.

Theorem 10.3 (Confidence sets and Scheffé intervals)

(a) $P\{(C'\mu - C'\hat\mu)'(C'DC)^{-1}(C'\mu - C'\hat\mu) \le r\hat\sigma^2 F_\alpha(r, n-k)\} = 1 - \alpha$.

(b) $P\Big\{|\mathbf{c}'\mu - \mathbf{c}'\hat\mu| \le \sqrt{\textstyle\sum_i c_i^2 \hat\sigma^2/n_i}\sqrt{rF_\alpha(r,n-k)} \ \text{ for all } \mathbf{c} \in \operatorname{col}(C)\Big\} = 1 - \alpha$.

Proof. (a) $C'\hat\mu \sim N(C'\mu, \sigma^2 C'DC)$, so $(C'\hat\mu - C'\mu)'(C'DC)^{-1}(C'\hat\mu - C'\mu)/\sigma^2 \sim \chi^2(r)$ (Theorem 4.18 in Volume I). It is independent of $(n-k)\hat\sigma^2/\sigma^2 \sim \chi^2(n-k)$, and the ratio of the two, each divided by its degrees of freedom, is $F(r, n-k)$.

(b) For $\mathbf{c} = C\mathbf{a}$, with $\mathbf{d} = C'(\mu - \hat{\mu})$ and $M = C'DC$, the Cauchy–Schwarz inequality gives

$$\max_{\mathbf{c}\in\mathrm{col}(C),\mathbf{c}\neq\mathbf{0}} \frac{\{\mathbf{c}'(\mu-\hat{\mu})\}^2}{\mathbf{c}'D\mathbf{c}} = \max_{\mathbf{a}\neq\mathbf{0}} \frac{(\mathbf{a}'\mathbf{d})^2}{\mathbf{a}'M\mathbf{a}} = \max_{\mathbf{b}\neq\mathbf{0}} \frac{(\mathbf{b}'M^{-1/2}\mathbf{d})^2}{\mathbf{b}'\mathbf{b}} = \mathbf{d}'M^{-1}\mathbf{d}.$$

So the event in (b) is the same as the event in (a). ■

A linear combination $\sum c_i\mu_i$ with $\sum c_i = 0$ is a **contrast**; the simplest are the pairwise differences $\mu_i - \mu_l = \alpha_i - \alpha_l$. The contrasts form a space of dimension $k-1$, so:

Theorem 10.4 (Simultaneous intervals for contrasts)

(a) (Scheffé) With probability $1-\alpha$, for all $\mathbf{c}$ with $\sum c_i = 0$,

$$|\mathbf{c}'\alpha - \mathbf{c}'\hat{\alpha}| \le \sqrt{\textstyle\sum_i c_i^2\hat{\sigma}^2/n_i}\,\sqrt{(k-1)F_\alpha(k-1,n-k)}.$$

(b) In particular, with probability at least $1-\alpha$, for all $i \neq l$,

$$|(\alpha_i-\alpha_l)-(\hat{\alpha}_i-\hat{\alpha}_l)| \le \sqrt{(n_i^{-1}+n_l^{-1})\hat{\sigma}^2}\sqrt{(k-1)F_\alpha(k-1,n-k)}.$$

(c) (Bonferroni) With $m = k(k-1)/2$ and $\alpha^* = \alpha/m$, with probability at least $1-\alpha$, for all $i \neq l$,

$$|(\alpha_i-\alpha_l)-(\hat{\alpha}_i-\hat{\alpha}_l)| \le \sqrt{(n_i^{-1}+n_l^{-1})\hat{\sigma}^2}\; t_{\alpha^*/2}(n-k).$$

Proof. (a) is Theorem 10.3(b) with $\mathrm{col}(C)$ the $(k-1)$-dimensional space of contrasts;

(b) is a sub-event.

(c) For a single pair, $r=1$ in Theorem 10.3, and $F_{\alpha^*}(1,n-k) = t_{\alpha^*/2}(n-k)^2$, so each interval has coverage $1-\alpha^*$. By the Bonferroni inequality $P(A_1\cap\cdots\cap A_m) \ge 1-\sum_i(1-P(A_i)) = 1-m\alpha^* = 1-\alpha$. ■

When there are few comparisons, Bonferroni intervals are shorter; Scheffé intervals cover all contrasts at once. For example, with $n=20$ and $\alpha=0.05$: for $k=3$ the multipliers are $\sqrt{(k-1)F_\alpha} = 2.680$ and $t_{\alpha^*/2} = 2.654$; for $k=7$, 4.182 versus 3.760.

Testing equality of treatments

Theorem 10.5 (F test)

Let $SSB = \sum_i n_i(\bar{X}_i - \bar{X})^2$ (between groups) and $SSW = \sum_{i,j}(X_{ij}-\bar{X}_i)^2 = (n-k)\hat{\sigma}^2$ (within groups). The LRT of $H_0: \mu_1 = \cdots = \mu_k$ (i.e. $\alpha_1 = \cdots = \alpha_k = 0$) rejects when

$$F_n = \frac{SSB/(k-1)}{SSW/(n-k)} \ge F_\alpha(k-1,n-k).$$

Proof. The MLEs are $\hat{\theta}^\Omega = (\bar{x}_1,\dots,\bar{x}_k,\hat{\sigma}^{2,\Omega})$ with $n\hat{\sigma}^{2,\Omega} = SSW$, and $\hat{\theta}^{\Omega_0} = (\bar{x},\dots,\bar{x},\hat{\sigma}^{2,0})$ with $n\hat{\sigma}^{2,0} = \sum_{i,j}(x_{ij}-\bar{x})^2 = SSW + SSB$. So

$$2\{l(\hat{\theta}^\Omega) - l(\hat{\theta}^{\Omega_0})\} = n\log\frac{\hat{\sigma}^{2,0}}{\hat{\sigma}^{2,\Omega}} = n\log\Big(1+\frac{SSB}{SSW}\Big),$$

increasing in F_n. For the null distribution, write $\mathbf{X} = (X_{ij})$ as an n-vector and let A map it to the vector with entries $\bar{X}_i - \bar{X}$. Then A is symmetric and idempotent, $SSB = \mathbf{X}'A\mathbf{X}$, and $\mathrm{tr}A = \sum_{i,j}(1/n_i - 1/n) = k-1$. Under H_0, $\mathbf{Z} = (\mathbf{X}-\bar{\mu}\mathbf{1})/\sigma \sim N_n(\mathbf{0},I)$ and $A\mathbf{1} = \mathbf{0}$, so

$SSB/\sigma^2 = \mathbf{Z}'A\mathbf{Z} \sim \chi^2(k-1)$. Since SSB is a function of the group means, it is independent of $SSW/\sigma^2 \sim \chi^2(n-k)$ (Theorem 10.2). Hence $F_n \sim F(k-1, n-k)$ under H_0. ■

One sometimes uses the unweighted effects $\tilde{\alpha}_i = \mu_i - \tilde{\mu}$, $\tilde{\mu} = \sum \mu_i/k$. Since $\tilde{\alpha}_1 = \cdots = \tilde{\alpha}_k = 0$ is again $\mu_1 = \cdots = \mu_k$, the same F test applies, and for contrasts $\sum c_i\tilde{\alpha}_i = \sum c_i\mu_i = \sum c_i\alpha_i$.

Example 10.1

Yields (in kg) of a crop under three fertilizers:

fertilizer	observations	n_i	$\bar{x}_i$
A	20, 22, 19, 23	4	21
B	25, 27, 26	3	26
C	22, 21, 24, 23, 25	5	23

Here $n = 12$, $\bar{x} = (84 + 78 + 115)/12 = 23.083$, $SSB = 4(2.083)^2 + 3(2.917)^2 + 5(0.083)^2 = 42.92$, and $SSW = 10 + 2 + 10 = 22$. So $F = (42.92/2)/(22/9) = 8.78 > F_{0.05}(2, 9) = 4.26$: the fertilizers differ. Bonferroni intervals ($m = 3$, $t_{0.05/6}(9) = 2.933$, $\hat{\sigma}^2 = 2.444$) for $\mu_B - \mu_A$: $5 \pm 2.933\sqrt{2.444(1/3 + 1/4)} = 5 \pm 3.50$, which excludes 0.

10.2 Two-way analysis of variance

With two factors A (levels $i = 1, \ldots, a$) and B (levels $j = 1, \ldots, b$), each combination (i, j) is a **cell**, observed n_{ij} times:

$$X_{ijk} = \mu_{ij} + e_{ijk}, \qquad e_{ijk} \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2), \qquad k = 1, \ldots, n_{ij}.$$

This is a one-way model with ab treatments, so as in Section 10.1:

Theorem 10.6

(a) The UMVUEs are $\hat{\mu}_{ij} = \bar{X}_{ij}$ and $\hat{\sigma}^2 = \sum_{i,j,k}(X_{ijk} - \bar{X}_{ij})^2/(n - ab)$, $n = \sum n_{ij}$.
(b) The $\bar{X}_{ij}$ are independent $N(\mu_{ij}, \sigma^2/n_{ij})$.
(c) $\hat{\sigma}^2$ is independent of the cell means.
(d) $(n - ab)\hat{\sigma}^2/\sigma^2 \sim \chi^2(n - ab)$.
(e) Confidence sets and Scheffé intervals are as in Theorem 10.3, with $D = \operatorname{diag}(1/n_{ij})$ and $n - ab$ degrees of freedom.

The effects are defined by

$$\bar{\mu}_{\cdot\cdot} = \sum_{i,j} \frac{n_{ij}}{n}\mu_{ij}, \quad \bar{\mu}_{i\cdot} = \sum_{j} \frac{n_{ij}}{n_{i\cdot}}\mu_{ij}, \quad \bar{\mu}_{\cdot j} = \sum_{i} \frac{n_{ij}}{n_{\cdot j}}\mu_{ij},$$

$$\alpha_i = \bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot\cdot}, \qquad \beta_j = \bar{\mu}_{\cdot j} - \bar{\mu}_{\cdot\cdot}, \qquad \gamma_{ij} = \mu_{ij} - \bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot j} + \bar{\mu}_{\cdot\cdot}.$$

α_i and β_j are the **main effects** of A and B; γ_{ij}, the part of μ_{ij} not explained additively, is the **interaction**. The model is **balanced** if $n_{ij} \equiv r$. Then the model reads

$$X_{ijk} = \bar{\mu}_{\cdot\cdot} + \alpha_i + \beta_j + \gamma_{ij} + e_{ijk}, \qquad \sum_i \alpha_i = \sum_j \beta_j = \sum_i \gamma_{ij} = \sum_j \gamma_{ij} = 0.$$

Theorem 10.7 (Pairwise comparisons of main effects)

In the balanced model, $\hat\alpha_i = \bar X_{i\cdot\cdot} - \bar X_{\cdots}$, $n_{i\cdot} = rb$, $n = rab$, and

(a) with probability at least $1-\alpha$, for all $i \neq l$, $|(\alpha_i - \alpha_l) - (\hat\alpha_i - \hat\alpha_l)| \le \sqrt{2\hat\sigma^2/(rb)}\sqrt{(a-1)F_\alpha(a-1, n-ab)}$;

(b) with $m = a(a-1)/2$, $\alpha^* = \alpha/m$, the same holds with $t_{\alpha^*/2}(n-ab)$ in place of $\sqrt{(a-1)F_\alpha(a-1,n-ab)}$.

Proof. $\alpha_i - \alpha_l = \sum_j(\mu_{ij} - \mu_{lj})/b$ is a contrast in the μ_{ij}, with $\sum c^2/n_{ij} = 2/(rb)$; the space of such contrasts has dimension $a-1$. Apply Theorem 10.6(e) and the Bonferroni argument of Theorem 10.4. ∎

Theorem 10.8 (Test for interaction)

In the balanced model let

$$SS_{AB} = \sum_{i,j} r(\bar X_{ij\cdot} - \bar X_{i\cdot\cdot} - \bar X_{\cdot j\cdot} + \bar X_{\cdots})^2, \qquad SSE = \sum_{i,j,k}(X_{ijk} - \bar X_{ij\cdot})^2.$$

The LRT of $H_0^{AB}: \gamma_{ij} = 0$ for all i,j (the additive model) rejects when

$$F_n = \frac{SS_{AB}/\{(a-1)(b-1)\}}{SSE/(n-ab)} \ge F_\alpha((a-1)(b-1), n-ab).$$

Proof. Under Ω, $\hat\mu_{ij} = \bar x_{ij\cdot}$ and $n\hat\sigma^{2,\Omega} = SSE$. Under H_0^{AB}, $\mu_{ij} = \bar\mu + \alpha_i + \beta_j$, and in the balanced case the least squares fit is $\hat\mu^0_{ij} = \bar x_{i\cdot\cdot} + \bar x_{\cdot j\cdot} - \bar x_{\cdots}$. By orthogonality of the decomposition, $n\hat\sigma^{2,0} = SSE + SS_{AB}$, so $2\{l(\hat\theta^\Omega) - l(\hat\theta^{\Omega_0})\} = n\log(1 + SS_{AB}/SSE)$, increasing in F_n. For the null distribution, the map $\Pi : (x_{ijk}) \mapsto (\bar x_{ij\cdot} - \bar x_{i\cdot\cdot} - \bar x_{\cdot j\cdot} + \bar x_{\cdots})$ is symmetric and idempotent, $SS_{AB} = \mathbf{X}'\Pi\mathbf{X}$, and

$$\operatorname{tr}\Pi = \sum_{i,j,k}\Big(\frac1r - \frac{1}{rb} - \frac{1}{ra} + \frac{1}{rab}\Big) = ab - a - b + 1 = (a-1)(b-1).$$

Under H_0^{AB}, Π annihilates the mean vector, so $SS_{AB}/\sigma^2 \sim \chi^2((a-1)(b-1))$. It depends only on the cell means, so it is independent of $SSE/\sigma^2 \sim \chi^2(n-ab)$. ∎

Theorem 10.9 (Tests for main effects)

In the balanced model, with $SS_A = \sum_i br(\bar X_{i\cdot\cdot} - \bar X_{\cdots})^2$ and $SS_B = \sum_j ar(\bar X_{\cdot j\cdot} - \bar X_{\cdots})^2$:

(a) the LRT of $H_0^A: \alpha_1 = \cdots = \alpha_a = 0$ rejects when $\dfrac{SS_A/(a-1)}{SSE/(n-ab)} \ge F_\alpha(a-1, n-ab)$;

(b) the LRT of $H_0^B: \beta_1 = \cdots = \beta_b = 0$ rejects when $\dfrac{SS_B/(b-1)}{SSE/(n-ab)} \ge F_\alpha(b-1, n-ab)$.

The proofs follow that of Theorem 10.8. The results are summarized in the two-way ANOVA table; δ is the noncentrality parameter of the next section.

source	SS	df	MS	F	δ
A	SS_A	$a-1$	MS_A	MS_A/MSE	$\sum br\alpha_i^2/\sigma^2$
B	SS_B	$b-1$	MS_B	MS_B/MSE	$\sum ar\beta_j^2/\sigma^2$
$A\times B$	SS_{AB}	$(a-1)(b-1)$	MS_{AB}	MS_{AB}/MSE	$\sum r\gamma_{ij}^2/\sigma^2$
error	SSE	$n-ab$	MSE		
total	SST	$n-1$			

10.3 Power of F tests

Motivation. A test is useful only if it can detect effects of a relevant size. Under the alternative the F statistic has a noncentral F distribution, whose noncentrality parameter determines the power and the required sample size.

Under alternatives the F statistics have **noncentral** distributions.

Theorem 10.10 (Noncentral chi-square)

Let $X_i\sim N(\mu_i,1)$, $i=1,\dots,r$, be independent, $Y=X_1^2+\cdots+X_r^2$ and $\delta=\mu_1^2+\cdots+\mu_r^2$. Then

(a) $K_Y(t)=-\frac{r}{2}\log(1-2t)+\frac{\delta}{2}\cdot\frac{2t}{1-2t}=\sum_{k\ge1}\frac{t^k}{k!}2^{k-1}(k-1)!\,(r+k\delta)$ for $t<1/2$;

(b) with $p_{P(\delta/2)}(k)$ the Poisson$(\delta/2)$ pmf, $f_Y(y)=\sum_{k\ge0}p_{P(\delta/2)}(k)\,f_{\chi^2(r+2k)}(y)$.

In particular the law of Y depends on $(\mu_1,\dots,\mu_r)$ only through δ. It is the **noncentral chi-square** $\chi^2(r;\delta)$.

Proof. (a) For $X\sim N(\mu,1)$ and $t<1/2$, completing the square,

$$Ee^{tX^2}=\int\frac{1}{\sqrt{2\pi}}\exp\Big\{-\frac{1}{2}(1-2t)\Big(x-\frac{\mu}{1-2t}\Big)^2+\frac{t\mu^2}{1-2t}\Big\}dx=(1-2t)^{-1/2}\exp\Big(\frac{\mu^2t}{1-2t}\Big).$$

Multiply over i and take logs. Expanding $-\log(1-2t)=\sum(2t)^k/k$ and $\frac{2t}{1-2t}=\sum_{k\ge1}(2t)^k$ gives the series.

(b) By (a), $Y\overset{d}{=}V+Z^2$ with $V\sim\chi^2(r-1)$ independent of $Z\sim N(\sqrt{\delta},1)$. With $\theta=\sqrt{\delta}$,

$$f_{Z^2}(x)=\frac{1}{2\sqrt{x}}\frac{1}{\sqrt{2\pi}}e^{-\frac12x-\frac12\theta^2}\big(e^{\theta\sqrt{x}}+e^{-\theta\sqrt{x}}\big)=\sum_{k\ge0}p_{P(\delta/2)}(k)f_{\chi^2(1+2k)}(x),$$

by expanding $e^{\theta\sqrt{x}}+e^{-\theta\sqrt{x}}=2\sum_k\theta^{2k}x^k/(2k)!$ and using $\Gamma(\frac12)\Gamma(2k+1)=2^{2k}\Gamma(k+\frac12)\Gamma(k+1)$. Convolving with $f_{\chi^2(r-1)}$ and using $\chi^2(r-1)\oplus\chi^2(1+2k)=\chi^2(r+2k)$ gives (b). ∎

Theorem 10.11

If $\mathbf{X}\sim N(\mu,I)$ and A is symmetric idempotent, then $\mathbf{X}'A\mathbf{X}\sim\chi^2(r;\delta)$ with $r=\operatorname{tr}A$ and $\delta=\mu'A\mu$.

Proof. Write $A=P\begin{pmatrix}I_r&0\\0&0\end{pmatrix}P'$ with P orthogonal, and $\mathbf{Z}=P'\mathbf{X}\sim N(\eta,I)$, $\eta=P'\mu$. Then $\mathbf{X}'A\mathbf{X}=Z_1^2+\cdots+Z_r^2\sim\chi^2(r;\eta_1^2+\cdots+\eta_r^2)$, and $\eta_1^2+\cdots+\eta_r^2=\mu'A\mu$. ∎

If $V_1\sim\chi^2(r_1;\delta)$ and $V_2\sim\chi^2(r_2)$ are independent, $(V_1/r_1)/(V_2/r_2)$ has the **noncentral** F distribution $F(r_1,r_2;\delta)$.

Example 10.2 (Power of the one-way F test)

In the one-way model, $SSB = \mathbf{X}'A\mathbf{X}$ with A idempotent of trace $k-1$, and A maps the mean vector (μ_i) to (α_i). By Theorem 10.11,

$$SSB/\sigma^2 \sim \chi^2(k-1;\delta), \qquad \delta = \sum_i n_i \alpha_i^2/\sigma^2,$$

so $F_n \sim F(k-1, n-k; \delta)$, and the power is $P\{F(k-1,n-k;\delta) \geq F_\alpha(k-1,n-k)\}$. It increases in δ: writing $F_n \overset{d}{=} \frac{(V+Z^2)/(k-1)}{W/(n-k)}$ with $V \sim \chi^2(k-2)$, $Z \sim N(\sqrt{\delta},1)$, $W \sim \chi^2(n-k)$ independent, the power is $P_\theta(Z^2 \geq Y)$ with Y independent of Z and $\theta = \sqrt{\delta}$, and

$$\frac{d}{d\theta} P_\theta(Z^2 \geq y) = e^{-y/2}\phi(\theta)\left(e^{\sqrt{y}\theta} - e^{-\sqrt{y}\theta}\right) > 0 \qquad (\theta > 0).$$

In the same way, in the balanced two-way model the interaction test has power $P\{F((a-1)(b-1), n-ab; \delta_{AB}) \geq F_\alpha\}$ with $\delta_{AB} = \sum r\gamma_{ij}^2/\sigma^2$, increasing in δ_{AB}.

10.4 Regression analysis

Motivation. Regression is the same linear-model theory as ANOVA with continuous regressors. Least squares estimates, their normal-theory distributions and the F-test for the slope follow from the results of the previous sections.

The normal linear regression model (Chapters 4 and 6) is

$$\mathbf{Y} = X\beta + \mathbf{e}, \qquad \mathbf{e} \sim N_n(\mathbf{0}, \sigma^2 I), \qquad \beta \in \mathbb{R}^{p+1},\ \sigma^2 > 0,\ \operatorname{rank} X = p+1.$$

Theorem 10.12

(a) The UMVUEs of $\mathbf{c}'\beta$ and σ^2 are $\mathbf{c}'\hat\beta = \mathbf{c}'(X'X)^{-1}X'\mathbf{Y}$ and $\hat\sigma^2 = \|\mathbf{Y} - X\hat\beta\|^2/(n-p-1)$.

(b) $\hat\beta \sim N_{p+1}(\beta, \sigma^2(X'X)^{-1})$.

(c) $\hat\beta$ and $\hat\sigma^2$ are independent.

(d) $(n-p-1)\hat\sigma^2/\sigma^2 \sim \chi^2(n-p-1)$.

Proof. The density $\exp\{-\mathbf{y}'\mathbf{y}/(2\sigma^2) + \mathbf{y}'X\beta/\sigma^2 - \beta'X'X\beta/(2\sigma^2) - \frac{n}{2}\log(2\pi\sigma^2)\}$ is a full-rank exponential family, so $(X'\mathbf{Y}, \mathbf{Y}'\mathbf{Y})$ is complete sufficient. Since $X'\mathbf{Y} = (X'X)\hat\beta$ and $\mathbf{Y}'\mathbf{Y} = (n-p-1)\hat\sigma^2 + \hat\beta'(X'X)\hat\beta$, $(\hat\beta, \hat\sigma^2)$ is a one-to-one function of it, and the unbiased estimators $\mathbf{c}'\hat\beta$, $\hat\sigma^2$ (Theorem 6.16) are UMVUEs. (b)–(d) are Theorem 4.19 in Volume I. ∎

Theorem 10.13 (Confidence sets for β)

For a $(p+1)\times r$ matrix C of rank r:

(a) $P\{(C'\beta - C'\hat\beta)'(C'(X'X)^{-1}C)^{-1}(C'\beta - C'\hat\beta) \leq r\hat\sigma^2 F_\alpha(r, n-p-1)\} = 1-\alpha$;

(b) $P\left\{|\mathbf{c}'\beta - \mathbf{c}'\hat\beta| \leq \sqrt{\mathbf{c}'(X'X)^{-1}\mathbf{c}\,\hat\sigma^2}\sqrt{rF_\alpha(r,n-p-1)}\ \ \forall \mathbf{c} \in \operatorname{col}(C)\right\} = 1-\alpha$.

The proof is that of Theorem 10.3.

Often only some coefficients matter; e.g. with an intercept, whether the slopes $\beta_1, \ldots, \beta_p$ are zero. Partition

$$\mathbf{Y} = X_0\beta_0 + X_1\beta_1 + \mathbf{e}, \qquad \operatorname{rank} X_0 = p_0,\ \operatorname{rank}(X_0, X_1) = p_0 + p_1,$$

and let $\Pi_0 = X_0(X_0'X_0)^{-1}X_0'$, $X_{1|0} = (I-\Pi_0)X_1$, $\Pi_{1|0} = X_{1|0}(X_{1|0}'X_{1|0})^{-1}X_{1|0}'$ and $\Pi_{0,1} = X(X'X)^{-1}X'$. From Theorem 6.17, $\Pi_{0,1} = \Pi_0 + \Pi_{1|0}$ and $\Pi_0\Pi_{1|0} = 0$.

Theorem 10.14 (Testing a subset of coefficients)

For $H_0 : \beta_1 = \mathbf{0}$ vs $H_1 : \beta_1 \neq \mathbf{0}$:

(a) $\min_{\beta_0}\|\mathbf{Y} - X_0\beta_0\|^2 = \min_{\beta_0,\beta_1}\|\mathbf{Y} - X_0\beta_0 - X_1\beta_1\|^2 + \mathbf{Y}'\Pi_{1|0}\mathbf{Y}$.

(b) With $R(1|0) = \mathbf{Y}'\Pi_{1|0}\mathbf{Y}$ and $SSE = \mathbf{Y}'(I-\Pi_0-\Pi_{1|0})\mathbf{Y}$, the LRT rejects when

$$F_n = \frac{R(1|0)/p_1}{SSE/(n-p_0-p_1)} \geq F_\alpha(p_1, n-p_0-p_1).$$

(c) $F_n \sim F(p_1, n-p_0-p_1; \delta)$ with $\delta = \beta_1'X_1'\Pi_{1|0}X_1\beta_1/\sigma^2$, and the power increases in δ.

Proof. (a) The minima are $\mathbf{Y}'(I-\Pi_0)\mathbf{Y}$ and $\mathbf{Y}'(I-\Pi_{0,1})\mathbf{Y}$, and $I-\Pi_0 = (I-\Pi_{0,1}) + \Pi_{1|0}$.

(b) The MLEs of σ^2 are SSE/n under Ω and $(SSE + R(1|0))/n$ under H_0, so $2\{l(\hat\theta^\Omega) - l(\hat\theta^{\Omega_0})\} = n\log(1 + R(1|0)/SSE)$, increasing in F_n.

(c) $\Pi_{1|0}$ is idempotent with $\mathrm{tr}\Pi_{1|0} = \mathrm{tr}\{(X_{1|0}'X_{1|0})^{-1}X_{1|0}'X_{1|0}\} = p_1$, so $R(1|0)/\sigma^2 \sim \chi^2(p_1;\delta)$ with $\delta = (X_0\beta_0 + X_1\beta_1)'\Pi_{1|0}(X_0\beta_0 + X_1\beta_1)/\sigma^2 = \beta_1'X_1'\Pi_{1|0}X_1\beta_1/\sigma^2$ since $\Pi_{1|0}X_0 = 0$. Also $SSE/\sigma^2 \sim \chi^2(n-p_0-p_1)$, and $\Pi_{1|0}(I-\Pi_0-\Pi_{1|0}) = 0$ gives independence. Under H_0, $\delta = 0$. Monotonicity of the power is shown as in Example 10.2. ■

Theorem 10.15 (General linear hypotheses)

For a $(p+1)\times r$ matrix C of rank r, the LRT of $H_0 : C'\beta = \mathbf{0}$ vs $H_1 : C'\beta \neq \mathbf{0}$ satisfies:

(a) with $\Pi_{1|0} = X(X'X)^{-1}C\{C'(X'X)^{-1}C\}^{-1}C'(X'X)^{-1}X'$,

$$\min_{C'\beta=\mathbf{0}}\|\mathbf{Y} - X\beta\|^2 = \min_{\beta}\|\mathbf{Y} - X\beta\|^2 + \mathbf{Y}'\Pi_{1|0}\mathbf{Y},$$

$$\mathbf{Y}'\Pi_{1|0}\mathbf{Y} = (C'\hat\beta)'\{C'(X'X)^{-1}C\}^{-1}(C'\hat\beta);$$

(b) with $SSE = \mathbf{Y}'(I - X(X'X)^{-1}X')\mathbf{Y}$, it rejects when $\dfrac{\mathbf{Y}'\Pi_{1|0}\mathbf{Y}/r}{SSE/(n-p-1)} \geq F_\alpha(r, n-p-1)$;

(c) this statistic is $F(r, n-p-1;\delta)$ with $\delta = \beta'C\{C'(X'X)^{-1}C\}^{-1}C'\beta/\sigma^2$.

Proof. Reduce to Theorem 10.14. Choose C_1 ($(p+1)\times r$) with orthonormal columns spanning $\mathrm{col}(C)$, and extend to an orthonormal basis $(C_0\ C_1)$ of $\mathbb{R}^{p+1}$; then $C = C_1B$ for an invertible B. With $D_0 = XC_0$, $D_1 = XC_1$, $\gamma_0 = C_0'\beta$, $\gamma_1 = C_1'\beta$,

$$X\beta = X(C_0\ C_1)(C_0\ C_1)'\beta = D_0\gamma_0 + D_1\gamma_1,$$

and $C'\beta = \mathbf{0} \iff \gamma_1 = \mathbf{0}$. Theorem 10.14 applies, and $\Pi_{1|0}$ is the projection onto the orthogonal complement of $\mathrm{col}(D_0)$ in $\mathrm{col}(X)$:

$$\{X\beta : \beta'X'XC_0 = \mathbf{0}\} = \{X\beta : X'X\beta \in \mathrm{col}(C)\} = \mathrm{col}(X(X'X)^{-1}C).$$

With $S = X(X'X)^{-1}C$, $\Pi_{1|0} = S(S'S)^{-1}S'$, which is the stated matrix, and $S'\mathbf{Y} = C'\hat\beta$ gives the quadratic form. ■

Linear models. The ANOVA models are regression models whose design matrix has dependent columns, made identifiable by linear constraints:

$$\mathbf{Y} = X\beta + \mathbf{e}, \quad C'\beta = \mathbf{0}, \quad \mathbf{e} \sim N_n(\mathbf{0}, \sigma^2 I), \quad \operatorname{rank} X = p + 1 - r < p + 1,$$

where the rows of X and C' together are linearly independent. Such a model is a **linear model**.

Example 10.3 (One-way model as a linear model)

In $X_{ij} = \bar{\mu} + \alpha_i + e_{ij}$, $\sum n_i \alpha_i = 0$, stack $\mathbf{Y} = (X_{11}, \dots, X_{1n_1}, \dots, X_{k1}, \dots, X_{kn_k})'$, $\beta = (\bar{\mu}, \alpha_1, \dots, \alpha_k)'$, $\mathbf{c} = (0, n_1, \dots, n_k)'$, and

$$X = \begin{pmatrix} \mathbf{1}_{n_1} & \mathbf{1}_{n_1} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{1}_{n_2} & \mathbf{0} & \mathbf{1}_{n_2} & \cdots & \mathbf{0} \\ \vdots & & & \ddots & \\ \mathbf{1}_{n_k} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{1}_{n_k} \end{pmatrix}.$$

This **design matrix** has rank $k < k + 1$; the constraint $\mathbf{c}'\beta = 0$ makes β identifiable. Since $X'X$ is singular, formulas must be written with projections rather than $(X'X)^{-1}$, but the estimation and testing theory goes through as in Theorem 10.15.

Chapter 11. Bayesian Inference

11.1 Prior and posterior distributions

The Bayes criteria of Chapters 8 and 9 weight the parameter values by a prior. Bayesian inference takes this literally: before the data are seen, uncertainty about $\theta \in \Omega$ is described by a **prior density** $\pi(\theta)$. Whether such a prior exists, and whether it is unique, is debatable, and this is the source of the long-standing controversy about Bayesian methods. Here we take the prior as given. To stress conditioning we write $f(x \mid \theta)$ for the model density.

Theorem 11.1 (Bayes' theorem)

If $X_1, \dots, X_n$ is a sample from $f(x \mid \theta)$ and θ has prior density $\pi(\theta)$, then given $\mathbf{X} = \mathbf{x}$ the density of θ is

$$\pi(\theta \mid \mathbf{x}) \propto L(\theta; \mathbf{x})\pi(\theta), \qquad L(\theta; \mathbf{x}) = \prod_{i=1}^{n} f(x_i \mid \theta).$$

Proof. By the definition of conditional density, $\pi(\theta \mid \mathbf{x}) = f(\mathbf{x}, \theta)/f_{\mathbf{X}}(\mathbf{x}) = f(\mathbf{x} \mid \theta)\pi(\theta)/f_{\mathbf{X}}(\mathbf{x})$. The normalizing constant is the marginal density $f_{\mathbf{X}}(\mathbf{x}) = \int_\Omega f(\mathbf{x} \mid \theta)\pi(\theta)\, d\theta$ (a sum in the discrete case), which does not involve θ. ∎

$\pi(\theta \mid \mathbf{x})$ is the **posterior density**; in words, posterior $\propto$ likelihood $\times$ prior. A prior that is not a probability density ($\int \pi = \infty$) is called **improper**; it can still be used if the posterior is proper.

Example 11.1 (Binomial–beta)

Bernoulli(θ) data with prior Beta(α, β): with $t = \sum x_i$,

$$\pi(\theta \mid \mathbf{x}) \propto \theta^t (1-\theta)^{n-t} \theta^{\alpha-1} (1-\theta)^{\beta-1},$$

so $\theta \mid \mathbf{x} \sim \text{Beta}(t+\alpha, n-t+\beta)$.

Example 11.2 (Poisson–gamma)

Poisson(θ) data with prior Gamma(α, β): $\pi(\theta \mid \mathbf{x}) \propto e^{-n\theta}\theta^t \theta^{\alpha-1} e^{-\theta/\beta}$, so $\theta \mid \mathbf{x} \sim \text{Gamma}(t+\alpha, (n+\beta^{-1})^{-1})$.

Example 11.3 (Exponential–gamma)

Exp(θ) data, with the rate $\lambda = 1/\theta$ given a Gamma(α, β) prior: $\pi(\lambda \mid \mathbf{x}) \propto \lambda^n e^{-\lambda t} \lambda^{\alpha-1} e^{-\lambda/\beta}$, so $\lambda \mid \mathbf{x} \sim \text{Gamma}(n+\alpha, (t+\beta^{-1})^{-1})$, i.e. $(t+\beta^{-1})/\theta \mid \mathbf{x} \sim \text{Gamma}(n+\alpha, 1)$. The posterior density of θ itself follows by the change of variables $\lambda = 1/\theta$: $\pi(\theta \mid \mathbf{x}) = \pi(\lambda \mid \mathbf{x})\theta^{-2}$.

Example 11.4 (Normal mean, known variance)

$N(\theta, \sigma^2)$ data, σ^2 known, prior $N(\mu_0, \sigma_0^2)$:

$$\pi(\theta \mid \mathbf{x}) \propto \exp\Big\{ -\frac{n}{2\sigma^2}(\theta - \bar{x})^2 - \frac{1}{2\sigma_0^2}(\theta - \mu_0)^2 \Big\}.$$

Completing the square, $\theta \mid \mathbf{x} \sim N(\mu(\mathbf{x}), \sigma^2(\mathbf{x}))$ with

$$\mu(\mathbf{x}) = \frac{n\sigma^{-2}\bar{x} + \sigma_0^{-2}\mu_0}{n\sigma^{-2} + \sigma_0^{-2}}, \qquad \sigma^2(\mathbf{x}) = (n\sigma^{-2} + \sigma_0^{-2})^{-1}.$$

Posterior precision is data precision plus prior precision.

In these examples the posterior belongs to the same family as the prior; such a prior is a **conjugate prior**.

Theorem 11.2 (Normal model, conjugate prior)

Let $X_1, \dots, X_n$ be $N(\mu, \sigma^2)$, and let the prior of $\theta = (\mu, \sigma^2)$ be

$$\mu \mid \sigma^2 \sim N(\mu_0, \sigma^2/\kappa_0), \qquad \nu_0\sigma_0^2/\sigma^2 \sim \chi^2(\nu_0).$$

Put $\kappa_n = n + \kappa_0$, $\nu_n = n + \nu_0$, $s^2 = \sum(x_i - \bar{x})^2/(n-1)$, and

$$\mu_n(\mathbf{x}) = \frac{n\bar{x} + \kappa_0\mu_0}{n + \kappa_0}, \qquad \nu_n\sigma_n^2(\mathbf{x}) = \nu_0\sigma_0^2 + (n-1)s^2 + (\kappa_0^{-1} + n^{-1})^{-1}(\bar{x} - \mu_0)^2.$$

Then

(a) $\mu \mid \sigma^2, \mathbf{x} \sim N(\mu_n(\mathbf{x}), \sigma^2/\kappa_n)$ and $\nu_n\sigma_n^2(\mathbf{x})/\sigma^2 \mid \mathbf{x} \sim \chi^2(\nu_n)$;

(b) $\dfrac{\mu - \mu_n(\mathbf{x})}{\sigma_n(\mathbf{x})/\sqrt{\kappa_n}} \,\Big|\, \mathbf{x} \sim t(\nu_n)$.

Proof. (a) Work with μ and $\tau = \sigma^{-2}$. The likelihood and prior are

$$L \propto \tau^{n/2} \exp\Big\{ -\frac{\tau}{2}\big[(n-1)s^2 + n(\mu - \bar{x})^2\big]\Big\},$$

$$\pi(\mu \mid \tau) \propto \tau^{1/2} e^{-\frac{\tau\kappa_0}{2}(\mu-\mu_0)^2}, \qquad \pi(\tau) \propto \tau^{\nu_0/2-1} e^{-\frac{\tau}{2}\nu_0\sigma_0^2}.$$

Completing the square in μ,

$$n(\mu - \bar{x})^2 + \kappa_0(\mu - \mu_0)^2 = \kappa_n(\mu - \mu_n(\mathbf{x}))^2 + \frac{n\kappa_0}{n + \kappa_0}(\bar{x} - \mu_0)^2,$$

and $\frac{n\kappa_0}{n+\kappa_0} = (\kappa_0^{-1} + n^{-1})^{-1}$. Hence

$$\pi(\mu, \tau \mid \mathbf{x}) \propto \tau^{1/2} e^{-\frac{\tau\kappa_n}{2}(\mu-\mu_n(\mathbf{x}))^2} \cdot \tau^{\nu_n/2-1} e^{-\frac{\tau}{2}\nu_n\sigma_n^2(\mathbf{x})},$$

which is the stated normal–chi-square form.

(b) By (a), given $\mathbf{x}$, $Z = \frac{\mu - \mu_n(\mathbf{x})}{\sigma/\sqrt{\kappa_n}} \sim N(0,1)$ conditionally on σ^2; this law does not involve σ^2, so Z is independent of $V = \nu_n\sigma_n^2(\mathbf{x})/\sigma^2 \sim \chi^2(\nu_n)$. The ratio $Z/\sqrt{V/\nu_n}$ is the stated quantity, which is $t(\nu_n)$. ∎

Without conjugacy, the marginal $f_{\mathbf{X}}(\mathbf{x})$ is often intractable, and the posterior is computed by Markov chain Monte Carlo. Since the theory is cleanest with conjugate priors, we use them below.

11.2 Bayesian inference

Bayesian inference summarizes the posterior. Point estimates are the **posterior mean**, **median** or **mode**; precision is measured by the posterior variance or quantiles. An interval $I(\mathbf{x})$ with $P(\theta \in I(\mathbf{x}) \mid \mathbf{X} = \mathbf{x}) = 1 - \alpha$ is a $1-\alpha$ **credible interval**. Unlike a confidence interval, its probability statement is about θ given the observed data.

Example 11.5 (Normal mean, known variance)

In Example 11.4 the posterior is normal, so its mean, median and mode coincide:

$$\hat{\theta} = \frac{n\sigma^{-2}\bar{x} + \sigma_0^{-2}\mu_0}{n\sigma^{-2} + \sigma_0^{-2}},$$

a weighted average of the prior mean and $\bar{x}$, with weights equal to the prior precision σ_0^{-2} and the data precision $n\sigma^{-2}$. The $1-\alpha$ credible interval is $\hat{\theta} \pm z_{\alpha/2}\sigma(\mathbf{x})$. For instance, with $\sigma = 10$, $n = 25$, $\bar{x} = 104$ and prior $N(100, 5^2)$: $n\sigma^{-2} = 0.25$, $\sigma_0^{-2} = 0.04$, $\hat{\theta} = (26 + 4)/0.29 = 103.45$, $\sigma(\mathbf{x}) = 0.29^{-1/2} = 1.857$, and the 95% credible interval is 103.45 ± 3.64.

Example 11.6 (Normal model, conjugate prior)

By Theorem 11.2(b), when $\nu_n > 1$ the posterior of μ is a shifted, scaled t, so the posterior mean, median and mode equal $\hat{\mu} = \mu_n(\mathbf{x})$, and the $1 - \alpha$ credible interval is $\hat{\mu} \pm t_{\alpha/2}(\nu_n)\sigma_n(\mathbf{x})/\sqrt{\kappa_n}$.
For σ^2, $Y = \nu_n\sigma_n^2(\mathbf{x})/\sigma^2 \mid \mathbf{x} \sim \chi^2(\nu_n)$ gives

$$\hat{\sigma}^{2,\text{mean}} = \nu_n\sigma_n^2(\mathbf{x})E[Y^{-1}] = \frac{\nu_n}{\nu_n - 2}\sigma_n^2(\mathbf{x}), \qquad \hat{\sigma}^{2,\text{median}} = \frac{\nu_n}{\chi^2_{0.5}(\nu_n)}\sigma_n^2(\mathbf{x}),$$

and, from the posterior density of σ^2, $\hat{\sigma}^{2,\text{mode}} = \frac{\nu_n}{\nu_n+2}\sigma_n^2(\mathbf{x})$. As $n \to \infty$ all three factors tend to 1. The $1 - \alpha$ credible interval is

$$\frac{\nu_n\sigma_n^2(\mathbf{x})}{\chi^2_{\alpha/2}(\nu_n)} \le \sigma^2 \le \frac{\nu_n\sigma_n^2(\mathbf{x})}{\chi^2_{1-\alpha/2}(\nu_n)}.$$

Example 11.7 (Binomial–beta)

With a $\text{Beta}(\alpha_1, \alpha_2)$ prior, $\theta \mid \mathbf{x} \sim \text{Beta}(t + \alpha_1, n - t + \alpha_2)$. The posterior mean is

$$\hat{\theta}^{\text{mean}} = \frac{t + \alpha_1}{n + \alpha_1 + \alpha_2} = \frac{n}{n + \alpha_1 + \alpha_2} \cdot \frac{t}{n} + \frac{\alpha_1 + \alpha_2}{n + \alpha_1 + \alpha_2} \cdot \frac{\alpha_1}{\alpha_1 + \alpha_2},$$

again a weighted average of the sample proportion and the prior mean, and the mode is $\frac{(t+\alpha_1-1)_+}{(t+\alpha_1-1)_+ + (n-t+\alpha_2-1)_+}$. With $B_p(r_1, r_2)$ the upper p quantile of $\text{Beta}(r_1, r_2)$, the credible interval is $[B_{1-\alpha/2}, B_{\alpha/2}]$ for $\text{Beta}(t+\alpha_1, n-t+\alpha_2)$. Since $\text{Beta}(r_1, r_2) \overset{d}{=} \frac{(r_1/r_2)F}{1+(r_1/r_2)F}$ with $F \sim F(2r_1, 2r_2)$, these quantiles come from F tables. For large n, the posterior is approximately $N(\hat{\theta}, \hat{\theta}(1-\hat{\theta})/(n+\alpha_1+\alpha_2))$, giving the approximate interval $\hat{\theta} \pm z_{\alpha/2}\sqrt{\hat{\theta}(1-\hat{\theta})/(n+\alpha_1+\alpha_2)}$. For example, $t = 12$ successes in $n = 40$ trials with a uniform prior ($\alpha_1 = \alpha_2 = 1$) give $\hat{\theta}^{\text{mean}} = 13/42 = 0.310$ and the approximate 95% interval 0.310 ± 0.140.

Bayesian testing. To decide between $H_0 : \theta \in \Omega_0$ and $H_1 : \theta \in \Omega_1$, compare the posterior probabilities through the **posterior odds**

$$\frac{P(\theta \in \Omega_0 \mid \mathbf{X} = \mathbf{x})}{P(\theta \in \Omega_1 \mid \mathbf{X} = \mathbf{x})} = \frac{P(\theta \in \Omega_0 \mid \mathbf{X} = \mathbf{x})}{1 - P(\theta \in \Omega_0 \mid \mathbf{X} = \mathbf{x})}.$$

Example 11.8 (Simple hypotheses)

For $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$ with prior probabilities π_0, π_1,

$$\frac{P(\theta = \theta_0 \mid \mathbf{x})}{P(\theta = \theta_1 \mid \mathbf{x})} = \frac{f(\mathbf{x} \mid \theta_0)\pi_0}{f(\mathbf{x} \mid \theta_1)\pi_1}.$$

Rejecting H_0 when the posterior odds are below 1 is exactly the Bayes test of Theorem 9.1. For $N(\mu, 1)$ data and μ_0 vs μ_1, the posterior odds are $\exp\{n(\mu_0 - \mu_1)(\bar{x} - \frac{\mu_0+\mu_1}{2}) + \log(\pi_0/\pi_1)\}$.

Example 11.9 (One-sided hypotheses)

(a) In the normal model with conjugate prior, for $H_0 : \mu \le \mu_0$ vs $H_1 : \mu > \mu_0$, Theorem 11.2(b) gives

$$P(\mu \le \mu_0 \mid \mathbf{x}) = F_{t(\nu_n)}\Big(\frac{\mu_0 - \mu_n(\mathbf{x})}{\sigma_n(\mathbf{x})/\sqrt{\kappa_n}}\Big),$$

with $F_{t(\nu)}$ the $t(\nu)$ cdf, and the posterior odds follow.
(b) In the binomial–beta model, for $H_0 : \theta \le \theta_0$, $P(\theta \le \theta_0 \mid \mathbf{x})$ is the Beta$(t + \alpha_1, n - t + \alpha_2)$ cdf at θ_0, computable from F tables, or for large n approximately $\Phi((\theta_0 - \hat\theta)/\sqrt{\hat\theta(1-\hat\theta)/(n + \alpha_1 + \alpha_2)})$.

11.3 Bayes and minimax estimators

Bayesian inference uses only the observed data, through the posterior. Frequentist inference instead evaluates a procedure over repeated samples, e.g. by its MSE $E[(\hat\theta(\mathbf{X}) - \theta)^2 \mid \theta]$. As in Chapter 8, two criteria free of the unknown θ are the Bayes MSE and the maximum MSE:

$$r(\pi, \hat\theta) = \int_\Omega E[(\hat\theta - \theta)^2 \mid \theta]\pi(\theta)\, d\theta, \qquad \max_{\theta \in \Omega} E[(\hat\theta - \theta)^2 \mid \theta].$$

Definition (Bayes and minimax estimators)

(a) $\hat\theta^\pi$ is a **Bayes estimator** for the prior π if $r(\pi, \hat\theta^\pi) = \min_{\hat\theta} r(\pi, \hat\theta)$.
(b) $\hat\theta^m$ is a **minimax estimator** if $\max_\theta E[(\hat\theta^m - \theta)^2 \mid \theta] = \min_{\hat\theta} \max_\theta E[(\hat\theta - \theta)^2 \mid \theta]$.

Theorem 11.3 (Bayes estimator under squared error)

The Bayes estimator minimizes the posterior expected squared error and is the posterior mean:

$$\hat\theta^\pi(\mathbf{x}) = \arg\min_a E[(a - \theta)^2 \mid \mathbf{X} = \mathbf{x}] = E(\theta \mid \mathbf{X} = \mathbf{x}).$$

Proof. $r(\pi, \hat\theta)$ is the expectation of $(\hat\theta(\mathbf{X}) - \theta)^2$ under the joint law of $(\mathbf{X}, \theta)$, so by iterated expectation

$$r(\pi, \hat\theta) = E\big[E\{(\hat\theta(\mathbf{X}) - \theta)^2 \mid \mathbf{X}\}\big].$$

If $\hat\theta^\pi(\mathbf{x})$ minimizes the inner conditional expectation for each $\mathbf{x}$, then $E\{(\hat\theta(\mathbf{X}) - \theta)^2 \mid \mathbf{X}\} \ge E\{(\hat\theta^\pi(\mathbf{X}) - \theta)^2 \mid \mathbf{X}\}$ for every $\hat\theta$, and taking expectations gives $r(\pi, \hat\theta) \ge r(\pi, \hat\theta^\pi)$. Finally, $E[(a - \theta)^2 \mid \mathbf{x}] = \mathrm{Var}(\theta \mid \mathbf{x}) + (a - E[\theta \mid \mathbf{x}])^2$ is minimized at $a = E[\theta \mid \mathbf{x}]$. ∎

The same argument works for a general **loss function** $L(\theta, \hat\eta)$: the **Bayes risk** $r(\pi, \hat\eta) = E[L(\theta, \hat\eta(\mathbf{X}))]$ is minimized by the estimator that minimizes the **posterior expected loss** $E[L(\theta, \hat\eta) \mid \mathbf{X} = \mathbf{x}]$ for

each $\mathbf{x}$. Minimax estimators are defined with the **risk function** $R_{\hat\eta}(\theta) = E[L(\theta,\hat\eta) \mid \theta]$ in place of the MSE.

Example 11.10

(a) Binomial–beta with squared error: $\hat\theta^\pi = \frac{t+\alpha_1}{n+\alpha_1+\alpha_2}$.
(b) Normal model, conjugate prior, loss $L = (\hat\mu-\mu)^2/\sigma^2$: the Bayes estimator minimizes $E[(\hat\mu-\mu)^2\sigma^{-2} \mid \mathbf{x}]$, so $\hat\mu^\pi = E(\mu\sigma^{-2} \mid \mathbf{x})/E(\sigma^{-2} \mid \mathbf{x})$. Since $E(\mu \mid \sigma^2, \mathbf{x}) = \mu_n(\mathbf{x})$ does not depend on σ^2, $E(\mu\sigma^{-2} \mid \mathbf{x}) = \mu_n(\mathbf{x})E(\sigma^{-2} \mid \mathbf{x})$, and $\hat\mu^\pi = \mu_n(\mathbf{x}) = (n\bar x + \kappa_0\mu_0)/(n+\kappa_0)$.
(c) Same model, loss $L = (\hat\sigma^2/\sigma^2 - 1)^2$: the Bayes estimator is $E(\sigma^{-2} \mid \mathbf{x})/E(\sigma^{-4} \mid \mathbf{x})$. With $Y = \nu_n\sigma_n^2(\mathbf{x})/\sigma^2 \sim \chi^2(\nu_n)$, $EY = \nu_n$ and $EY^2 = 2\nu_n + \nu_n^2$, so

$$\hat\sigma^{2,\pi} = \nu_n\sigma_n^2(\mathbf{x})\frac{\nu_n}{2\nu_n+\nu_n^2} = \frac{\nu_n}{\nu_n+2}\sigma_n^2(\mathbf{x}),$$

the posterior mode.

Theorem 11.4 (Bayes estimators with constant risk are minimax)

If an estimator $\hat\eta^*$ has (a) constant risk, $R_{\hat\eta^*}(\theta) = R_{\hat\eta^*}(\theta_0)$ for all θ, and (b) is the Bayes estimator for some prior probability distribution π, then it is minimax.

Proof. By (a), $\max_\theta R_{\hat\eta^*}(\theta) = \int R_{\hat\eta^*}(\theta)\pi(\theta)\,d\theta = r(\pi,\hat\eta^*)$. By (b), for any $\hat\eta$,

$$r(\pi,\hat\eta^*) \le r(\pi,\hat\eta) = \int R_{\hat\eta}(\theta)\pi(\theta)\,d\theta \le \max_\theta R_{\hat\eta}(\theta).$$

Hence $\max_\theta R_{\hat\eta^*} \le \max_\theta R_{\hat\eta}$ for every $\hat\eta$. ■

Example 11.11 (Minimax estimation of a binomial proportion)

For Bernoulli(θ) data and squared error, consider estimators $a\hat\theta + b$ with $\hat\theta = t/n$. Their risk is

$$E[(a\hat\theta+b-\theta)^2 \mid \theta] = \frac{a^2\theta(1-\theta)}{n} + \{b+(a-1)\theta\}^2,$$

a quadratic in θ whose coefficients of θ^2 and θ vanish when $(1-a)^2 = a^2/n$ and $a^2/n - 2b(1-a) = 0$, i.e.

$$a = \frac{\sqrt n}{\sqrt n+1}, \qquad b = \frac{1}{2(\sqrt n+1)}.$$

The resulting estimator $\hat\theta^* = \frac{t+\sqrt n/2}{n+\sqrt n}$ has constant risk $\frac{1}{4(\sqrt n+1)^2}$, and by Example 11.10(a) it is the Bayes estimator for the prior Beta($\sqrt n/2, \sqrt n/2$). By Theorem 11.4 it is minimax. Its maximum risk is smaller than that of $\hat\theta$, which is $1/(4n)$, but $\hat\theta$ has smaller risk except for θ near $1/2$.

When the minimax estimator is not Bayes for any proper prior, it may still be a limit of Bayes estimators.

Theorem 11.5 (Least favourable sequence of priors)

If for some estimator $\hat\eta^*$ there are priors π_k with Bayes estimators $\hat\eta^{\pi_k}$ such that

$$\lim_{k\to\infty} r(\pi_k,\hat\eta^{\pi_k}) \ge \max_\theta R_{\hat\eta^*}(\theta),$$

then $\hat\eta^*$ is minimax.

Proof. For any $\hat{\eta}$ and every k, $r(\pi_k, \hat{\eta}^{\pi_k}) \le r(\pi_k, \hat{\eta}) = \int R_{\hat{\eta}} \pi_k \le \max_\theta R_{\hat{\eta}}(\theta)$. Letting $k \to \infty$, $\max_\theta R_{\hat{\eta}} \ge \lim_k r(\pi_k, \hat{\eta}^{\pi_k}) \ge \max_\theta R_{\hat{\eta}^*}$. ■

Example 11.12 (The sample mean is minimax)

For $N(\mu, \sigma^2)$ data and loss $(\hat{\mu}-\mu)^2/\sigma^2$, the Bayes estimator under the conjugate prior is $\mu_n(\mathbf{x})$ (Example 11.10(b)). Given σ^2 and $\mathbf{x}$, $\mu \sim N(\mu_n(\mathbf{x}), \sigma^2/\kappa_n)$, so $E[(\mu - \mu_n(\mathbf{x}))^2/\sigma^2 \mid \sigma^2, \mathbf{x}] = 1/\kappa_n$ and the Bayes risk is $1/(n+\kappa_0)$. Take $\kappa_0 = 1/k$; then $r(\pi_k, \hat{\mu}^{\pi_k}) = 1/(n+1/k) \to 1/n$. The sample mean has constant risk $E[(\bar{X} - \mu)^2/\sigma^2] = 1/n$. By Theorem 11.5, $\bar{X}$ is minimax for this loss.

11.4 Selected exercises

Motivation. These are textbook exercises for this chapter (numbered as in the book). They add two conjugate families to those of Section 11.1, compute a Bayes estimator under a relative loss, and show that a weighted loss makes the sample proportion both Bayes and minimax.

Throughout, Gamma(α, β) has density $\theta^{\alpha-1}e^{-\theta/\beta}/\{\Gamma(\alpha)\beta^\alpha\}$, $\theta > 0$, as in Example 11.2.

Exercise 11.1 (Pareto data, gamma prior)

Let $X_1, \dots, X_n$ have density $f(x;\theta) = \theta x^{-\theta-1}$ for $x > 1$ (Pareto$(1,\theta)$, $\theta > 0$), and let $\theta \sim$ Gamma(α, β). Find the posterior distribution of θ.

Solution.

Step 1. With $t = \sum_i \log x_i$ (positive, since $x_i > 1$), the likelihood is $L(\theta; \mathbf{x}) = \theta^n \prod_i x_i^{-\theta-1} \propto \theta^n e^{-\theta t}$ as a function of θ.

Step 2. By Theorem 11.1,

$$\pi(\theta \mid \mathbf{x}) \propto \theta^n e^{-\theta t} \cdot \theta^{\alpha-1} e^{-\theta/\beta} = \theta^{n+\alpha-1} e^{-\theta(t+\beta^{-1})},$$

the kernel of a gamma density. Hence $\theta \mid \mathbf{x} \sim \text{Gamma}(n+\alpha, (t+\beta^{-1})^{-1})$. (This is Example 11.3 in disguise: $\log X_i$ is exponential with rate θ.) ■

Exercise 11.2 (Multinomial data, Dirichlet prior)

Let $\mathbf{X} = (X_1, \dots, X_k)' \sim \text{Multi}(n, (p_1, \dots, p_k)')$ with $\sum_j p_j = 1$, and let $\theta = (p_1, \dots, p_{k-1})'$ have the Dirichlet$(\alpha_1, \dots, \alpha_k)$ prior, whose density on $\{p_j > 0,\ \sum_{j<k} p_j < 1\}$ is

$$\pi(\theta) = \frac{\Gamma(\alpha_1 + \cdots + \alpha_k)}{\Gamma(\alpha_1)\cdots\Gamma(\alpha_k)} \prod_{j=1}^{k} p_j^{\alpha_j - 1}, \qquad p_k = 1 - \sum_{j<k} p_j.$$

Find the posterior distribution of θ.

Solution.

Step 1. The likelihood is $L(\theta; \mathbf{x}) = \frac{n!}{x_1!\cdots x_k!} \prod_j p_j^{x_j} \propto \prod_j p_j^{x_j}$, with $p_k = 1 - \sum_{j<k} p_j$.

Step 2. By Theorem 11.1, on the same simplex,

$$\pi(\theta \mid \mathbf{x}) \propto \prod_{j=1}^{k} p_j^{x_j} \prod_{j=1}^{k} p_j^{\alpha_j - 1} = \prod_{j=1}^{k} p_j^{x_j + \alpha_j - 1},$$

the kernel of the Dirichlet$(x_1+\alpha_1,\dots,x_k+\alpha_k)$ density. Two densities with proportional kernels on the same support coincide, so $\theta \mid \mathbf{x} \sim \text{Dirichlet}(x_1+\alpha_1,\dots,x_k+\alpha_k)$. For $k=2$ this is Example 11.1; each α_j acts as a prior count added to the observed count x_j. ■

Exercise 11.10 (A Bayes estimator under relative squared error)

In Exercise 11.1, find the Bayes estimator of θ for the loss $L(\theta,\hat\theta) = (\hat\theta/\theta - 1)^2$, assuming $n+\alpha > 2$.

Solution.

Step 1. As noted after Theorem 11.3, the Bayes estimator minimizes the posterior expected loss for each $\mathbf{x}$. Here

$$E[(a/\theta - 1)^2 \mid \mathbf{x}] = a^2 E(\theta^{-2} \mid \mathbf{x}) - 2aE(\theta^{-1} \mid \mathbf{x}) + 1,$$

a quadratic in a minimized at $a^* = E(\theta^{-1} \mid \mathbf{x})/E(\theta^{-2} \mid \mathbf{x})$.

Step 2. If $\theta \sim \text{Gamma}(a_1, b)$ and $k < a_1$, then

$$E\theta^{-k} = \frac{\int_0^\infty \theta^{a_1-k-1}e^{-\theta/b}\,d\theta}{\Gamma(a_1)b^{a_1}} = \frac{\Gamma(a_1-k)}{\Gamma(a_1)}\,b^{-k}.$$

Hence $E\theta^{-1} = \{(a_1-1)b\}^{-1}$ and $E\theta^{-2} = \{(a_1-1)(a_1-2)b^2\}^{-1}$, both finite when $a_1 > 2$.

Step 3. With the posterior parameters $a_1 = n+\alpha$ and $b = (t+\beta^{-1})^{-1}$ of Exercise 11.1,

$$\hat\theta^\pi = (a_1-2)b = \frac{n+\alpha-2}{\sum_i \log x_i + \beta^{-1}}.$$

It is smaller than the posterior mean $(n+\alpha)b$: relative loss penalizes overestimation of a small θ heavily. ■

Exercise 11.12 (A weighted loss for a proportion)

Let $X_1,\dots,X_n$ be Bernoulli(θ), $0<\theta<1$, with loss $L(\theta,\hat\theta) = (\hat\theta-\theta)^2/\{\theta(1-\theta)\}$.
(a) Find the Bayes estimator for the uniform prior $U(0,1)$.
(b) Find a minimax estimator of θ.

Solution.

(a)

Step 1. With $t = \sum x_i$, the posterior is Beta$(t+1, n-t+1)$ (Example 11.1), so the posterior expected loss of an action a is proportional to

$$Q(a) = \int_0^1 (a-\theta)^2\theta^{t-1}(1-\theta)^{n-t-1}\,d\theta.$$

Step 2. If $0<t<n$, $Q(a) = a^2B(t,n-t) - 2aB(t+1,n-t) + B(t+2,n-t)$ is finite for all a and is minimized at

$$a^* = \frac{B(t+1,n-t)}{B(t,n-t)} = \frac{\Gamma(t+1)\Gamma(n)}{\Gamma(t)\Gamma(n+1)} = \frac{t}{n}.$$

Step 3. If $t=0$, the integrand behaves like $a^2\theta^{-1}$ near $\theta=0$, so $Q(a)=\infty$ unless $a=0$, while $Q(0) = \int_0^1 \theta(1-\theta)^{n-1}d\theta < \infty$. Hence $a^* = 0 = t/n$. By symmetry ($\theta \leftrightarrow 1-\theta$) $a^* = 1 = t/n$ when $t=n$.

Step 4. Thus the Bayes estimator is $\hat\theta^\pi = \bar X$ for every sample.

(b) The risk of $\bar{X}$ is

$$E\Big[\frac{(\bar{X}-\theta)^2}{\theta(1-\theta)}\;\Big|\;\theta\Big] = \frac{\mathrm{Var}(\bar{X}\mid\theta)}{\theta(1-\theta)} = \frac{\theta(1-\theta)/n}{\theta(1-\theta)} = \frac{1}{n},$$

which is constant in θ. By (a) $\bar{X}$ is the Bayes estimator for the proper prior $U(0,1)$, so by Theorem 11.4 it is minimax. Compare Example 11.11: under unweighted squared error $\bar{X}$ is not minimax, but the weight $1/\{\theta(1-\theta)\}$ equalizes its risk. ■